

# Estimation and Inference on Two-Threshold Value Functionals\*

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July 27, 2026

Version 6

## Abstract

We study estimation and inference for value functionals of the form  $\Theta_0 = \mathbb{E}[\mathbb{1}\{g_0(X) \geq 0\} \mathbb{1}\{h_0(X) \geq 0\} v_0(X)]$ , where  $g_0$  and  $h_0$  are two nonparametric contrast functions estimated from the same experiment. Such *two-threshold* functionals arise whenever a planner imposes a pointwise admissibility screen on top of a benefit margin: leading examples are the share of the population that benefits in the short run but is harmed in the long run under a surrogate-index treatment rule (Athey et al., 2024), and treatment-path welfare components of an optimal dynamic regime.

Our primary contribution is the *second-order* shape calculus of such functionals, which we derive in full and verify numerically. The first shape derivative is a sum of two boundary (Hausdorff) integrals over the two margins. The second contains, in addition to two margin curvature forms, a codimension-two *corner* integral over  $\{g_0 = 0\} \cap \{h_0 = 0\}$  together with two corner *diagonal* terms weighted by the cosine of the intersection angle. We then give a complete order accounting of the terms this generates in a sieve plug-in expansion: the corner contributes  $O(K_n/n)$ , the margins  $O(K_n^{1+1/d}/n)$ , and we show that the margin bound can be attained in suitably phased mesh-resonant examples, while its exact order for a fixed general geometry remains open. The accounting implies that margin-by-margin quadratic debiasing à la Chen and Gao (2026) leaves a corner cross term whose upper bound is rate-relevant on the window  $(d+1)/2 < s \leq d-1$ , nonempty for  $d \geq 4$ ; matching relevance requires the stated optional noncancellation condition. We therefore construct corner-aware

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\*This section extends the single-threshold framework of Chen et al. (2025) and its debiased-estimation companion to value functionals defined by the intersection of two estimated decision boundaries. It is written to be self-contained; notation follows the JMP draft `may_2026.tex`.

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split-sample and leave-one-out corrections, with a tuning-free half-sample jackknife representation.

The inferential results are stated as *conditional* theorems and we are explicit about what is and is not proved. Under estimator-specific third-order and series-linearization remainder conditions that we do not derive from primitive smoothness assumptions, the corner-aware split-sample estimator attains  $n^{-s/(2s+1)}$  under  $s > (d+1)/2$ ; this threshold controls the displayed leading terms but is not, by itself, a primitive sufficient condition for those remainders. A centered normal limit requires strict undersmoothing and, for the plug-in, strict  $s > d$ ; we give the explicit sieve-dimension windows. We also correct an error in the previous version’s *doubly debiased* estimator, which combined the quadratic correction with a cross-fitted Riesz correction: as constructed there, the Riesz score reintroduces exactly the own-half second-order diagonals the quadratic correction removes. We derive the score of the composite split-sample functional, which is  $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A)$  rather than  $\frac{1}{2}D\Theta(\hat{\gamma}^A)$ , and give a three-fold construction that repairs it. Monte Carlo experiments calibrated to the Banerjee et al. (2015) graduation experiment, following the Wasserstein-GAN calibration protocol of Chen and Ritzwoller (2023), illustrate the corner mechanism; their doubly debiased cells evaluate the superseded construction and we read them accordingly.

**Status of the results.** Theorem 3.1 (the second shape derivative), Proposition 2.1, the polarization identity (37), the transversal smooth-face support-edge term of Remark 1.12, and the jackknife finite-difference identity of Lemma 4.2 are established, and verified numerically to ten significant digits (Appendix C). The order accounting of Section 3.2 is a set of *upper bounds* whose exponent arithmetic we have checked symbolically; Remark 3.4 identifies where those bounds are and are not attained. Theorems 2.2, 4.6, 5.3 and 5.8 are *conditional*: each displays, in one place, every remainder condition it uses — Assumption 1.9 collects the main Taylor, empirical-process, moment, and series-linearization conditions, while Assumption 4.1 collects the additional jackknife conditions — and Appendix B, §B.4 lists exactly which of those we have not verified from primitive conditions. The reader should treat the four theorems as specifications of what must be proved, complete as to their hypotheses but not as to their proofs. In particular, Version 4’s proposed  $C^1$  modulus for  $\gamma \mapsto D^2\Theta(\gamma)$  was not merely unproved: it is false because  $D^2\Theta$  depends on the curvature of  $\gamma$ . Version 5 therefore imposes the two actual evaluation-point remainders directly and gives a  $C^2$  modulus only as an optional sufficient condition.

A drafting convention that matters for reading Section 4: we distinguish throughout between the *ideal* split-sample estimators, built from the closed-form  $D^2\Theta$  of Theorem 3.1, and their *feasible* finite-difference (jackknife) counterparts. The decomposition identities hold

exactly *within* each family and only up to a fourth-order error *across* them; v3 conflated the two.

**Revision note (v6).** This version repairs the inferential gaps left in v5. (i) In-sample variance consistency now requires empirical stability of the estimated representer against the true residuals; population  $sd$ -norm convergence alone is not used as an empirical LLN. (ii) The DML product norms, projection, evaluation-fold convention, and  $K_n \rightarrow \infty$  regime are defined explicitly. (iii) The DD theorem requires the sum of the two composite representers to approach the ordinary representer and imposes a conditional Lyapunov condition for the actual fold score; equality of variances is no longer treated as a CLT. (iv) The empirical-measure component of the composite functional is evaluated on the held-out fold and expanded rather than called “common.” (v) Corner noncancellation is optional and invoked only for matching lower bounds. (vi) The remaining common-weight LOO formula and unjustified DD  $\asymp$  statements are corrected. (vii) The support-edge formula is restricted to transversal intersections with smooth support faces, and the mesh-resonance calculations are described as finite-sample evidence rather than asymptotic lower bounds.

**Revision note (v5).** This version corrects nine remaining defects in v4. (i) The  $C^1$  second-derivative modulus is replaced by direct, estimator-specific remainder conditions. A  $C^2$  modulus is stated only as a sufficient condition, because curvature is not continuous in the  $C^1$  topology. (ii) The fourth-derivative condition for the jackknife is retained as an explicit high-level assumption; spline degree ensures existence but does not make its constant uniform in  $K_n$ . (iii) Ratio consistency of the Riesz variance now requires *Riesz-weighted* residual consistency, not merely an unweighted  $L^2$  condition. (iv) The stochastic expansions are written as arm-wise sums, consistently with the four regressions actually estimated. (v) The corrected DD theorem is studentized by the variance of its actual composite score; no unproved equivalence with the ordinary two-band score is used. (vi) The exact corner centering identity is stated only for the oracle derivative statistic; evaluation at the estimated midpoint and the jackknife each carry an explicit remainder. (vii) Stale  $\asymp$ , “negligible,” and cross-reference claims are corrected. (viii) The population-Gram series expansion now includes the empirical-LS remainder instead of being written as an exact identity. (ix) Joint residual nondegeneracy and the moment condition needed for variance ratio consistency are imposed explicitly.

**Revision note (v4).** This version repairs internal defects of v3 that were mathematical, not merely unproved. (i) v3 asserted  $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$  with  $\hat{\Theta}_{SS}$  the ideal derivative-

based estimator. That is false: the polarization identity holds exactly for the *jackknife* family, and across families there is a fourth-order gap. Section 4 now carries both families with their own exact identities ((34) and (36)). (ii) Theorem 2.2(i) controlled only the linear and quadratic terms while the expansion contains a third-order remainder; Assumption 1.9 now collects every remainder condition used anywhere in the paper, and each theorem cites it. (iii) The jackknife centred CLT needs its own sieve window  $a < 3d/(4d + 7)$ , not merely the smoothness condition (33); Theorem 4.6(iii) supplies it. (iv) The second-moment “exact sandwich” of v3 was written for a single contrast regression; the paper’s estimator runs *four separate arm regressions*, so the exact expression is an arm-wise sum of sandwiches (23), the localization error is additive rather than relative, and the  $h_0 = -\tau_Y$  convention flips a sign. (v) Assumption 5.1(c) claimed Hausdorff-distance plus region-measure convergence was “equivalent” to normal-graph convergence. It is not — a surface can be Hausdorff-close while oscillating, with divergent surface measure — so (c) now *requires* a normal-graph representation with  $C^1$  control, plus a convention for empty margins. (vi) The SS expansion also carries an evaluation-point remainder  $\{D^2\Theta(\bar{\gamma}) - D^2\Theta(\gamma_0)\}[\Delta, \Delta]$ , distinct from the Taylor remainder; both are now controlled by a single *second-derivative modulus* condition. Version 5 replaces that invalid  $C^1$  modulus by direct conditions on both actual remainders; see Assumption 1.9(a). (vii) v3’s proof of Theorem 4.6 justified variance consistency by claiming the correction is  $o_p(\text{se})$ . In the SS smoothness window it need not be; consistency follows instead from the asymptotically linear representation. (viii) The corner noncancellation condition was imposed on population coefficients, which does not prevent the spline-weighted sequence from cancelling; it is now imposed on the actual normalized sequence  $\frac{n}{K_n}\mathcal{B}_{C,n}$ , and the full-bracket and cross-term conditions are stated separately, since neither implies the other. (ix) We add a condition that no earlier version or critique identified: in the empirical-measure case the plug-in carries a *stochastic-equicontinuity* term  $\frac{1}{\sqrt{n}}\mathbb{G}_n[m_{\bar{\gamma}} - m_{\gamma_0}]$  of order  $(K_n/n)^{3/4}$ , which imposes  $a < d/(3d - 2)$  and binds for  $d \geq 3$  (Remark 1.11). (x) Assorted repairs: uniform overlap; the stacked standard-deviation norm defined on the summed influence rather than as a scalar; the bilinear extension of  $D^2\Theta$  stated by polarization; “shape (Gâteaux) derivative” replacing “pathwise derivative”; arm-residual consistency added to Proposition 2.4; the composite representer conditions and the empirical-measure fold convention added to Section 5.2;  $\asymp$  demoted to  $O$  in Proposition 4.10; the LOO claim softened; and the internal contradiction between the abstract, the main text, and Appendix B.1 over mesh resonance resolved in favour of the upper-bound reading.

Relative to v1–v2 the corrections recorded in the v3 note stand: the invalidity of the old doubly debiased estimator and its composite-score repair, the separation of rate statements from centred CLTs, the margin/corner order distinction, and the two incorrect claims of

Remarks 5.4 and 5.5. The Monte Carlo tables are reproduced unchanged from v1; Section 6.5 states what they do and do not support.

# 1 Setup and Examples

## 1.1 Data and estimands

The observed data are an i.i.d. sample  $\{Z_i = (X_i, W_i, S_i, Y_i)\}_{i=1}^n$ , where  $X_i \in \mathcal{X} \subset \mathbb{R}^d$  are covariates with density  $f_0$  on a bounded rectangular support,  $W_i \in \{0, 1\}$  is a binary treatment with  $(S_i(0), S_i(1), Y_i(0), Y_i(1)) \perp W_i \mid X_i$  and overlap  $p_0(x) := \mathbb{E}[W_i \mid X_i = x] \in (0, 1)$ ,  $S_i$  is a short-run outcome (or surrogate), and  $Y_i$  is a long-run outcome. Define the arm means and contrasts

$$\mu_{S,0}(x, w) := \mathbb{E}[S_i \mid X_i = x, W_i = w], \quad \tau_S(x) := \mu_{S,0}(x, 1) - \mu_{S,0}(x, 0), \quad (1)$$

$$\mu_{Y,0}(x, w) := \mathbb{E}[Y_i \mid X_i = x, W_i = w], \quad \tau_Y(x) := \mu_{Y,0}(x, 1) - \mu_{Y,0}(x, 0). \quad (2)$$

Throughout we write  $\gamma_0 := (g_0, h_0)$  for a generic pair of identified contrast functions built from  $(\mu_{S,0}, \mu_{Y,0})$  — in the leading example  $g_0 = \tau_S$  and  $h_0 = -\tau_Y$  — and consider the *two-threshold value functional*

$$\Theta(\gamma_0) := \int_{\mathcal{X}} \mathbb{1}\{g_0(x) \geq 0\} \mathbb{1}\{h_0(x) \geq 0\} v_0(x) f(x) dx, \quad (3)$$

where  $v_0$  is a known, user-chosen value weight and  $f$  is the covariate density of the target population. As in the single-threshold analysis we distinguish the *known-density* case ( $f$  given, integral computed numerically) and the *empirical* case ( $f = f_0$  unknown, integral replaced by the sample average  $n^{-1} \sum_i \mathbb{1}\{\cdot\} v_0(X_i)$ ); the first-order asymptotic theory is identical in both, because the additional sampling error of the empirical measure is  $O_p(n^{-1/2})$ , which is negligible relative to the slower boundary-driven rate of (3). It is *not* negligible at the sample sizes of Section 6, and we therefore carry it explicitly in the variance formula (15) below.

## 1.2 Leading examples

**Example 1.1** (Surrogate-induced harm share). *A planner who observes only the short-run outcome adopts the surrogate-index rule “treat iff  $\tau_S(x) \geq 0$ .” The long-run criterion would*

instead require  $\tau_Y(x) \geq 0$ . With  $g_0 = \tau_S$ ,  $h_0 = -\tau_Y$ , and  $v_0 \equiv 1$ , (3) becomes

$$\theta_0 = \mathbb{P}_f(\tau_S(X) \geq 0, \tau_Y(X) \leq 0),$$

the mass of the population that the surrogate rule treats but that is harmed in the long run — a direct diagnostic of the surrogate-index approach of [Athey et al. \(2024\)](#). This is the functional we take to the calibrated Monte Carlo of Section 6.

**Example 1.2** (Admissibility-constrained policy value). *In the two-item-checklist planner problem of the JMP draft, a hard admissibility screen  $g_0(x) \geq 0$  (e.g. no expected severe side effect) is imposed before the benefit margin  $h_0(x) \geq 0$ ; treated-population aggregates take the form (3) with  $v_0$  equal to a utility, cost, or eligibility weight.*

**Example 1.3** (Dynamic treatment-path welfare). *The  $(1, 1)$  path-welfare component of an optimal two-stage dynamic regime,  $V_{11}^* = \mathbb{E}[Y^{(1,1)} \mathbb{1}\{\kappa(S) \geq 0\} \mathbb{1}\{\delta(X^{(1)}) \geq 0\}]$ , has the same two-threshold structure, with the additional complication that the first-stage contrast  $\kappa$  is itself a functional of the second-stage threshold. The static analysis of this section is the template for that case; the extra “moving-boundary-within-a-moving-boundary” terms are treated in the DTR section of the draft.*

### 1.3 Geometry and maintained assumptions

Write  $w(x) := v_0(x)f(x)$  for the boundary weight, and define the two *margins* and the *corner*

$$\begin{aligned} \mathcal{M}_g &:= \{x : g_0(x) = 0, h_0(x) > 0\}, & \mathcal{M}_h &:= \{x : h_0(x) = 0, g_0(x) > 0\}, \\ \mathcal{C} &:= \{x : g_0(x) = h_0(x) = 0\}. \end{aligned} \tag{4}$$

**Assumption 1.4** (Model). *(a)  $\{Z_i\}$  is i.i.d.;  $\mathcal{X}$  is bounded rectangular; and overlap is uniform,  $p_0(x) \in [\underline{p}, 1 - \underline{p}]$  for some  $\underline{p} > 0$  — pointwise  $p_0(x) \in (0, 1)$  does not suffice, because the inverse-propensity weights of (25) and the arm-wise Gram matrices  $R_w$  of (23) must be bounded uniformly; (b)  $\mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda^s(\mathcal{X})$  (Hölder class) with  $s > 1$ , for  $w \in \{0, 1\}$ ; (c) the conditional second moments  $\sigma_S^2(x, w), \sigma_Y^2(x, w)$  of the regression residuals  $\epsilon_{S,i} := S_i - \mu_{S,0}(X_i, W_i)$ ,  $\epsilon_{Y,i} := Y_i - \mu_{Y,0}(X_i, W_i)$  are bounded and bounded away from zero, and the conditional covariance matrix of  $(\epsilon_{S,i}, \epsilon_{Y,i})$  has eigenvalues bounded above and bounded away from zero uniformly in  $(x, w)$ . In particular, the conditional correlation  $\rho_{SY}(x, w) := \text{Corr}(\epsilon_{S,i}, \epsilon_{Y,i} \mid X_i = x, W_i = w)$  is well defined and uniformly bounded away from  $\pm 1$ . Moreover,  $\mathbb{E}[|\epsilon_{o,i}|^{2+\varsigma} \mid X_i, W_i]$  is bounded for some  $\varsigma > 0$  and  $o \in \{S, Y\}$ ; (d)  $v_0$  is bounded and continuously differentiable, and  $f$  is bounded with bounded gradient, absolutely*

continuous w.r.t.  $f_0$  with bounded Radon–Nikodym derivative bounded away from zero on a neighborhood of  $\mathcal{M}_g \cup \mathcal{M}_h$ .

**Assumption 1.5** (Regular margins and transversal corner).  $g_0$  and  $h_0$  are twice continuously differentiable in neighborhoods of  $\{g_0 = 0\}$  and  $\{h_0 = 0\}$  respectively, with (a)  $\|\nabla g_0\| \geq \underline{c} > 0$  on  $\{g_0 = 0\}$  and  $\|\nabla h_0\| \geq \underline{c} > 0$  on  $\{h_0 = 0\}$ ; (b) (transversality) on  $\mathcal{C}$ , the angle  $\omega(x)$  between  $\nabla g_0(x)$  and  $\nabla h_0(x)$ , defined by  $\cos \omega(x) = \langle \nabla g_0, \nabla h_0 \rangle / (\|\nabla g_0\| \|\nabla h_0\|)$ , satisfies  $|\sin \omega(x)| \geq \underline{c}_T > 0$ ; (c)  $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$  and  $\mathcal{H}^{d-2}(\mathcal{C}) < \infty$ ; (d) (no support edge) the level sets  $\{g_0 = 0\}$  and  $\{h_0 = 0\}$  meet  $\partial\mathcal{X}$  only where  $w$  vanishes, so that no margin terminates at the edge of the covariate support and the boundary calculus below involves only the margins and their common corner.

Under Assumption 1.5, the closures of  $\mathcal{M}_g$  and  $\mathcal{M}_h$  in the interior of  $\mathcal{X}$  are  $(d-1)$ -dimensional  $C^2$  submanifolds-with-boundary whose common relative boundary is the  $(d-2)$ -dimensional corner  $\mathcal{C}$  (a finite set of points when  $d = 2$ ; empty when  $d = 1$  or when the constraints never bind jointly). Transversality rules out tangential intersections, at which the corner would inflate to a higher-dimensional set and the second-order calculus below would fail.

We separate the core nondegeneracy condition from an optional corner noncancellation condition because two-sided statements are false without the relevant restriction: a functional with  $v_0 \equiv 0$  or empty margins has no boundary irregularity, while a signed corner coefficient can cancel even when a corner exists.

**Assumption 1.6** (Parameter space and nondegeneracy). (a) (Parameter space.)  $\Lambda_c^s(\mathcal{X})$  denotes the ball of radius  $c$  in the Hölder space of order  $s$  on  $\mathcal{X}$ , and the statistical model is

$$\mathcal{P}_c^s := \left\{ P : \mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda_c^s \text{ for } w \in \{0, 1\}, \text{ and } P \text{ satisfies} \right. \\ \left. \text{Assumptions 1.4–1.5 and (b) below with common constants} \right\}.$$

Rates and minimax statements are uniform over  $\mathcal{P}_c^s$ . (b) (Nonvanishing boundary weight.)

$$\mathcal{H}^{d-1}(\mathcal{M}_g) \vee \mathcal{H}^{d-1}(\mathcal{M}_h) \geq \underline{m} > 0 \quad \text{and} \quad \inf_{\mathcal{M}_g \cup \mathcal{M}_h} w \geq \underline{w} > 0.$$

At least one margin must be nondegenerate; this is what makes  $\Theta_0$  irregular and, with the upper bound  $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$  of Assumption 1.5(c), delivers  $\sigma_n^2 \asymp K_n^{1/d}$ . Without it the representer norm can be  $O(1)$  and  $\Theta_0$  is  $\sqrt{n}$ -estimable. It is deliberately “ $\vee$ ” and not “ $\wedge$ ”: the single-threshold submodel used for the minimax lower bound in Theorem 2.2(ii)

has  $\mathcal{M}_h = \emptyset$ , and requiring both margins to be nondegenerate would exclude it from  $\mathcal{P}_c^s$  and break the nesting argument.

**Assumption 1.7** (Optional corner noncancellation). *Let  $\mathcal{B}_{\mathcal{C},n}$  and  $\mathcal{B}_{\mathcal{C},n}^\times$  denote, respectively, the full corner coefficient (27) and its cross-term component (28) defined in Section 3.2; both are deterministic sequences depending on the sieve through the normalized second-moment functions  $\mathcal{V}_{\bullet,n}/K_n$ . When  $\mathcal{C} \neq \emptyset$ , either of the following may be imposed separately:*

$$(c1) \quad \liminf_n \left| \frac{n}{K_n} \mathcal{B}_{\mathcal{C},n} \right| > 0, \quad (c2) \quad \liminf_n \left| \frac{n}{K_n} \mathcal{B}_{\mathcal{C},n}^\times \right| > 0. \quad (5)$$

(c1) is used for lower bounds on the plug-in's corner bias, (c2) for the bias that survives margins-only debiasing (Proposition 4.10). Neither implies the other, and neither follows from pointwise nonvanishing of  $\varrho_{gh}$ : the bracket in (27) is signed, its three components may offset, contributions from different corner points may offset, and the spline-dependent weights  $\mathcal{V}_{\bullet,n}(x)/K_n$  vary across corner points and with  $n$ . Each is needed only for its corresponding lower bound; no upper bound below uses either. If  $\mathcal{C} = \emptyset$ , neither condition is invoked and both corner coefficients are identically zero.

**Remark 1.8** ( $\mathcal{C}$  is not automatically material). *Assumption 1.7 is a genuine optional restriction, not a formality. It is not part of  $\mathcal{P}_c^s$  and is invoked only when a matching corner lower bound is claimed. At an orthogonal corner ( $\cos \omega = 0$ ) with  $\rho_{SY} = 0$  the bracket vanishes identically; with  $d = 2$  and several corner points the terms carry the sign of  $\cos \omega$  at each point, which typically varies; and even with a fixed nonzero pointwise coefficient the  $n$ -dependent weights could conspire to cancel along a subsequence, which is why (5) is imposed on the normalized sequence rather than on population quantities. In the calibrated design of Section 6 the ten corners do not cancel (Table 4), but this is a property of that design.*

**Assumption 1.9** (Remainders). *Write  $\bar{\Delta}$  for a generic first-stage error,  $\eta$  for a generic direction, and  $\text{se}_n := \sqrt{K_n^{1/d}/n}$ .*

(a) (Taylor and evaluation-point remainders.) *With  $R_3 := \frac{1}{2}\{D^2\Theta(\gamma_\xi) - D^2\Theta(\gamma_0)\}[\bar{\Delta}, \bar{\Delta}]$  the exact mean-value remainder of the second-order expansion (17), and  $R_3^{SS}$  the total remainder of the corrected estimator defined in (38), the relevant one is  $O_p(r_n^*)$  where a rate is claimed and  $o_p(\text{se}_n)$  where a centred limit is claimed. This is a direct, estimator-specific condition; no continuity of  $D^2\Theta$  in the  $C^1$  topology is presumed.*

(b) (Stochastic equicontinuity; empirical-measure case only.) *With  $m_\gamma(x) := \mathbb{1}\{g(x) \geq 0\}\mathbb{1}\{h(x) \geq 0\}v_0(x)$ ,*

$$n^{-1/2} \left| \mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}] \right| = o_p(\text{se}_n) \quad \text{for centered limits,} \quad \mathbb{G}_n := \sqrt{n}(\mathbb{P}_n - P). \quad (6)$$



For rate statements the corresponding requirement is  $n^{-1/2}|\mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}]| = O_p(r_n^*)$ .

(c) (Lyapunov.) For some  $\varsigma > 0$ ,  $n^{-\varsigma/2}\mathbb{E}[|v_S^*\epsilon_S + v_Y^*\epsilon_Y|^{2+\varsigma}]/\sigma_n^{2+\varsigma} \rightarrow 0$ .

(d) (Series linearization; sieve results only.) For every full- and split-sample series fit, the population-Gram expansion (21) has remainders  $(r_g, r_h)$ . After these are substituted in the relevant linear, margin, corner, and finite-difference expressions, their total contribution is  $O_p(r_n^*)$  for a rate statement and  $o_p(\text{se}_n)$  for a centered limit. The corresponding second-moment remainders in (27) are smaller than the displayed leading orders. This condition is imposed directly; an empirical least-squares coefficient does not equal its population-Gram linearization identically.

Collecting the remainder conditions in one place is deliberate. Every theorem below cites Assumption 1.9 explicitly, so that no statement silently omits a term of its own expansion — a defect of v3's Theorem 2.2(i), which controlled the linear and quadratic terms of (17) but not  $R_3$ .

**Remark 1.10** (Why the remainder is imposed directly). *Version 4 tried to deduce Assumption 1.9(a) from*

$$|\{D^2\Theta(\gamma) - D^2\Theta(\gamma_0)\}[\eta, \eta]| \lesssim \varpi(\|\gamma - \gamma_0\|_{C^1}) \|\eta\|_\infty \|\eta\|_{C^1}.$$

That inequality is false in general. Already in one dimension, with  $g_0(x) = x$ , constant weight, and  $g_\delta(x) = x + \delta^3\{\cos(x/\delta^2) - 1\}$ , one has  $\|g_\delta - g_0\|_{C^1} = O(\delta)$  and the same regular zero set, but  $g_\delta''(0) = -\delta^{-1}$ . The single-threshold specialization of  $D^2\Theta(\gamma)[1, 1]$  contains  $g''(0)/g'(0)^3$  and therefore diverges.

A possible sufficient route is instead a curvature-controlling modulus,

$$|\{D^2\Theta(\gamma) - D^2\Theta(\gamma_0)\}[\eta, \eta]| \leq \varpi_2(\|\gamma - \gamma_0\|_{C^2}) \|\eta\|_\infty \|\eta\|_{C^1}, \quad \varpi_2(0+) = 0, \quad (7)$$

under uniform geometry and a common modulus of continuity for the Hessians. But verifying (7) uniformly requires curvature control that is not part of the stated Hölder class at low  $s$ , and  $C^2$  consistency of a growing sieve imposes additional inverse-inequality costs. We therefore do not use (7) to advertise a primitive smoothness threshold: the two actual remainders are imposed directly in Assumption 1.9(a).

**Remark 1.11** (The equicontinuity term, and why it is new). Condition (6) is not a technicality and was absent from v1–v3. Writing  $\hat{\Theta} - \Theta(\hat{\gamma}) = n^{-1/2}\mathbb{G}_n[m_{\gamma_0}] + n^{-1/2}\mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}]$ , the first term is the  $O_p(n^{-1/2})$  empirical-measure error carried explicitly in (15). The second

is a genuine empirical-process term that earlier versions absorbed into “ $O_p(n^{-1/2})$ ” without argument. The class  $\{m_\gamma : \gamma \in \mathcal{S}_{K_n}^2\}$  is VC of index  $O(K_n)$ , and  $\|m_{\hat{\gamma}} - m_{\gamma_0}\|_{L^2}^2 \lesssim \|\hat{\gamma} - \gamma_0\|_\infty$  because the two regions differ in a band of that width around the margins, so the standard maximal inequality gives

$$n^{-1/2} |\mathbb{G}_n[m_{\hat{\gamma}} - m_{\gamma_0}]| = O_p\left(\|\hat{\gamma} - \gamma_0\|_\infty^{1/2} \sqrt{K_n \log K_n/n}\right) = \tilde{O}_p((K_n/n)^{3/4}), \quad (8)$$

using  $\|\hat{\gamma} - \gamma_0\|_\infty \asymp \sqrt{K_n/n}$  up to logs. This is  $o(\text{se}_n)$  iff

$$a < \frac{d}{3d-2}, \quad \text{compatible with } a > \frac{d}{2s+1} \text{ iff } s > \frac{3(d-1)}{2}. \quad (9)$$

At  $d = 2$  the edge is  $a < 1/2$ , which coincides with the off-diagonal edge  $d/(d+2)$  and with  $s > 3/2 = (d+1)/2$ , so it does not bind in the designs of Section 6. For  $d \geq 3$  it is the binding upper edge ( $3/7$  at  $d = 3$  against  $3/5$ ), and it demands  $s > 3$  rather than  $s > 2$ . Two caveats. The known-density case has no such term, the integral being computed against  $f$  rather than estimated. And (8) uses a global VC bound; a local-complexity argument exploiting that only the  $O(\kappa^{-(d-1)})$  basis functions meeting the margins are active could plausibly improve it, which we have not attempted. We state (6) as a condition and (8) as the sufficient bound we can currently supply.

**Remark 1.12** (Assumption 1.5(d) is substantive). Condition (d) is not a technicality. Suppose a margin exits through the relative interior of a smooth face of  $\partial\mathcal{X}$  at a point where  $w > 0$ , the intersection is transversal, and  $|\sin \omega_\partial| \geq c_\partial > 0$ . Then  $\mathcal{M}_g$  has a second kind of relative boundary — a support edge — and the second derivative (18) acquires an extra  $\mathcal{H}^{d-2}$  integral over  $\{g_0 = 0\} \cap \partial\mathcal{X}$ . Because the support is fixed while the level set flows, only the “induced motion of the cut” mechanism (step (4) of Appendix A) operates: the edge contributes a  $u^2$  diagonal term

$$- \int_{\{g_0=0\} \cap \partial\mathcal{X}} \frac{u^2 w \cos \omega_\partial}{\|\nabla g_0\|^2 |\sin \omega_\partial|} d\mathcal{H}^{d-2}, \quad \cos \omega_\partial := \frac{\langle \nabla g_0, \nu \rangle}{\|\nabla g_0\|},$$

with  $\nu$  the outward unit normal of  $\mathcal{X}$ , and no cross term (there is no second perturbed surface to cross with). It vanishes when the margin meets  $\partial\mathcal{X}$  orthogonally. The first derivative (10) is unaffected. Tangential contact ( $\sin \omega_\partial = 0$ ), intersection at a corner of the rectangular support, and changes in contact topology are not covered by this display and require separate local analysis. Appendix C, item 6, verifies this display to ten digits. We flag it because the principal calibrated design of Section 6 violates (d): see Section 6.5, item (L3), which reports twelve such crossings.

## 2 First-Order Theory

### 2.1 Pathwise derivative: two boundary integrals

For directions  $(u, v)$  (perturbations of  $(g_0, h_0)$  induced by perturbations of the four arm means), let  $\gamma_t := (g_0 + tu, h_0 + tv)$ .

**Proposition 2.1** (First shape derivative). *Under Assumptions 1.4–1.5, for directions  $(u, v) \in C^1(\mathcal{X})^2$ ,*

$$D\Theta(\gamma_0)[u, v] := \frac{d}{dt}\Theta(\gamma_t)\Big|_{t=0} = \int_{\mathcal{M}_g} \frac{u(x)w(x)}{\|\nabla g_0\|} d\mathcal{H}^{d-1}(x) + \int_{\mathcal{M}_h} \frac{v(x)w(x)}{\|\nabla h_0\|} d\mathcal{H}^{d-1}(x). \quad (10)$$

*Proof sketch.* Apply the generalized Leibniz rule (Reynolds transport theorem; Delfour and Zolésio, 2001, Thm. 4.2) to the moving region  $\Omega_t = \{g_t \geq 0\} \cap \{h_t \geq 0\}$  with fixed integrand  $w$ . The boundary  $\partial\Omega_0$  decomposes, up to the  $\mathcal{H}^{d-1}$ -null corner, into the two faces  $\mathcal{M}_g$  and  $\mathcal{M}_h$ ; on each face the normal velocity is that of the corresponding level set,  $-u/\|\nabla g_0\|$  and  $-v/\|\nabla h_0\|$  respectively, and the coarea representation of the surface measure yields (10).  $\square$

Three features of (10) drive everything that follows.

(i) *Irregularity.* Each term is a  $(d-1)$ -dimensional Hausdorff integral: a bounded linear functional of  $(u, v)$  on  $C^1$  but *not* on  $L^2(f)$ . Hence, exactly as in the single-threshold case, no  $L^2$ -Riesz representer (influence function) exists whenever  $w$  does not vanish on  $\mathcal{M}_g \cup \mathcal{M}_h$ , and  $\Theta_0$  is not  $\sqrt{n}$ -estimable. The welfare-type exception — boundary weights that vanish on their own margins, as for total dynamic welfare  $V^*$  in the DTR section — is the regular special case.

(ii) *Two bands.* The derivative aggregates information about *two different contrasts* on *two different thin sets*. The variance of a plug-in estimator therefore has two band contributions, and, because  $\hat{g}$  and  $\hat{h}$  are estimated from the same observations, a within-unit covariance term appears whenever  $\rho_{SY} \neq 0$ .

(iii) *The corner is invisible at first order.*  $\mathcal{C}$  has  $\mathcal{H}^{d-1}$ -measure zero and does not appear in (10); it enters only the second-order expansion (Section 3), which is precisely why it has been ignorable in first-order analyses and why it becomes relevant for debiasing.

### 2.2 Plug-in estimator, rate, and minimax optimality

Let  $\hat{\mu}_S, \hat{\mu}_Y$  be stacked sieve least-squares estimators of the arm means on tensor-product B-spline bases of dimension  $K_n$  (per arm) and degree exceeding  $s$  — required for the approximation rate  $\|b_g\|_\infty \lesssim K_n^{-s/d}$  used throughout, and raised further to degree 5 in Assump-

tion 4.1(b) once the jackknife is introduced — let  $(\hat{g}, \hat{h})$  be the implied contrast estimates, and let

$$\hat{\Theta} := \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{\hat{g}(X_i) \geq 0\} \mathbb{1}\{\hat{h}(X_i) \geq 0\} v_0(X_i)$$

be the (empirical-measure) plug-in estimator; in the known-density case the sample average is replaced by numerical integration against  $f$ .

**Theorem 2.2** (Plug-in: rate, minimax optimality, and a centered CLT). *Let Assumptions 1.4–1.6 hold together with the sieve regularity conditions of Chen et al. (2025, Thm. 3) applied to each of the four arm regressions, and write  $K_n \asymp n^a$  and  $\sigma_n^2 := \|v_{K_n}^*\|_{sd}^2$ , with  $v_{K_n}^*$  the sieve-Riesz representer of (10) defined in (12). Under Assumption 1.6(b),  $\sigma_n^2 \asymp K_n^{1/d}$ .*

- (i) (Rate.) *Let  $a = d/(2s+1)$ ,  $s \geq d$ , and impose Assumption 1.9(a),(b),(d) in their  $O_p(r_n^*)$  form. Then  $\hat{\Theta} - \Theta_0 = O_p(r_n^*)$ ,  $r_n^* = n^{-s/(2s+1)}$ , with a constant depending on  $\mathcal{P}_c^s$  only through  $(c, \underline{c}, \underline{c}_T, \underline{w}, \underline{m}, \underline{p})$  and the sieve constants — hence uniformly over  $\mathcal{P}_c^s$ , provided the cited sieve regularity conditions and Assumption 1.9 hold uniformly over the class, which we assume but have not verified.*
- (ii) (Minimax.)  *$r_n^*$  is a minimax lower-bound rate over  $\mathcal{P}_c^s$ , so the plug-in rate in (i) is sharp.*
- (iii) (Centered CLT.) *Impose Assumption 1.9(a)–(d) in their  $o_p(\text{se}_n)$  form and*

$$\frac{d}{2s+1} < a < \min \left\{ \frac{d}{2d+1}, \frac{d}{3d-2} \right\}, \quad (11)$$

*where the second edge is present only in the empirical-measure case (Remark 1.11); since  $d/(2d+1) < d/(3d-2)$  iff  $d < 3$ , the quadratic edge binds for  $d = 2$  and the equicontinuity edge for  $d \geq 3$ . The window is nonempty if and only if  $s > d$  strictly (known density, or  $d = 2$ ), respectively  $s > \max\{d, 3(d-1)/2\}$  (empirical measure,  $d \geq 3$ ). Then  $\sqrt{n}(\hat{\Theta} - \Theta_0)/\sigma_n \xrightarrow{d} \mathcal{N}(0, 1)$ , and the attained rate is  $\text{se}_n = n^{-(1-a/d)/2}$ , slower than  $r_n^*$  but approaching it as  $a \downarrow d/(2s+1)$ .*

*Proof sketch. Rate.* The expansion (17) has a linear term that is a sum of two submanifold integrals of sieve LS errors; each is handled by the single-threshold argument (Chen and Gao, 2026, Thm. 8; Chen et al., 2025, Thm. 3) with the truncation indicators  $\mathbb{1}\{h_0 > 0\}$ ,  $\mathbb{1}\{g_0 > 0\}$  carried through as fixed bounded weights. Its stochastic part is  $O_p(\sqrt{K_n^{1/d}/n}) = O_p(r_n^*)$  at  $a = d/(2s+1)$  and its bias part is  $O(K_n^{-s/d}) = O(r_n^*)$ . The quadratic term's dominant piece is the margin stochastic diagonal, bounded by  $K_n^{1+1/d}/n$ , which Table 1 shows is  $O(r_n^*)$  at  $a = d/(2s+1)$  exactly when  $s \geq d$ . The third-order and equicontinuity remainders are

supplied by Assumption 1.9(a),(b),(d); v3 omitted these from this part of the statement. *Minimax (nesting)*. Consider the subclass of  $\mathcal{P}_c^s$  in which  $h_0 \geq \underline{\eta} > 0$  on the support, so that  $\mathbb{1}\{h_0 \geq 0\} \equiv 1$  and  $\Theta$  reduces to the single-threshold value functional  $V(g_0)$ . This subclass has  $\mathcal{M}_h = \mathcal{C} = \emptyset$ ; it lies in  $\mathcal{P}_c^s$  because Assumption 1.6(b) asks only that *one* margin be nondegenerate and no maintained condition requires  $\mathcal{C} \neq \emptyset$ . (Had (b) been stated with “ $\wedge$ ” the subclass would be empty and the argument would fail — see the remark following Assumption 1.6.) Every estimator of  $\Theta_0$  is then an estimator of  $V(g_0)$ , and the minimax lower bound of Chen and Gao (2026, Thm. 5) / Chen et al. (2025, Thm. 5) applies to that submodel. *Centered CLT*. Jointly asymptotically normal linear terms follow from the Cramér–Wold device applied to the stacked representer, with the Lyapunov condition Assumption 1.9(c) — which v3 invoked from Assumption 5.1(g), a hypothesis of a different theorem. For the limit to be *centered*, the linear bias, the quadratic diagonal and the equicontinuity term must all be  $o(\text{se}_n)$ :  $K_n^{-s/d} = o(\text{se}_n)$  iff  $a > d/(2s+1)$ ;  $K_n^{1+1/d}/n = o(\text{se}_n)$  iff  $a < d/(2d+1)$ ; and (8) is  $o(\text{se}_n)$  iff  $a < d/(3d-2)$ . These are compatible iff  $2s+1 > 2d+1$ , respectively  $2s+1 > 3d-2$ .  $\square$

**Remark 2.3** (Why the rate-optimal  $K_n$  does not give a centered limit). *v1 and v2 asserted asymptotic normality at  $K_n \asymp n^{d/(2s+1)}$  under  $s \geq d$ . This is not right, for two separate reasons, both visible in (11). First, at the rate-optimal  $K_n$  the sieve approximation bias contributes  $D\Theta[b] = O(K_n^{-s/d}) = O(r_n^*)$  to the linear term, exactly the order of its standard deviation  $\sqrt{K_n^{1/d}/n}$ ; the limit is normal but not centered. This is the usual undersmoothing requirement and it is not special to two thresholds. Second, at  $s = d$  the upper bound  $K_n^{1+1/d}/n$  for the margin quadratic diagonal is exactly of order  $r_n^*$ . Thus the maintained bounds deliver  $O(r_n^*)$ , not  $o(r_n^*)$ : enough for the rate statement (i), not for a centered limit without an additional cancellation argument. Strict  $s > d$  is needed, with  $K_n$  chosen inside (11). In the revision note’s language, “negligible under  $s \geq d$ ” should read “no larger than the target rate under  $s \geq d$ ”; negligibility needs the strict inequality. Remark 3.4 notes that this conclusion is conservative if the margin bound is not attained.*

## 2.3 Two-band sieve-Riesz variance

Let  $\psi(x, w)$  denote the stacked per-arm sieve regressors of one outcome equation and  $b(x) := (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$  the contrast basis. The (infeasible) sieve-Riesz representer of (10) restricted to the sieve space is, for each outcome equation  $o \in \{S, Y\}$ ,

$$v_{K_n,o}^*(x, w) := \psi(x, w)' R_{K_n,o}^{-1} D_o \Theta(\gamma_0) [\psi], \quad R_{K_n,o} := \mathbb{E}[\psi(X_i, W_i) \psi(X_i, W_i)'], \quad (12)$$

where  $D_o\Theta[\psi]$  stacks the coordinatewise boundary integrals of (10) over the relevant margin. Feasible versions replace population by empirical moments and the Hausdorff integrals by  $\varepsilon$ -band sample averages. Because an  $\varepsilon$ -band average of a function  $\varphi$  over  $\{|\hat{g}| < \varepsilon\}$  converges by the coarea formula to  $\int_{\{g_0=0\}} \varphi f_0 / \|\nabla g_0\| \, d\mathcal{H}^{d-1}$  — against the *sampling* density  $f_0$ , not against  $w = v_0 f$  — the band average must be reweighted by  $w/f_0$ :

$$\begin{aligned}\widehat{D_g\Theta}[b] &= \frac{+1}{2\varepsilon_S} \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{|\hat{g}(X_i)| < \varepsilon_S, \hat{h}(X_i) > 0\} \frac{v_0(X_i)f(X_i)}{\hat{f}_0(X_i)} b(X_i), \\ \widehat{D_h\Theta}[b] &= \frac{-1}{2\varepsilon_Y} \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{|\hat{\tau}_Y(X_i)| < \varepsilon_Y, \hat{g}(X_i) \geq 0\} \frac{v_0(X_i)f(X_i)}{\hat{f}_0(X_i)} b(X_i).\end{aligned}\tag{13}$$

In the empirical-measure case  $f = f_0$  and the weight reduces to  $v_0(X_i)$  — *no density estimate is required* — and in Example 1.1, where  $v_0 \equiv 1$ , it reduces to 1, which is the case implemented in Section 6. (v1 stated (13) without the weight, i.e. only for that case.) The known-density case  $f \neq f_0$  does require  $\hat{f}_0$ , and then Proposition 2.4 needs the extra condition

$$\inf_{\mathcal{N}(\mathcal{M}_g \cup \mathcal{M}_h)} f_0 > 0, \quad \left\| \hat{f}_0 - f_0 \right\|_{\infty, \mathcal{N}(\mathcal{M}_g \cup \mathcal{M}_h)} = o_p(1),\tag{14}$$

on a neighborhood  $\mathcal{N}(\cdot)$  of the margins, which in turn requires smoothness of  $f_0$  that Assumption 1.4(d) does not impose (bounded gradient gives a uniform rate only for  $d$  small). We flag this rather than develop it: all results below are stated for the empirical-measure case, and the known-density case carries (14) as an additional maintained condition. The displays are instantiated for the harm-share convention of Example 1.1: for a generic pair  $(g, h)$  both bands carry a + sign in  $(g, h)$  coordinates, and the − sign of the second display is the Jacobian of  $h_0 = -\tau_Y$  (raising  $\tau_Y$  shrinks the harmed region).

The variance estimator sums the *per-unit* influence contributions of the two outcome equations *before* squaring, and adds the empirical-measure term:

$$\widehat{\text{Var}}(\hat{\Theta}) = \frac{\hat{\sigma}_n^2}{n} + \frac{\hat{\zeta}^2}{n}, \quad \hat{\sigma}_n^2 := \frac{1}{n} \sum_{i=1}^n \left( \widehat{v}_S^*(X_i, W_i) \hat{\epsilon}_{S,i} + \widehat{v}_Y^*(X_i, W_i) \hat{\epsilon}_{Y,i} \right)^2,\tag{15}$$

where  $\hat{\zeta}^2 := n^{-1} \sum_i (\mathbb{1}\{\hat{g}_i \geq 0\} \mathbb{1}\{\hat{h}_i \geq 0\} v_0(X_i) - \hat{\Theta})^2$  reduces to  $\hat{\Theta}(1 - \hat{\Theta})$  when  $v_0 \equiv 1$ . Summing the two influence contributions before squaring means the within-unit covariance of the short- and long-run residuals ( $\rho_{SY} \neq 0$ ) is captured automatically. Here the stacked standard-deviation norm is defined on the *summed* per-unit influence,

$$\|(v_S, v_Y)\|_{sd}^2 := \mathbb{E}[\{v_S(X_i, W_i)\epsilon_{S,i} + v_Y(X_i, W_i)\epsilon_{Y,i}\}^2],\tag{16}$$

not as a scalar  $\mathbb{E}[v^2\sigma^2]$ : the within-unit correlation  $\rho_{SY}$  enters  $\sigma_n^2$  through the cross term of (16) and would otherwise be lost.  $v_{K_n}^*$  denotes the stacked  $(S, Y)$  representer. Adding the two components of (15) ignores their covariance — the same  $X_i$  enter the empirical measure and the first-stage residuals — which is of smaller order than either and which we do not estimate. Because  $\sigma_n^2 \asymp K_n^{1/d} \rightarrow \infty$  while  $\hat{\varsigma}^2 = O(1)$ , the second term of (15) is asymptotically negligible; it is 6–12% of the reported standard error at the sample sizes of Section 6, and omitting it produces a spurious SE/SD ratio of 0.87–0.95.

**Proposition 2.4** (Ratio consistency). *Let Assumptions 1.4–1.6 hold, impose the Lyapunov condition Assumption 1.9(c), and suppose: (a)  $\varepsilon_n \downarrow 0$  with  $\varepsilon_n^{-1}(\|\hat{g} - g_0\|_\infty + \|\hat{h} - h_0\|_\infty) = o_p(1)$  and  $n\varepsilon_n/(K_n \log K_n) \rightarrow \infty$ ; (b) the empirical Gram matrix satisfies  $\|\hat{R}_{K_n} R_{K_n}^{-1} - I\|_{\text{op}} = o_p(1)$ , which for tensor B-splines holds when  $K_n \log K_n/n \rightarrow 0$ ; (c) the feasible band vectors converge in the representer norm,  $\|\hat{v}_{K_n}^* - v_{K_n}^*\|_{sd} = o_p(\sigma_n)$ ; (c') writing  $\delta_{o,i} := \hat{\epsilon}_{o,i} - \epsilon_{o,i}$  for the residual error induced by the four arm regressions, the error is small in the Riesz-weighted norm,*

$$\frac{1}{n} \sum_{i=1}^n \{\hat{v}_S^*(X_i, W_i)\delta_{S,i} + \hat{v}_Y^*(X_i, W_i)\delta_{Y,i}\}^2 = o_p(\sigma_n^2).$$

*A sufficient condition is uniform consistency of all four arm means together with*

$$n^{-1} \sum_i \{(\hat{v}_{S,i}^*)^2 + (\hat{v}_{Y,i}^*)^2\} = O_p(\sigma_n^2).$$

*Unweighted residual  $L^2$  consistency is not sufficient because the representer concentrates on shrinking margin tubes. As before, consistency of the contrasts  $\hat{g}, \hat{h}$  alone does not imply arm-residual consistency;*

*(c'') the estimated representer is stable against the true in-sample residuals,*

$$\frac{1}{n} \sum_{i=1}^n \left[ \{\hat{v}_S^* - v_S^*\}(X_i, W_i)\epsilon_{S,i} + \{\hat{v}_Y^* - v_Y^*\}(X_i, W_i)\epsilon_{Y,i} \right]^2 = o_p(\sigma_n^2).$$

*This is distinct from the population norm in (c): because  $\hat{v}^*$  is constructed from the same observations, population sd-norm convergence does not imply this empirical statement. It follows, for example, after cross-fitting the representer or from a uniform operator LLN for the residual-weighted sieve sandwich matrix; (d) (known-density case only) (14) holds. Then  $\hat{\sigma}_n^2/\sigma_n^2 \xrightarrow{p} 1$ , and whenever  $\sqrt{n}(\hat{\Theta} - \Theta_0)/\sigma_n$  has a centered normal limit (Theorem 2.2(iii)) so does the studentized statistic  $\sqrt{n}(\hat{\Theta} - \Theta_0)/\hat{\sigma}_n$ .*

*Proof sketch.* Let  $q_i := v_S^*(X_i, W_i)\epsilon_{S,i} + v_Y^*(X_i, W_i)\epsilon_{Y,i}$  and let  $\hat{q}_i$  be its feasible analogue. The Lyapunov condition implies  $n^{-1} \sum_i q_i^2/\sigma_n^2 \rightarrow_p 1$ . Conditions (c')–(c'') give  $n^{-1} \sum_i (\hat{q}_i - q_i)^2 =$



$o_p(\sigma_n^2)$ , so Cauchy–Schwarz yields  $n^{-1} \sum_i \hat{q}_i^2 / \sigma_n^2 \rightarrow_p 1$ . This empirical step is precisely what population representer convergence in (c) could not supply by itself.  $\square$

Conditions (b) and (c) are new relative to v1–v2, which imposed only (a). They are not decoration: (13) estimates a  $K_n$ -vector from the  $\asymp n\varepsilon_n$  observations that fall in a band and then inverts a  $K_n \times K_n$  Gram matrix against it, so an empirical-process argument with the usual logarithmic factor is required, and *consistency of  $\hat{v}^*$  in the standard-deviation norm* is what makes  $\hat{\sigma}_n^2$  estimate  $\sigma_n^2$  — a product-rate condition of the kind used in Theorem 5.3 does not imply it. We have not verified (b)–(c'') from primitive conditions; see Appendix B, §B.4.

Ratio consistency is the relevant notion because  $\sigma_n^2 \rightarrow \infty$  in the irregular regime. The proof adapts the single-threshold argument to each band, using Evans and Garipey (2015, Thm. 3.13(iii)) for the  $\varepsilon$ -band-to-Hausdorff convergence on each truncated margin. The single-threshold statement requires only  $n\varepsilon_n \rightarrow \infty$ ; we impose the stronger  $n\varepsilon_n/K_n \rightarrow \infty$  because (13) estimates a  $K_n$ -vector from the  $\asymp n\varepsilon_n$  observations that fall in the band, and the feasible representer inverts a  $K_n \times K_n$  Gram matrix against it. This is a sufficient-side strengthening, not an imported result. The requirement  $\varepsilon_n \rightarrow 0$  is essential: a bandwidth that stabilizes at a fixed positive fraction of  $\text{sd}(\hat{\tau})$  estimates a smeared functional, not  $D\Theta$ , and Proposition 2.4 does not apply to it. See Section 6.5, item (L5).

## 3 Second-Order Theory: the Corner

### 3.1 The second shape derivative

The plug-in expansion

$$\hat{\Theta} - \Theta_0 = D\Theta(\gamma_0)[\hat{\gamma} - \gamma_0] + \frac{1}{2}D^2\Theta(\gamma_0)[\hat{\gamma} - \gamma_0, \hat{\gamma} - \gamma_0] + R_3, \quad (17)$$

with  $R_3$  a third-order remainder, is what limits the smoothness requirement of Theorem 2.2. The new object is  $D^2\Theta$ .

**Theorem 3.1** (Second derivative: two margins and a corner). *Under Assumptions 1.4–1.5, with  $n_g := \nabla g_0 / \|\nabla g_0\|$ ,  $H_g := \text{div}(n_g)$  the mean curvature of  $\{g_0 = 0\}$  (and symmetrically for  $h$ ),  $\Theta$  is twice shape (Gâteaux) differentiable along  $C^1$  directions with*

$$D^2\Theta(\gamma_0)[(u, v), (u, v)] = Q_g[u, u] + Q_h[v, v] + 2 \int_C \frac{u(x) v(x) w(x)}{\|\nabla g_0\| \|\nabla h_0\| |\sin \omega(x)|} d\mathcal{H}^{d-2}(x), \quad (18)$$



where the margin quadratic forms are

$$Q_g[u, u] = - \int_{\mathcal{M}_g} \left[ \frac{u (n_g \cdot \nabla u) w}{\|\nabla g_0\|^2} + \frac{u}{\|\nabla g_0\|} \left\{ n_g \cdot \nabla \left( \frac{u w}{\|\nabla g_0\|} \right) + \frac{u w}{\|\nabla g_0\|} H_g \right\} \right] d\mathcal{H}^{d-1} \\ - \int_{\mathcal{C}} \frac{u^2(x) w(x) \cos \omega(x)}{\|\nabla g_0\|^2 |\sin \omega(x)|} d\mathcal{H}^{d-2}(x), \quad (19)$$

and  $Q_h$  is the mirror image (swap  $g \leftrightarrow h$ ,  $u \leftrightarrow v$ , and restrict to  $\mathcal{M}_h$ ).

*Proof sketch; full derivation in Appendix A.* Differentiate each term of (10) along  $\gamma_t$ . Three effects arise for the  $\mathcal{M}_g$ -term: (i) the integrand  $u w / \|\nabla g_t\|$  changes (the  $n_g \cdot \nabla u$  term); (ii) the level set  $\{g_t = 0\}$  flows with normal velocity  $-u / \|\nabla g_0\|$ , producing the tangential-gradient and curvature terms of the moving-level-set lemma of Chen and Gao (2026, Lem. PathD–Level) applied with weight  $u w / \|\nabla g_0\|$ ; (iii) the *truncation*  $\{h_t > 0\}$  of  $\mathcal{M}_g$  moves. Effect (iii) has two parts: the direct motion of the cut in the  $v$ -direction, an edge integral over  $\mathcal{C}$  with within-manifold coarea factor  $\|P_T \nabla h_0\| = \|\nabla h_0\| |\sin \omega|$  (this is the cross term, and it appears once from each margin, giving the factor 2 in (18)); and the induced motion of the edge caused by the flow of the manifold itself across the fixed cut — pulling the cut condition back through the normal flow perturbs it by  $-tu \|\nabla h_0\| \cos \omega / \|\nabla g_0\|$ , which yields the diagonal corner term in (19). Transversality (Assumption 1.5(b)) keeps all corner factors bounded.  $\square$

When  $d = 2$  the corner is a finite set of points and  $\mathcal{H}^0$  is counting measure: the cross term is  $2 \sum_{x \in \mathcal{C}} u(x)v(x)w(x) / (\|\nabla g_0\| \|\nabla h_0\| |\sin \omega|)(x)$ .

**Remark 3.2** (Bilinear extension). *Theorem 3.1 is stated on the diagonal,  $D^2\Theta[\eta, \eta]$ , but is used below as a symmetric bilinear form on pairs of distinct directions —  $D^2\Theta[u^A, u^B]$  in Theorem 4.6,  $D^2\Theta[e_A, e_B]$  in Theorem 5.8,  $C[\Delta_g, \Delta_h]$  throughout. The extension is by polarization,*

$$D^2\Theta[\eta_1, \eta_2] := \frac{1}{4} \{ D^2\Theta[\eta_1 + \eta_2, \eta_1 + \eta_2] - D^2\Theta[\eta_1 - \eta_2, \eta_1 - \eta_2] \}, \quad (20)$$

valid on the direction space  $C^1(\mathcal{X})^2$ , on which the right-hand side of (18) is a bounded symmetric quadratic form under Assumptions 1.4–1.5. Explicitly, writing  $\eta_k = (u_k, v_k)$ , the polarized form is  $Q_g[u_1, u_2] + Q_h[v_1, v_2] + C[u_1, v_2] + C[u_2, v_1]$ , with  $Q_g$  and  $C$  polarized in the same way. All bilinear manipulations below — in particular the split-sample identity (31) — are manipulations of (20).

**Remark 3.3** (Numerical verification). *Theorem 3.1 is verified to ten significant digits in Appendix C on closed-form  $d = 2$  designs that between them activate every term of (18)–(19):*

a radially weighted disk (the transport and curvature terms of  $Q_g$ ); a disk with a non-constant direction  $u$  (the  $u n_g \cdot \nabla u$  term); a disk cut by an oblique chord and the intersection of two disks (both corner diagonals and the corner cross term, at unequal  $u, v$ , at one-sided perturbations, and at both signs of  $\cos \omega$ ). The same appendix verifies the polarization identity (37) and the support-edge term of Remark 1.12. This is the part of the analysis that is checked rather than asserted.

### 3.2 Orders of the second-order terms

Write  $\Delta_g := \hat{g} - g_0$ ,  $\Delta_h := \hat{h} - h_0$  for sieve LS first stages with  $K_n$  basis functions per arm. Decompose each error into an approximation bias, an arm-wise population-Gram linear term, and a series linearization remainder. Put  $q_1 = 1, q_0 = -1$  and

$$s_{w,i}(x) := \mathbb{1}\{W_i = w\} \psi(x)' R_w^{-1} \psi(X_i).$$

Then, for the harm-share convention  $g = \tau_S$  and  $h = -\tau_Y$ ,

$$\begin{aligned} \Delta_g &= b_g + u_g + r_g, \quad u_g(x) := \frac{1}{n} \sum_{i=1}^n \sum_{w=0}^1 q_w s_{w,i}(x) \epsilon_{S,i}, \\ \Delta_h &= b_h + u_h + r_h, \quad u_h(x) := -\frac{1}{n} \sum_{i=1}^n \sum_{w=0}^1 q_w s_{w,i}(x) \epsilon_{Y,i}. \end{aligned} \tag{21}$$

with  $\|b_g\|_\infty \lesssim K_n^{-s/d}$ ,  $\|\nabla b_g\|_\infty \lesssim K_n^{-(s-1)/d}$ , and the same bounds for  $b_h$ , with  $R_w$  as defined below. Within an arm the  $S$  and  $Y$  regressions use the same basis and design matrix, so their arm-specific weights  $s_{w,i}$  are common; weights are not common *across* arms. This within-arm sharing is what generates the within-unit coupling below. Assumption 1.9(d) controls  $(r_g, r_h)$ ; all sandwich identities and entries in Table 1 refer to the displayed leading linear terms. Without that remainder, (21) would incorrectly identify the random empirical Gram matrix with its population counterpart.

Two scaling facts, uniform in  $w$ , for tensor-product B-splines with mesh  $\varkappa = K_n^{-1/d}$  drive the whole accounting:

$$\mathcal{V}_w(x) := \psi(x)' R_w^{-1} \psi(x) \asymp K_n, \quad \|R_w^{-1/2} \nabla \psi(x)\| \lesssim K_n^{1/2+1/d}, \tag{22}$$

(each derivative of a basis function costs a factor  $\varkappa^{-1} = K_n^{1/d}$ ), whence  $|\psi(x)' R_w^{-1} \partial_j \psi(x)| \lesssim K_n^{1+1/d}$  by Cauchy–Schwarz. Only the first display is two-sided; the second bounds the norm of a *signed* vector, and v2’s “ $b' G^{-1} \nabla b \asymp K_n^{1+1/d}$ ” must be read that way. The distinction is

not cosmetic: Remark 3.4 shows that the integrated version of the signed quantity can be an order smaller.

The estimator runs *four separate arm regressions* on the common basis  $\psi$ , so the exact second moments are arm-wise sums of sandwiches, not a single one. Write  $\pi_1(x) := p_0(x)$ ,  $\pi_0 := 1 - \pi_1$ , and

$$R_w := \mathbb{E}[\pi_w(X)\psi(X)\psi(X)'], \quad \Omega_{oo'}^w := \mathbb{E}[\pi_w(X)\varsigma_{oo'}(X, w)\psi(X)\psi(X)'],$$

for  $o, o' \in \{S, Y\}$ , with  $\varsigma_{SS} = \sigma_S^2$ ,  $\varsigma_{YY} = \sigma_Y^2$  and  $\varsigma_{SY} = \rho_{SY}\sigma_S\sigma_Y$ . Since  $\mathbb{1}\{W_i = 1\}\mathbb{1}\{W_i = 0\} = 0$ , cross-arm covariances vanish and, with  $g_0 = \tau_S$  and  $h_0 = -\tau_Y$  as in Example 1.1,

$$\mathcal{V}_{gh,n}(x) := n \mathbb{E}[u_g(x)u_h(x)] = - \sum_{w \in \{0,1\}} \psi(x)' R_w^{-1} \Omega_{SY}^w R_w^{-1} \psi(x), \quad (23)$$

and  $\mathcal{V}_{gg,n}, \mathcal{V}_{hh,n}$  likewise with  $\Omega_{SS}^w, \Omega_{YY}^w$  and no leading minus. *The minus sign is the Jacobian of  $h_0 = -\tau_Y$ : the estimation error of  $h$  is the negative of that of  $\tau_Y$ , so a positive  $\rho_{SY}$  produces a negative  $\mathcal{V}_{gh,n}$ .* v3 omitted this.

Under uniform overlap and Assumption 1.4(c),  $\mathcal{V}_{gg,n}(x) \asymp \mathcal{V}_{hh,n}(x) \asymp K_n$  and  $|\mathcal{V}_{gh,n}(x)| \lesssim K_n$  uniformly. If in addition the  $\varsigma$ 's and  $\pi_w$  are Lipschitz, local support of  $\psi$  gives the *additive* approximation

$$\begin{aligned} \mathcal{V}_{gg,n}(x) &= \varrho_{gg}(x) \mathcal{V}(x) + O(\varkappa K_n), & |\mathbb{E}[u_g(x)(n_g \cdot \nabla u_g)(x)]| &\lesssim \frac{K_n^{1+1/d}}{n}, \\ \mathcal{V}_{gh,n}(x) &= -\varrho_{gh}(x) \mathcal{V}(x) + O(\varkappa K_n), \end{aligned} \quad (24)$$

where  $\mathcal{V}(x) := \psi(x)' R^{-1} \psi(x)$  with  $R := \mathbb{E}[\psi\psi']$ , and

$$\varrho_{gg}(x) := \sum_{w \in \{0,1\}} \frac{\sigma_S^2(x, w)}{\pi_w(x)}, \quad \varrho_{gh}(x) := \sum_{w \in \{0,1\}} \frac{\rho_{SY}(x, w) \sigma_S(x, w) \sigma_Y(x, w)}{\pi_w(x)}, \quad (25)$$

$\varrho_{hh}$  symmetrically. The error term in (24) is *additive*,  $O(\varkappa K_n)$ , not relative: a relative  $1 + O(\varkappa)$  statement is vacuous — indeed false — near points where the signed  $\varrho_{gh}$  vanishes, which is exactly where the sign of the corner contribution turns over. Under conditional homoskedasticity and constant propensity the error is identically zero.

Three corrections are recorded here. v1 wrote the covariance display with an undefined normalization and “ $\asymp$ ” applied to a signed quantity; v2 replaced it by an exact identity that holds only under homoskedasticity; v3 wrote a single-sandwich form appropriate to one pooled contrast regression rather than the four arm regressions actually used, with a

relative error term and without the  $h_0 = -\tau_Y$  sign. The consequences: the sign of  $\mathcal{V}_{gh,n}(x)$  is *not* that of  $\rho_{SY}(x, w)$  — it is the opposite, under the harm-share convention — and a nonzero conditional correlation at a corner does not by itself guarantee a nonzero *integrated* corner bias, which is why any matching lower-bound claim imposes Assumption 1.7 on the normalized sequence rather than deriving it from  $\rho_{SY} \neq 0$ . The stochastic orders below survive as upper bounds; the constants do not.

Taking expectations term by term in (18) gives the *upper bounds* in Table 1. The essential point — and the correction to v1 — is that the margin forms  $Q_g, Q_h$  contain a factor  $\nabla u$  while the corner forms do not. *The margin diagonals are therefore bounded by  $K_n^{1/d}$  times the corner diagonals*, not by the same quantity. Every entry of Table 1 is an  $O(\cdot)$ , and this is all the theorems below use; Remark 3.4 discusses which entries are actually attained.

Table 1: Upper bounds on the second-order terms and the smoothness needed for each to be  $O(r_n^*)$ ,  $r_n^* = n^{-s/(2s+1)}$ , at  $K_n \asymp n^{d/(2s+1)}$ . “Diagonal” terms enter the estimator error linearly and must be compared with  $r_n^*$  itself; “off-diagonal” (cross-half /  $i \neq j$ ) terms are mean-zero degenerate  $U$ -statistics. Derivations of the off-diagonal orders are in Appendix B. Entries are bounds, not exact orders: see Remark 3.4.

| Component                                                                   | Term                           | Bound                        | bound is $O(r_n^*)$ iff |
|-----------------------------------------------------------------------------|--------------------------------|------------------------------|-------------------------|
| <i>Stochastic diagonals</i> (removed by SS/LOO debiasing)                   |                                |                              |                         |
| Margin                                                                      | $Q_g[u_g, u_g], Q_h[u_h, u_h]$ | $K_n^{1+1/d}/n$              | $s \geq d$              |
| Corner                                                                      | all three corner pieces        | $K_n/n$                      | $s \geq d - 1$          |
| <i>Approximation-bias diagonals</i> (removed by undersmoothing)             |                                |                              |                         |
| Margin                                                                      | $Q_g[b_g, b_g]$                | $K_n^{-(2s-1)/d}$            | $s > 1$                 |
| Corner                                                                      | $C[b_g, b_h]$ etc.             | $K_n^{-2s/d} = r_n^{*2}$     | always                  |
| <i>Bias <math>\times</math> stochastic cross terms</i> (removed by neither) |                                |                              |                         |
| Margin                                                                      | $Q_g[b_g, u_g]$                | $K_n^{(1-s)/d} \sqrt{K_n/n}$ | $s \geq (d+1)/2$        |
| Corner                                                                      | $C[b_g, u_h]$                  | $K_n^{-s/d} \sqrt{K_n/n}$    | $s \geq (d-1)/2$        |
| <i>Cross-half off-diagonals</i> (mean zero; degenerate)                     |                                |                              |                         |
| Margin                                                                      | gradient part                  | $n^{-1} K_n^{(d+3)/(2d)}$    | $s \geq (d+1)/2$        |
| Corner                                                                      | —                              | $n^{-1} K_n^{(d+2)/(2d)}$    | $s \geq d/2$            |
| Margin                                                                      | curvature part                 | $n^{-1} K_n^{(d+1)/(2d)}$    | $s \geq (d-1)/2$        |

Three consequences follow, and they replace the corresponding discussion of v1.

(a) *The plug-in’s smoothness requirement comes from the margin diagonal.*  $K_n^{1+1/d}/n = n^{-(2s-d)/(2s+1)}$  is  $O(r_n^*)$  if and only if  $s \geq d$  — precisely the condition in Theorem 2.2. In v1 this requirement was attributed to  $K_n/n$ , which would have given  $s \geq d - 1$ , and the one-derivative discrepancy was left unexplained. With the bound of Table 1 the condition

$s \geq d$  is exactly what is needed for the bound to be no larger than  $r_n^*$ ; strict  $s > d$  is needed for negligibility at the rate-optimal  $K_n$ . We do *not* claim it is necessary; see Remark 3.4.

(b) *The corner upper bound can be rate-relevant on a nonempty window.*  $K_n/n = n^{-(2s+1-d)/(2s+1)}$  is  $O(r_n^*)$  if and only if  $s \geq d - 1$ . Since split-sample debiasing (Section 4) extends the guarantee down to  $s > (d + 1)/2$ , this *upper bound* exceeds  $r_n^*$  exactly on

$$\frac{d+1}{2} < s \leq d - 1, \quad \text{nonempty iff } d \geq 4. \quad (26)$$

Under Assumption 1.7(c1) the realized full corner coefficient has this order; without (c1), (26) identifies only potential rate relevance. For  $d \in \{2, 3\}$  the window is empty, so at the  $d = 2$  of our designs corner-awareness is a *finite-sample constants* question, not an asymptotic rate question. This is a weaker claim than v1’s abstract made, and it is the correct one.

(c) *The corner coefficient has three pieces, not one.* Collecting the corner terms of (18)–(19) and using (24), the corner’s contribution to  $\mathbb{E}[\frac{1}{2}D^2\Theta[\Delta, \Delta]]$  is

$$\mathcal{B}_{C,n} = \int_{\mathcal{C}} \frac{w(x)}{2 |\sin \omega(x)|} \left[ \frac{2V_{gh}(x)}{\|\nabla g_0\| \|\nabla h_0\|} - \cos \omega(x) \left( \frac{V_g(x)}{\|\nabla g_0\|^2} + \frac{V_h(x)}{\|\nabla h_0\|^2} \right) \right] d\mathcal{H}^{d-2}(x), \quad (27)$$

with cross-term component

$$\mathcal{B}_{C,n}^\times := \int_{\mathcal{C}} \frac{V_{gh}(x) w(x)}{\|\nabla g_0\| \|\nabla h_0\| |\sin \omega(x)|} d\mathcal{H}^{d-2}(x), \quad (28)$$

where, all at  $x$ ,

$$V_g := \mathbb{E}[\Delta_g^2] = \mathcal{V}_{gg,n}/n + b_g^2 + \mathcal{R}_{V,g}, \quad V_{gh} := \mathbb{E}[\Delta_g \Delta_h] = \mathcal{V}_{gh,n}/n + b_g b_h + \mathcal{R}_{V,gh},$$

and  $V_h$  is symmetric. The  $\mathcal{R}_V$  terms collect the series linearization remainders controlled by Assumption 1.9(d); they were silently set to zero in v4. All three leading terms are  $O(K_n/n)$  up to their bias parts, so  $\mathcal{B}_{C,n} = O(K_n/n)$  under that assumption — and *only*  $O$ . The bracket is a *signed* difference, so its three components may offset; contributions from different corner points may offset; and the weights  $\mathcal{V}_{\bullet,n}(x)/K_n$  are  $n$ -dependent and vary across corner points. Matching lower bounds are exactly (5), which is why those are imposed on the normalized sequences  $\frac{n}{K_n}\mathcal{B}_{C,n}$  and  $\frac{n}{K_n}\mathcal{B}_{C,n}^\times$  rather than on population coefficients. v2 wrote “ $\mathcal{B}_C \asymp K_n/n$ ”; that is a two-sided claim the formula does not support on its own, and v3’s population-level noncancellation condition did not repair it. Two further readings matter. First, the sign of the corner bias is not determined by  $\text{sign}(\rho_{SY})$  alone — nor, by (23), is  $\text{sign}(V_{gh}(x))$  determined by  $\text{sign}(\rho_{SY}(x, w))$ . Second, the corner contribution to the *plug-in* bias does not

vanish when  $\rho_{SY} = 0$ : the two  $\cos \omega$  terms survive, and they vanish only when the margins meet orthogonally. In the notation of (27),  $v_1$  tracked only the first term of the bracket. Note carefully, however, that the two  $\cos \omega$  terms belong to  $Q_g$  and  $Q_h$  and are therefore removed by margin-by-margin debiasing; only the  $V_{gh}$  term survives it, and *that* term does vanish at  $\rho_{SY} = 0$ . Section 4 makes this precise (Proposition 4.10), and the two statements should not be conflated: (27) is the corner’s contribution to the plug-in’s bias, not the part that margins-only debiasing leaves behind.

**Remark 3.4** (Are these bounds attained? A caution about mesh resonance). *Table 1 reports bounds, and for the margin diagonal the bound need not be tight. For the four-arm decomposition, the leading  $g$ -margin diagonal is, exactly,*

$$\mathbb{E}[Q_g[u_g, u_g]]_{grad} = -\frac{1}{n} \int_{\mathcal{M}_g} \frac{w}{\|\nabla g_0\|^2} n_g \cdot \nabla \mathcal{V}_{gg,n} d\mathcal{H}^{d-1}, \quad (29)$$

because  $\mathbb{E}[u_g \partial_n u_g] = (2n)^{-1} \partial_n \mathcal{V}_{gg,n}$ , with  $\mathcal{V}_{gg,n}$  given by the arm-wise sandwich. Under homoskedasticity and constant propensity this is a constant multiple of the usual leverage function  $\mathcal{V}(x) := \psi(x)' R^{-1} \psi(x)$ . The effective leverage  $\mathcal{V}_{gg,n} \asymp K_n$  can oscillate at the mesh scale  $\varkappa = K_n^{-1/d}$ , so  $\|\nabla \mathcal{V}\|_\infty \asymp K_n^{1+1/d}$  pointwise — but (29) integrates it against a smooth weight over a  $(d-1)$ -dimensional surface, and an oscillating integrand can cancel. Whether it does depends on the arithmetic relationship between the margin’s normal direction and the tensor mesh. Writing  $\mathcal{V}$  as a Fourier series in the mesh period, an exact resonant pair on a straight  $d = 2$  margin of slope  $\alpha$  solves  $j + \ell\alpha = 0$ . Every rational slope has such integer pairs, not only rationals with a small denominator; a larger denominator merely pushes the resonance into a smaller Fourier coefficient and can delay its visibility at practical meshes. For irrational slopes there is no exact pair, but the order depends on Diophantine approximation, Fourier decay, the intercept/phase, and endpoint terms. The finite calculation in Appendix C, item 5, shows large phase-sensitive values for aligned and slope-one margins and much smaller values for several other slopes through  $K_n^{1/2} = 128$ . It is a diagnostic of possible cancellation, not a proof that “generic” irrationals have order  $K_n$ .

Three implications. (i)  $s \geq d$  in Theorem 2.2(i) is sufficient but we do not claim it necessary. If the margin diagonal is in fact  $O(K_n/n)$  under additional geometric/arithmetic conditions, the rate condition relaxes to  $s \geq d-1$  and the centered-CLT window (11) widens to  $d/(2s+1) < a < d/(2d-1)$ , nonempty iff  $s > d-1$  — one derivative cheaper throughout. Nothing below relies on this; we record it because it is the difference between the plug-in needing  $s > d$  and  $s > d-1$ . (ii) This particular mechanism does not reach the corner:  $V_g, V_h, V_{gh}$  in (27) involve  $\mathcal{V}$  undifferentiated and are evaluated pointwise on a  $(d-2)$ -set, so there is no oscillating derivative to average away. The two variance pieces  $V_g, V_h$  are  $\asymp K_n/n$

pointwise, whereas the signed covariance  $V_{gh}$  is only  $O(K_n/n)$  and can vanish. This does not make  $\mathcal{B}_C$  itself two-sided: as noted above, the corner coefficient can still cancel through the signed bracket or across corner points, which is a geometric and covariance phenomenon rather than a mesh phenomenon, and which Assumption 1.7 excludes by assumption. *v2's* claim that “the corner bound is the robust one” should be read narrowly: robust to mesh resonance, not to cancellation in general. (iii) Consequently the safe statement of the margin-versus-corner comparison is an inequality between the available upper bounds: the corner upper bound is  $K_n^{-1/d}$  times the margin upper bound. This does not order the realized signed integrals without separate lower bounds; on some oblique geometries cancellation can make the two comparable. Two caveats on the mechanism itself: the identity  $\mathbb{E}[u \partial_n u] = (\sigma^2/2n) \partial_n \mathcal{V}$  requires homoskedasticity (in general the object is  $n^{-1} b' G^{-1} \Omega_{gg} G^{-1} \nabla b$  by (23), not an exact gradient), and the evidence is a  $d = 2$  probe with straight margins. Establishing the exact order of (29) for curved margins and general  $d$  is open; we flag it because it is the one place where the accounting above could be conservative rather than wrong.

A margin-by-margin application of the single-threshold correction of Chen and Gao (2026) removes the margin diagonals *and* the two  $\cos \omega$  corner diagonals, but leaves the corner cross term  $\mathcal{B}_C^\times$  in place; Proposition 4.10 below records the implication.

## 4 Quadratic Debiasing

### 4.1 Split-sample estimator and a tuning-free representation

Randomly split the sample into halves  $A$  and  $B$  (stratified by  $W$ ), run the stacked sieve regressions on each half, and let  $\hat{\gamma}^A = (\hat{g}^A, \hat{h}^A)$ ,  $\hat{\gamma}^B$  denote the two estimated pairs,  $\bar{\gamma} := (\hat{\gamma}^A + \hat{\gamma}^B)/2$ , and  $\Delta := \hat{\gamma}^A - \hat{\gamma}^B$ . The split-sample (SS) debiased estimator is

$$\hat{\Theta}_{SS} := \Theta(\bar{\gamma}) - \frac{1}{8} D^2 \Theta(\bar{\gamma})[\Delta, \Delta]. \quad (30)$$

The construction works because the approximation bias is common to the two halves and cancels from  $\Delta$ : writing  $\hat{\gamma}^j = \gamma_0 + b + u^j + r^j$  with a common deterministic  $b$ , independent mean-zero population-Gram terms  $u^A, u^B$ , and the series linearization remainders of Assumption 1.9(d),

$$\bar{\Delta} := \bar{\gamma} - \gamma_0 = b + \bar{u} + \bar{r}, \quad \bar{u} = \frac{1}{2}(u^A + u^B), \quad \Delta = u^A - u^B + r^A - r^B, \quad (31)$$

where  $\bar{r} = (r^A + r^B)/2$ . For the leading terms, bilinearity gives  $D^2 \Theta[\bar{u}, \bar{u}] - \frac{1}{4} D^2 \Theta[\Delta, \Delta] = D^2 \Theta[u^A, u^B]$  when  $\Delta$  is replaced by  $u^A - u^B$ ; every term containing  $r^A$  or  $r^B$  is part of the



linearization remainder. Thus (30) removes the *stochastic* diagonals — margins and corner alike — and nothing else. It does not touch the bias diagonals or the bias  $\times$  stochastic cross terms; those are the third and fourth blocks of Table 1 and are controlled by smoothness, not by debiasing.

**Assumption 4.1** (Path regularity). (a) (Weight.)  $w = v_0 f \in C^3$  in a neighborhood of  $\{g_0 = 0\} \cup \{h_0 = 0\}$ . This is a restriction on the user-chosen  $v_0$  and on the target density  $f$ ; it is not a restriction on  $\mu_{S,0}, \mu_{Y,0}$  and therefore does not interact with  $s$ . (b) (Sieve order.) The first stages take values in a space  $\mathcal{S} \subset C^4(\mathcal{X})$  — for tensor B-splines, degree at least 5 (quintic or higher), since a degree- $p$  B-spline is only  $C^{p-1}$  at its knots. The order must in any case exceed  $s$  for the approximation rate  $\|b_g\|_\infty \lesssim K_n^{-s/d}$  used throughout, so the operative requirement is degree  $\geq \max\{5, \lceil s \rceil\}$ . (c) (Uniform geometry along the path.) With probability tending to one, every  $\gamma$  on each segment  $\{\bar{\gamma} + t\xi : |t| \leq \frac{1}{2}\}$ , for  $\xi \in \{\Delta, (\Delta_g, 0), (0, \Delta_h)\}$ , satisfies Assumption 1.5(a)–(d) with constants  $\underline{c}/2, \underline{c}_T/2$ . (d) (Fourth-derivative bound.) On each such segment  $t \mapsto \Theta(\bar{\gamma} + t\xi)$  is four times continuously differentiable with  $\sup_{|t| \leq 1/2} \left| \frac{d^4}{dt^4} \Theta(\bar{\gamma} + t\xi) \right| \leq C \|\xi\|_{C^1}^4$  for a deterministic constant  $C$  independent of  $n$ . This uniform inequality is a separate high-level condition; it is not implied by membership of the sieve in  $C^4$ . (e) (Split-contrast size.) For the three directions in (c),

$$\|\xi\|_{C^1} = O_p(K_n^{1/2+1/d} n^{-1/2}).$$

In particular, the common approximation bias has cancelled and the empirical-Gram remainder has not enlarged the leading stochastic  $C^1$  rate. This is also a high-level series condition.

Three comments. First, v2 stated the path conditions as a single requirement of a  $C^3$  neighborhood of  $\gamma_0$ , which implicitly forces  $\gamma_0 \in C^3$  and hence  $s \geq 3$  — flatly inconsistent with a theorem claiming performance over  $\Lambda_c^s$  with, say,  $1.5 < s < 3$  at  $d = 2$ . The smoothness a fourth-order finite difference needs is smoothness of the *path along which*  $\Theta$  is differenced, i.e. of the *fits* and of  $w$ , not of the truth: the level sets being deformed are those of  $\bar{\gamma} + t\Delta$ . Splitting the condition this way removes the inconsistency. Second, (b) is a necessary implementation constraint that v2 overlooked: cubic B-splines are only  $C^2$  at their knots, so the third and fourth derivatives that (d) presumes do not exist for the maintained implementation of Section 6. The order needed follows from the derivative count in Appendix A:  $D^k \Theta$  involves  $k$  derivatives of the level-set function and  $k - 1$  of the weight and of the perturbation direction —  $D^2$  already carries  $H_g$  (two derivatives of  $g$ ),  $\nabla w$ , and  $\nabla u$ . Hence  $D^4$  needs  $C^4$  level-set functions and  $C^3$  weight and direction. Since  $\bar{\gamma}$  and  $\Delta$  inhabit the same sieve space, that space must be  $C^4$ : degree 5, not degree 4 as an earlier draft of this revision had it. The calibrated Monte Carlo uses cubics and therefore

satisfies neither. Degree 5 guarantees existence only: because  $D^4\Theta$  depends on the  $C^4$  size of the base path and the  $C^3$  size of  $\Delta$ , inverse inequalities can make the constant in (d) grow with  $K_n$ . Establishing the displayed uniform bound requires an additional argument. Third,  $C^1$  consistency,  $K_n^{1/2+1/d}n^{-1/2} + K_n^{-(s-1)/d} = o(1)$ , stabilizes gradient nondegeneracy, transversality, and the normal-graph representation under the usual compact separation conditions. It does *not* by itself preserve the no-support-edge condition in Assumption 1.5(d). For that condition we additionally require either that the relevant level sets stay a fixed distance from  $\partial\mathcal{X}$  or that  $w$  vanish on a fixed neighborhood of  $\partial\mathcal{X}$ .

**Lemma 4.2** (Half-sample jackknife representation). *Under Assumption 4.1, approximating  $D^2\Theta(\bar{\gamma})[\Delta, \Delta]$  by the symmetric second difference at step  $\delta = \frac{1}{2}$  — i.e. evaluating  $\Theta$  at  $\bar{\gamma} \pm \frac{1}{2}\Delta = \hat{\gamma}^A, \hat{\gamma}^B$  — gives the closed form*

$$\hat{\Theta}_{SS}^{\text{jack}} = 2\Theta(\bar{\gamma}) - \frac{1}{2}(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B)). \quad (32)$$

Writing  $\phi(t) := \Theta(\bar{\gamma} + t\Delta)$ , the approximation error is exactly

$$\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS} = -\frac{1}{384}\phi^{(4)}(\xi) \quad \text{for some } |\xi| \leq \frac{1}{2}.$$

Under the fourth-derivative bound of Assumption 4.1,  $\phi^{(4)}(\xi) = O_p(\|\Delta\|_{C^1}^4) = O_p(K_n^{2+4/d}n^{-2})$ , which is  $o(r_n^*)$  at  $K_n \asymp n^{d/(2s+1)}$  under the sufficient condition

$$s > \frac{2(d+1)}{3}. \quad (33)$$

Moreover (32) is exact whenever  $t \mapsto \Theta(\bar{\gamma} + t\Delta)$  is a quadratic polynomial.

*Proof.*  $\phi(\frac{1}{2}) + \phi(-\frac{1}{2}) - 2\phi(0) = \frac{1}{4}\phi''(0) + \frac{1}{192}\phi^{(4)}(\xi)$  by Taylor's theorem with Lagrange remainder, valid because  $\phi \in C^4[-\frac{1}{2}, \frac{1}{2}]$  under Assumption 4.1. Substituting  $D^2\Theta(\bar{\gamma})[\Delta, \Delta] = \phi''(0)$  into (30) and rearranging gives (32) with the stated error. The rate statement uses  $\|\Delta\|_{C^1} = O_p(K_n^{1/2+1/d}n^{-1/2})$  and  $K_n \asymp n^{d/(2s+1)}$ , giving exponent  $(2d+2-3s)/(2s+1)$  relative to  $r_n^*$ .  $\square$

**Remark 4.3** (The jackknife rate remains high-level). *Condition (33) is strictly stronger than the  $s > (d+1)/2$  of Theorem 4.6:  $2(d+1)/3 > (d+1)/2$  for every  $d$ , e.g.  $s > 2$  against  $s > 1.5$  at  $d = 2$ . Conditional on Assumption 4.1(d)–(e), the tuning-free implementation (32) therefore does not inherit the full weak-smoothness range of the ideal estimator (30).*

*More importantly, Version 4 treated the inequality in Assumption 4.1(d) as if spline smoothness supplied it. It does not: a primitive fourth-shape-derivative bound will generally involve higher derivative norms and may carry additional powers of  $K_n$ . Until such a bound*

is proved, the honest statement is: (32) is exact for quadratic paths, has a fourth-order error in general, and has the rate and CLT windows stated here conditional on Assumption 4.1(d)–(e).

Representation (32) is how we implement (30): it requires *no* numerical differentiation step size, no explicit computation of curvatures, normals, or corner angles, and it automatically includes *every* term of (18) — both margin diagonals, both diagonal corner terms, and the cross corner term — because the functional itself is evaluated along the joint direction. It is a half-sample generalized jackknife: the correction subtracts the “curvature” of  $\Theta$  between the two independent half-sample fits.

**Remark 4.4** (The jackknife identity is not a curvature correction for non-smooth first stages). *Lemma 4.2 is a statement about a smooth path. If  $\hat{\gamma}^A, \hat{\gamma}^B$  are piecewise-constant (boosted trees, random forests), then  $\phi(t) = \Theta(\bar{\gamma} + t\Delta)$  is a step function of  $t$ : it has no derivatives at all,  $\phi^{(4)}$  does not exist, and Assumption 4.1 fails at every  $\gamma$  in the relevant neighborhood. Formula (32) remains a well-defined estimator — three evaluations of  $\Theta$  — but it is no longer an approximation to  $\frac{1}{8}D^2\Theta[\Delta, \Delta]$ , and neither Theorem 4.6 nor Theorem 5.8 covers it. In Table 2 this mechanism is confounded with a second one: those cells also implement the superseded doubly debiased construction, whose own second-order defect (Remark 5.6) predicts over-correction on smooth designs too. The two can be separated only by re-running the corrected estimator of Definition 5.7 with smooth and tree first stages; see Section 6.5, item (L7).*

## 4.2 Isolating the corner: two families, two exact identities

Perturbing one contrast at a time isolates the corner. Because the decomposition can be carried out either on the closed-form  $D^2\Theta$  or on finite differences, and because these do *not* agree exactly, we define both families and keep them separate. Write  $C[u, v] := \int_{\mathcal{C}} uvw / (\|\nabla g_0\| \|\nabla h_0\| |\sin \omega|) d\mathcal{H}^{d-2}$  for the corner cross form, and recall from (18) the polarized identity  $D^2\Theta[(u, v), (u, v)] = Q_g[u, u] + Q_h[v, v] + 2C[u, v]$ .

*Ideal family.* With  $Q_g, Q_h, C$  evaluated at  $\bar{\gamma}$ , set

$$\hat{\Theta}_{SS}^{\text{mar}} := \Theta(\bar{\gamma}) - \frac{1}{8} \{Q_g[\Delta_g, \Delta_g] + Q_h[\Delta_h, \Delta_h]\}, \quad \text{corner} := \frac{1}{4} C[\Delta_g, \Delta_h]. \quad (34)$$

Then, *exactly*,  $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \text{corner}$ , because  $\frac{1}{8}D^2\Theta[\Delta, \Delta] = \frac{1}{8}(Q_g + Q_h) + \frac{1}{4}C$ .

*Jackknife family.* With the second differences

$$\begin{aligned} q_{\text{full}} &:= 4[\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B) - 2\Theta(\bar{\gamma})], \\ q_g &:= 4[\Theta(\hat{g}^A, \bar{h}) + \Theta(\hat{g}^B, \bar{h}) - 2\Theta(\bar{\gamma})], \quad q_h := 4[\Theta(\bar{g}, \hat{h}^A) + \Theta(\bar{g}, \hat{h}^B) - 2\Theta(\bar{\gamma})], \\ \hat{\Theta}_{SS}^{\text{mar, jack}} &:= \Theta(\bar{\gamma}) - \frac{1}{8}(q_g + q_h), \quad \widehat{\text{corner}} := \frac{1}{8}(q_{\text{full}} - q_g - q_h), \end{aligned} \quad (35)$$

one has, again *exactly*,

$$\hat{\Theta}_{SS}^{\text{mar, jack}} - \widehat{\text{corner}} = \Theta(\bar{\gamma}) - \frac{1}{8}q_{\text{full}} = \hat{\Theta}_{SS}^{\text{jack}}. \quad (36)$$

**Remark 4.5** (A correction to v3). *v3 asserted  $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$ , mixing the ideal left-hand side with the finite-difference right-hand side. That is false unless  $t \mapsto \Theta(\bar{\gamma} + t\Delta)$  is exactly quadratic: by (36) the correct right-hand side is  $\hat{\Theta}_{SS}^{\text{jack}}$ , and Lemma 4.2 puts  $\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS} = -\frac{1}{384}\phi^{(4)}(\xi)$ . The two identities (34) and (36) are each exact within their own family; only the cross-family statement carries a fourth-order error. All statements below are labelled accordingly.*

Why  $q_g$  absorbs a corner term: perturbing  $g$  alone corresponds to the direction  $(\Delta_g, 0)$ , and  $D^2\Theta[(\Delta_g, 0), (\Delta_g, 0)] = Q_g[\Delta_g, \Delta_g]$  including the  $\cos \omega$  corner diagonal that (19) attaches to  $Q_g$ . Hence  $q_g, q_h$  absorb both corner diagonals and, by bilinearity,

$$q_{\text{full}} - q_g - q_h = 2C[\Delta_g, \Delta_h] + O_p(4\text{th-order}), \quad (37)$$

the corner *cross* form alone. So  $\widehat{\text{corner}}$  measures the cross term's stochastic diagonal, *not* the whole of (27). Identity (37) is verified numerically to full precision on thirteen closed-form designs in Appendix C; it matters because it determines exactly which part of the corner survives margins-only debiasing (Proposition 4.10).

**Theorem 4.6** (SS debiasing: a conditional weak-smoothness result). *Let Assumptions 1.4–1.6, 1.9, and 4.1 hold, let the four arm regressions use sieve dimension  $K_n \asymp n^a$  (up to logs), and suppose  $s > (d+1)/2$ . Define the total remainder of the corrected estimator,*

$$R_3^{SS} := \underbrace{\frac{1}{2}\{D^2\Theta(\gamma_\xi) - D^2\Theta(\gamma_0)\}[\bar{\Delta}, \bar{\Delta}]}_{\text{Taylor mean-value remainder}} - \underbrace{\frac{1}{8}\{D^2\Theta(\bar{\gamma}) - D^2\Theta(\gamma_0)\}[\Delta, \Delta]}_{\text{evaluation-point remainder}}, \quad (38)$$

whose sum is governed directly by Assumption 1.9(a). Then:

- (i) (Rate, ideal estimator.) At  $a = d/(2s+1)$ , if  $R_3^{SS} = O_p(r_n^*)$  then  $\hat{\Theta}_{SS} - \Theta_0 = O_p(r_n^*)$ ,  $r_n^* = n^{-s/(2s+1)}$ , which by Theorem 2.2(ii) is minimax-optimal.

(ii) (Centered CLT, ideal estimator.) If in addition  $R_3^{SS} = o_p(\text{se}_n)$  and

$$\frac{d}{2s+1} < a < \min \left\{ \frac{d}{d+2}, \frac{d}{3d-2} \right\}, \quad (39)$$

a window nonempty iff  $s > (d+1)/2$  (known density, or  $d = 2$ , where the two edges coincide at  $d/(d+2)$ ), respectively iff  $s > 3(d-1)/2$  (empirical measure with  $d \geq 3$ ), then  $\sqrt{n}(\hat{\Theta}_{SS} - \Theta_0)/\hat{\sigma}_n \xrightarrow{d} \mathcal{N}(0, 1)$  with  $\hat{\sigma}_n$  from (15) under Proposition 2.4.

(iii) (Conditional centered CLT, feasible jackknife estimator.) For the tuning-free  $\hat{\Theta}_{SS}^{\text{jack}}$  of (32), the fourth-order error of Lemma 4.2 must also be  $o(\text{se}_n)$ . Conditional on the high-level bounds in Assumption 4.1(d)–(e), this is ensured by its own sieve window:

$$\frac{d}{2s+1} < a < \min \left\{ \frac{3d}{4d+7}, \frac{d}{3d-2} \right\}, \quad (40)$$

nonempty iff  $s > 2(d+1)/3$  (and  $s > 3(d-1)/2$  in the empirical-measure case with  $d \geq 3$ ). Since  $3d/(4d+7) < d/(d+2)$  for every  $d$ , this is strictly inside (39). Neither sieve membership in  $C^4$  nor this smoothness threshold alone establishes Assumption 4.1(d)–(e); part (iii) is therefore not a primitive-condition result.

**Remark 4.7** (What v3 got wrong here). Two things. (a) v3 stated part (iii) as “the same conclusions hold under the additional smoothness condition (33)”. But (33) was derived by comparing the fourth-order error with  $r_n^*$  at the rate-optimal  $K_n$ ; a centred CLT compares it with  $\text{se}_n$  at an undersmoothed  $K_n$ , which is a different and  $a$ -dependent requirement. Without (40) one may choose  $a \in (3d/(4d+7), d/(d+2))$  satisfying every stated condition while the jackknife error dominates the standard error. (b) v3’s  $R_3$  was the Taylor remainder only. The estimator subtracts  $D^2\Theta(\bar{\gamma})[\Delta, \Delta]$  while the cancellation argument uses  $D^2\Theta(\gamma_0)$ ; the evaluation-point difference in (38) is a separate third-order term of the same order, not covered by the Taylor remainder. Version 5 imposes their sum directly in Assumption 1.9(a); the optional  $C^2$  sufficient bound (7) is discussed in Remark 1.10.

**Remark 4.8** ( $R_3^{SS}$  is an assumption, not a consequence). Assumption 1.9(a) is where the weak-smoothness claim actually rests, and we have not verified it from primitive conditions. Version 4’s candidate  $C^1$  moduli cannot be used: Remark 1.10 gives a counterexample even with a fixed, perfectly regular zero set.

For scale only, suppose the optional  $C^2$  modulus (7) were locally Lipschitz. A spline LS fit has

$$\|\bar{\Delta}\|_{C^2} = O_p(K_n^{1/2+2/d}n^{-1/2} + K_n^{-(s-2)/d}),$$

so even  $C^2$  consistency requires  $s > (d+3)/2$  at  $K_n \asymp n^{d/(2s+1)}$ . Its stochastic part, multiplied

by  $\|\bar{\Delta}\|_\infty \|\bar{\Delta}\|_{C^1}$ , is  $n^{-3/2}K_n^{3/2+3/d}$  and is  $o(r_n^*)$  only if  $s > 3(d+1)/4$ . Thus this conservative route requires at least

$$s > \max\left\{\frac{d+3}{2}, \frac{3(d+1)}{4}\right\},$$

not merely  $s > (d+1)/2$ . A sharper estimator-specific argument may exploit integration or cancellation and recover the weaker range, but it has not been supplied. Theorem 4.6 is therefore conditional:  $s > (d+1)/2$  is the range on which every other term in the error budget is controlled, while the remainder conditions in Assumption 1.9(a),(d) remain additional high-level requirements.

*Proof sketch.* Expand  $\Theta(\bar{\gamma})$  around  $\gamma_0$  to second order and substitute (31). The linear term  $D\Theta[\bar{\Delta}]$  is the average of two independent half-sample linear terms and delivers the CLT with the same  $\sigma_n$ . The quadratic term decomposes as

$$\frac{1}{2}D^2\Theta[\bar{\Delta}, \bar{\Delta}] - \frac{1}{8}D^2\Theta[\Delta, \Delta] = \underbrace{\frac{1}{2}D^2\Theta[b, b]}_{(i)} + \underbrace{D^2\Theta[b, \bar{u}]}_{(ii)} + \underbrace{\frac{1}{2}D^2\Theta[u^A, u^B]}_{(iii)} + R_{LS,SS},$$

the stochastic diagonals of the population-Gram terms having cancelled exactly. By Table 1: (i) is  $O(K_n^{-(2s-1)/d}) = o(r_n^*)$  under  $s > 1$ ; (ii) is  $O_p(K_n^{(1-s)/d} \sqrt{K_n/n})$ ; and (iii) is a mean-zero degenerate  $U$ -statistic of order  $n^{-1}K_n^{(d+3)/(2d)}$  (Appendix B). For (i), at  $a = d/(2s+1)$  both (ii) and (iii) are  $O_p(r_n^*)$  iff  $s \geq (d+1)/2$ . For (ii), relative to  $\sqrt{K_n^{1/d}/n}$  the term (ii) carries exponent  $a(d-2s+1)/(2d)$ , negative for *every*  $a > 0$  iff  $s > (d+1)/2$  — so it is a pure smoothness condition — while (iii) is  $o(\sqrt{K_n^{1/d}/n})$  iff  $a < d/(d+2)$ , and the sieve bias is  $o(\sqrt{K_n^{1/d}/n})$  iff  $a > d/(2s+1)$ ; the two bracket (39), nonempty iff  $2s+1 > d+2$ . Thus the binding constraint  $s > (d+1)/2$  comes from the margin terms (ii) and (iii); the corner analogues of both are smaller and impose no binding requirement beyond transversality. The remaining pieces are  $R_3^{SS}$  of (38), assumed negligible by Assumption 1.9(a) — see Remark 4.8 — the equicontinuity term of Assumption 1.9(b), the series-linearization term  $R_{LS,SS}$  controlled by Assumption 1.9(d), and, for the jackknife implementation, the fourth-order error of Lemma 4.2, whose own window is (40).

Ratio consistency of  $\hat{\sigma}_n$  follows from Proposition 2.4 *not* because the correction is small — inside (39) it need not be, and  $v3$ 's assertion that it is  $o_p(\sigma_n/\sqrt{n})$  was wrong — but because the above displays an *asymptotically linear* representation  $\hat{\Theta}_{SS} - \Theta_0 = \frac{1}{n} \sum_i \{v_S^* \epsilon_{S,i} + v_Y^* \epsilon_{Y,i}\} + o_p(\text{se}_n)$  with the *same* influence function as the plug-in. The whole point of the estimator is that the correction cancels the plug-in's quadratic diagonal; both may exceed  $\text{se}_n$  individually while their sum does not.  $\square$

**Remark 4.9** (What “ $s > (d+1)/2$ ” is, and is not). *v1 asserted that at the rate-optimal*

$K_n$  the condition  $s > (d+1)/2$  is “exactly”  $K_n^{1/2-s/d} \rightarrow 0$ . That is false:  $K_n^{1/2-s/d} = n^{(d-2s)/(2(2s+1))} \rightarrow 0$  iff  $s > d/2$ . The correct equivalent statement in terms of the estimated geometry is

$$\frac{K_n^{1/2+1/d}}{\sqrt{n}} \rightarrow 0 \iff s > \frac{d+1}{2} \quad \text{at } K_n \asymp n^{d/(2s+1)}, \quad (41)$$

i.e. uniform consistency of the gradient of the estimated surface (each derivative costing  $K_n^{1/d}$  on top of the  $\sqrt{K_n/n}$  pointwise stochastic size). It is not a coincidence that the same threshold appears in the proof of Theorem 4.6: both terms (ii) and (iii) there carry exactly one gradient factor.

**Proposition 4.10** (Margins-only debiasing is insufficient). *Under the conditions of Theorem 4.6, within each family separately,*

$$\hat{\Theta}_{SS}^{\text{mar}} - \hat{\Theta}_{SS} = \text{corner} \quad (\text{ideal, (34)}), \quad \hat{\Theta}_{SS}^{\text{mar,jack}} - \hat{\Theta}_{SS}^{\text{jack}} = \widehat{\text{corner}} \quad (\text{jackknife, (36)}),$$

both exact; the cross-family statement carries the fourth-order error of Lemma 4.2. By (37) the error of the margins-only estimator retains the corner cross term and nothing else of the corner, with expectation  $\mathcal{B}_{\mathcal{C},n}^\times$  of (28) splitting as

$$\mathcal{B}_{\mathcal{C},n}^\times = \underbrace{\mathcal{B}_{\mathcal{C},n}^{\times,\text{stoch}}}_{O(K_n/n)} + \underbrace{\mathcal{B}_{\mathcal{C},n}^{\times,\text{bias}}}_{O(K_n^{-2s/d})=O(r_n^{*2})} + R_{\text{LS},B}^\times.$$

The two  $\cos \omega$  corner diagonals of (19) are components of  $Q_g$  and  $Q_h$  and are removed by the margins-only correction. To state the diagnostic’s centering correctly, define the oracle statistic  $\Delta^u := u^A - u^B$  and  $\text{corner}_0 := \frac{1}{4}C_{\gamma_0}[\Delta_g^u, \Delta_h^u]$ . Because the common approximation bias cancels and the two population-Gram terms are independent and mean zero,

$$\mathbb{E}[\text{corner}_0] = \mathcal{B}_{\mathcal{C},n}^{\times,\text{stoch}}.$$

The actual ideal statistic uses the empirical-series errors and is evaluated at  $\bar{\gamma}$ , so  $\text{corner} = \text{corner}_0 + R_{\text{LS},C} + R_{\text{eval},C}$ ; the jackknife statistic additionally satisfies  $\widehat{\text{corner}} = \text{corner} + R_{4,C}$ , with  $R_{4,C} = O_p(\|\Delta\|_{C^1}^4)$  under Assumption 4.1(d). Thus neither feasible statistic is exactly unbiased without controlling its series-linearization, evaluation-point, and finite-difference remainders. Neither contains the approximation-bias product as a leading quadratic term. The bias part of the margins-only estimator is  $O(r_n^{*2}) = o(r_n^*)$  at the rate-optimal choice; it is made smaller by undersmoothing, not removed by debiasing. Consequently, provided Assumption 1.7(c2) holds — which neither (c1) nor pointwise nonvanishing of



$\varrho_{gh}$  implies, the three being logically independent restrictions —  $\hat{\Theta}_{SS}^{mar}$  has a stochastic cross-corner component  $\asymp K_n/n$ . This component is larger than  $r_n^*$  when  $s < d - 1$  and equal in order when  $s = d - 1$ . Since the SS guarantee of Theorem 4.6 extends down to  $s > (d + 1)/2$ , margins-only debiasing forfeits the rate guarantee on the strict window

$$\frac{d+1}{2} < s < d - 1,$$

nonempty whenever  $d \geq 4$ ; at  $s = d - 1$  it can still attain the rate but the corner term is not negligible. If the two contrast errors are conditionally uncorrelated at every corner (in particular, when  $\rho_{SY} = 0$  there), then  $\mathcal{B}_{C,n}^{\times, \text{stoch}} = 0$ : margins-only and corner-aware debiasing have the same expected stochastic centering to this order. They are not estimator-equivalent. Their exact difference is the random corner statistic displayed above; only its leading oracle stochastic component is mean zero under this condition. The statistic need not vanish and, before further cancellation, can remain  $O_p(K_n/n)$ , subject to the remainder qualifications in the preceding paragraph. For  $d \in \{2, 3\}$  the window is empty and the corner cross term is a finite-sample constants effect — of order at most  $K_n^{-1/d}$  times the margin diagonals' bound, and by Remark 3.4 it can be of the same order as a realized margin diagonal when the latter cancels. No generic-geometry classification is claimed.

*Proof.* The two family identities follow directly from (34) and (36). For the residual bias, apply the decomposition in the proof of Theorem 4.6 to

$$D^2\Theta[\bar{\Delta}, \bar{\Delta}] - \frac{1}{4}D^2\Theta[(\Delta_g, 0), (\Delta_g, 0)] - \frac{1}{4}D^2\Theta[(0, \Delta_h), (0, \Delta_h)] :$$

by bilinearity the  $Q_g$  and  $Q_h$  stochastic diagonals cancel in full — including their corner components — leaving  $C[\bar{\Delta}_g, \bar{\Delta}_h]$ , whose expectation is  $\mathcal{B}_C^\times$ . Splitting  $\bar{\Delta} = b + \bar{u} + \bar{r}$  and using  $\mathbb{E}[\bar{u}] = 0$  separates its leading terms into the stated stochastic and bias parts, of orders  $O(K_n/n)$  and  $K_n^{-2s/d}$  by (24); all terms containing  $\bar{r}$  form  $R_{LS,B}^\times$ . The lower bound for the first is Assumption 1.7(c2). The oracle corner statistic omits the bias part because it uses  $\Delta^u = u^A - u^B$ . Replacing the leading difference  $\Delta^u$  by the empirical-series difference gives  $R_{LS,C}$ , replacing  $C_{\gamma_0}$  by  $C_{\bar{\gamma}}$  gives  $R_{\text{eval},C}$ , and replacing the ideal second derivative by the three finite differences in (35) gives  $R_{4,C}$ .  $\square$

**Remark 4.11** (Relation to v1). *v1's Proposition on margins-only debiasing stated  $\mathbb{E}[\widehat{\text{corner}}] \asymp 2C[b_g, b_h] + \rho_{SY}K_n/n$ . The identification of the leftover with the  $\rho_{SY}$ -driven cross term is correct — an earlier revision of the present draft wrongly reassigned the  $\cos\omega$  diagonals to it, and (37) settles the matter. What does not belong is the bias-product term in  $\mathbb{E}[\widehat{\text{corner}}]$ , whose presence contradicted v1's own (correct) description of Table 4. Proposition 4.10 keeps*

the two objects separate: the residual bias of  $\hat{\Theta}_{SS}^{mar}$  contains both parts; the SS estimate of it targets only the stochastic part up to  $R_{LS,C}$  and  $R_{eval,C}$  and, for the jackknife family,  $R_{4,C}$ . The empirical “remainder” column of Table 4 is precisely  $\mathcal{B}_C^{\times, \text{bias}}$ , and its near-constancy in  $n$  at fixed  $K$  is the corresponding prediction.

The pair (Theorem 4.6, Proposition 4.10) is the central methodological message of the two-threshold extension: *quadratic debiasing must be corner-aware*. Because the jackknife form (32) perturbs both contrasts jointly, it is corner-aware by construction — the theory identifies what would go wrong with the seemingly natural surface-by-surface application of the single-threshold correction.

**Remark 4.12** (Leave-one-out analogue). *The LOO estimator replaces  $\frac{1}{8}D^2\Theta[\Delta, \Delta]$  by*

$$\frac{1}{2n^2} \sum_{i=1}^n D^2\Theta(\hat{\gamma})[\zeta_i^{(-i)}, \zeta_i^{(-i)}], \quad \zeta_i^{(-i)}(x) := \begin{pmatrix} \sum_w q_w s_{w,i}^{(-i)}(x) \hat{\epsilon}_{S,i}^{(-i)} \\ - \sum_w q_w s_{w,i}^{(-i)}(x) \hat{\epsilon}_{Y,i}^{(-i)} \end{pmatrix},$$

where  $s_{w,i}^{(-i)}$  is the arm- $w$  leave-one-out series weight (and is zero unless  $W_i = w$ ). There is no single weight shared across treatment arms; the common weight is only across the two outcomes within the observed arm, as in (21). With this arm-specific convention, whose corner cross component carries the product  $\hat{\epsilon}_{S,i}^{(-i)} \hat{\epsilon}_{Y,i}^{(-i)}$  — an explicit estimate of the  $\rho_{SY}$ -weighted part of (27) — while its corner diagonal components carry  $(\hat{\epsilon}_{S,i}^{(-i)})^2$  and  $(\hat{\epsilon}_{Y,i}^{(-i)})^2$ , which is why the LOO correction, like the SS correction, captures the  $\cos \omega$  terms as well. We expect it to satisfy the analogue of Theorem 4.6 under the corresponding LOO conditions of Chen and Gao (2026), but we do not claim it: the LOO construction carries its own  $R_{\text{diag}}$  remainder, distinct from (38), which we have not analysed. We focus on SS in the simulations because (32) is tuning-free.

## 5 Sieve DML and the Doubly Debiased Estimator

### 5.1 Cross-fitted two-band sieve-Riesz correction

The debiasing of Section 4 targets the *second-order* remainder but retains sieve LS first stages. The complementary device — the sieve-Riesz DML of the extended single-threshold draft — targets the *first-order* term on the sieve space and thereby accommodates generic machine-learning first stages. Let  $(I_\kappa)_{\kappa=1}^K$  partition the sample into equal-sized folds,  $\hat{\mu}_{-\kappa}$  be *any* first-stage learner of the four arm means fit off fold  $\kappa$ , and  $\hat{v}_{K_n, o, -\kappa}^*$  the feasible two-band sieve-Riesz representer (12)–(13) computed *off fold*  $\kappa$  — both its Gram matrix and its bands

must use only  $I_{-\kappa}$ . For stacked functions write

$$\|a\|_{P,2}^2 := \mathbb{E} \left[ \sum_{o \in \{S,Y\}} a_o(X, W)^2 \right], \quad \langle a, b \rangle_P := \mathbb{E} \left[ \sum_{o \in \{S,Y\}} a_o(X, W) b_o(X, W) \right].$$

Let  $\Pi_{K_n}$  be the  $\langle \cdot, \cdot \rangle_P$ -orthogonal projection of the four arm-mean functions onto their stacked sieve space. For the empirical measure case define the held-out-fold value

$$\Theta_\kappa(\gamma) := \frac{K}{n} \sum_{i \in I_\kappa} m_\gamma(X_i), \quad m_\gamma(x) := \mathbb{1}\{g(x) \geq 0\} \mathbb{1}\{h(x) \geq 0\} v_0(x);$$

in the known-density case put  $\Theta_\kappa(\gamma) := \Theta(\gamma)$ . The cross-fitted sieve-debiased estimator is

$$\tilde{\Theta} := \frac{1}{K} \sum_{\kappa=1}^K \left[ \Theta_\kappa(\hat{\gamma}_{-\kappa}) + \frac{K}{n} \sum_{i \in I_\kappa} \sum_{o \in \{S,Y\}} \hat{v}_{K_n, o, -\kappa}^*(X_i, W_i) (O_{o,i} - \hat{\mu}_{o, -\kappa}(X_i, W_i)) \right], \quad (42)$$

where  $O_{S,i} := S_i$  and  $O_{Y,i} := Y_i$ . The correction annihilates the first-order estimation error of  $\hat{\mu}_{-\kappa}$  on the sieve space: it is the local influence function of the irregular functional, well defined even though no global influence function exists.

**Assumption 5.1** (Two-band local robustness). *For each fold  $\kappa$ : (a)  $\|\hat{\mu}_{-\kappa} - \mu_0\|_\infty = o_p(1)$  and  $\|\hat{\gamma}_{-\kappa} - \gamma_0\|_\infty = o_p(\varepsilon_n)$ ; (b) the band sequence satisfies  $\varepsilon_n \downarrow 0$ ,  $n\varepsilon_n/(K_n \log K_n) \rightarrow \infty$ , and  $\varepsilon_n^{-1} \|\hat{\gamma}_{-\kappa} - \gamma_0\|_\infty = o_p(1)$ ; (c) (boundary stability, normal-graph form) for both  $\varphi \in \{g, h\}$  with  $\mathcal{M}_\varphi := \{\varphi_0 = 0\}$  nonempty, the estimated level set admits, w.p.  $\rightarrow 1$ , the normal-graph representation*

$$\{\hat{\varphi}_{-\kappa} = 0\} = \{x + r_n(x) n_\varphi(x) : x \in \mathcal{M}_\varphi\}, \quad \|r_n\|_{C^1(\mathcal{M}_\varphi)} = o_p(1), \quad (43)$$

with  $\mathcal{M}_\varphi$  of positive reach and the graph map's Jacobian bounded above and below uniformly; if  $\mathcal{M}_\varphi = \emptyset$  we require instead  $\mathbb{P}(\{\hat{\varphi}_{-\kappa} = 0\} \neq \emptyset) \rightarrow 0$ . Condition (43) implies  $d_H \rightarrow 0$  and  $\text{Leb}(\{\hat{\varphi} \geq 0\} \triangle \{\varphi_0 \geq 0\}) \rightarrow 0$ , but is strictly stronger than both together (Remark 5.2); (d) the product rate  $\|\hat{v}_{K_n, -\kappa}^* - v_{K_n}^*\|_{P,2} \cdot \|\hat{\mu}_{-\kappa} - \mu_0\|_{P,2} = o_p(\sqrt{K_n^{1/d}/n})$ ; (d') (representer consistency)  $\|\hat{v}_{K_n, -\kappa}^* - v_{K_n}^*\|_{sd} = o_p(\sigma_n)$  and

$$\frac{K}{n} \sum_{i \in I_\kappa} \sum_{o \in \{S,Y\}} \{\hat{v}_{K_n, o, -\kappa}^*(X_i, W_i)\}^2 = O_p(\sigma_n^2);$$

(e) with  $\Delta_\mu := \hat{\mu}_{-\kappa} - \mu_0$  and the projection defined above, the out-of-span boundary remainder

$$R_{5,\kappa} := D_\mu \Theta(\mu_0) [(I - \Pi_{K_n}) \Delta_\mu] = o_p(\sqrt{K_n^{1/d}/n}).$$

On the sieve span,  $D_\mu \Theta(\mu_0) [\Pi_{K_n} \Delta_\mu] = \langle v_{K_n}^*, \Pi_{K_n} \Delta_\mu \rangle_P$  by the definition of  $v_{K_n}^*$ , so there is no additional undefined “representer approximation bias” term; (f) (leading nonlinear remainder) writing  $\Delta := \hat{\gamma}_{-\kappa} - \gamma_0$ ,

$$\|\Delta\|_\infty \|\Delta\|_{C^1} + \|\Delta\|_\infty^2 = o_p(\sqrt{K_n^{1/d}/n}),$$

which bounds  $|D^2 \Theta[\Delta, \Delta]|$  by (18)–(19), together with a genuine third-order remainder condition  $R_3 = o_p(\sqrt{K_n^{1/d}/n})$ ;

(g) the Lyapunov condition Assumption 1.9(c) holds, and  $\liminf \sigma_n^2/K_n^{1/d} > 0$ , the latter supplied by Assumption 1.6(b);

(h) the number of folds  $K$  is fixed,  $K_n \rightarrow \infty$ , and, in the empirical-measure case, Assumption 1.9(b) holds foldwise for  $m_{\hat{\gamma}_{-\kappa}} - m_{\gamma_0}$ .

**Remark 5.2** (Two corrections to v2’s assumption list). Condition (c) as stated in v2 was  $\mathcal{H}^{d-1}(\{\hat{g} = 0\} \triangle \{g_0 = 0\}) = o_p(1)$ . That is not merely strong, it is unsatisfiable: two distinct hypersurfaces have essentially disjoint  $\mathcal{H}^{d-1}$ -supports, so the symmetric difference has measure  $\approx 2\mathcal{H}^{d-1}(\{g_0 = 0\})$  no matter how close they are — two parallel segments  $10^{-10}$  apart already fail it. v2 also stated (c) for  $g$  only. v3 replaced it by Hausdorff distance plus region symmetric-difference measure and called the pair “equivalent” to normal-graph convergence. That is also wrong, in the opposite direction: a surface can be arbitrarily Hausdorff-close to  $\mathcal{M}_\varphi$  while oscillating at a fine scale, so that its surface measure diverges and its normals do not converge, all with region symmetric-difference measure tending to zero. Since the band construction (13) converges to a Hausdorff integral — carrying  $\|\nabla g_0\|^{-1}$  and hence the normal direction — neither metric controls what is needed. Hence (c) now requires the normal-graph representation (43) with  $C^1$  control of the displacement, which is what makes the coarea change of variables valid with a Jacobian tending to one; the empty-margin case needs a separate convention because Hausdorff distance to  $\emptyset$  is not defined. Condition (f) in v2 read  $\|\Delta\|_{C^1}^2 \|\Delta\|_\infty$  and called it the “second-order remainder”; under  $\Delta = t\eta$  that quantity is  $O(t^3)$ , whereas the term it is supposed to control,  $\frac{1}{2}D^2\Theta[\Delta, \Delta]$ , is generically  $O(t^2)$ . A cubic bound cannot control a quadratic term, so Theorem 5.3 did not follow from v2’s list. The corrected (f) has the right homogeneity, and for a sieve LS first stage it reduces to  $K_n^{1+1/d}/n$ , consistent with Table 1. Condition (d’) is new: the product rate in (d) can hold with an oracle learner ( $\|\hat{\mu} - \mu_0\| = 0$ ) for an arbitrarily wrong  $\hat{v}^*$ , which would still ruin  $\hat{\sigma}_n$ .

**Theorem 5.3** (Two-band sieve DML). *Under Assumptions 1.4–1.6, 1.9, and 5.1,*

$$\frac{\sqrt{n}(\tilde{\Theta} - \Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0, 1), \quad \sigma_n^2 \asymp K_n^{1/d},$$

and the two-band variance estimator (15), computed with cross-fitted residuals, is ratio-consistent. As in the single-threshold case, the irregularity weakens the product-rate requirement relative to regular DML: the tolerance is  $\sqrt{K_n^{1/d}/n} \gg n^{-1/2}$ . In the empirical-measure case the held-out values in (42) add the ordinary term  $n^{-1} \sum_i \{m_{\gamma_0}(X_i) - \Theta_0\} = O_p(n^{-1/2})$ ; it is  $o_p(\sigma_n/\sqrt{n})$  because  $K_n \rightarrow \infty$ . Thus the displayed normalization and the two-band variance deliberately cover the irregular regime in Assumption 5.1(h), not a fixed- $K_n$  regular limit.

*Proof sketch.* Foldwise linearization of the held-out plug-in and conditioning on the training sample give

$$\tilde{\Theta} - \Theta_0 = \frac{1}{n} \sum_{i=1}^n \sum_o v_{K_n, o}^*(X_i, W_i) \epsilon_{o, i} + \frac{1}{n} \sum_{i=1}^n \{m_{\gamma_0}(X_i) - \Theta_0\} + o_p(\sigma_n/\sqrt{n}),$$

with the second sum absent under known integration. The in-span first-stage error cancels by the Riesz identity; (d) and (d') control use of the estimated representer, (e) controls the uncanceled out-of-span derivative, and (f) controls the moving-boundary nonlinear remainder. Condition (h) makes the ordinary empirical-process sum negligible, and (g) gives the triangular-array CLT. Proposition 2.4, applied conditionally fold by fold together with (d'), supplies ratio consistency.  $\square$

Assumption 5.1 is longer than the corresponding list in v1, which cited only sup-norm consistency, a product rate, and sieve approximation bias. Those three do not control a boundary functional: (b) and (c) are what make the  $\varepsilon$ -band a consistent surrogate for a Hausdorff integral at an *estimated* surface, (d') is what makes the variance estimate consistent, (f) is the nonlinear (moving-boundary) remainder that has no analogue in regular DML, and (g) is what a CLT for a triangular array with  $\sigma_n^2 \rightarrow \infty$  requires. We state them because the two conditions that bind in our experiments are (e) and (f), not the product rate. As in Theorem 2.2(iii), the conclusion is a *centered* limit only because (e) forces the *out-of-span boundary derivative* below  $\sqrt{K_n^{1/d}/n}$ , which is again an undersmoothing requirement.

**Remark 5.4** (Least-squares orthogonality; why raw AIPW corrections miscalibrate). *Suppose the first stage is the sieve LS estimator on the same basis used for the representer, and consider the non-cross-fitted estimator: a full-sample fit with in-sample residuals. Then the normal equations  $\sum_i \psi(X_i, W_i) \hat{\epsilon}_i = 0$  hold exactly, and since  $\hat{v}^*$  lies in the span of  $\psi$*

the correction vanishes identically. This is an algebraic in-sample orthogonality statement; it does not remove the out-of-span or nonlinear bias of the sieve plug-in. It helps explain a phenomenon documented in our earlier experiments: adding a raw  $\varepsilon$ -band AIPW correction  $(2\varepsilon)^{-1} \mathbb{1}\{|\hat{\tau}| < \varepsilon\} \xi_i$  on top of a full-sample sieve plug-in can over-correct. The projected (sieve-Riesz) correction is exactly zero, whereas the raw band weight generally lies outside the regression span and is not annihilated by the normal equations. Its discrepancy can contain both bias and high-variance noise; without an additional conditional-centering argument it should not be called “pure noise,” although its second moment can scale as  $1/\varepsilon$ .

This exactness does not survive cross-fitting, contrary to what v1 asserted. In (42) the residuals  $O_{o,i} - \hat{\mu}_{o,-\kappa}(X_i, W_i)$  for  $i \in I_\kappa$  are out-of-fold: the normal equations hold on  $I_{-\kappa}$ , not on  $I_\kappa$ . Conditional on the training sample the correction decomposes as

$$\frac{K}{n} \sum_{i \in I_\kappa} \hat{v}_{-\kappa}^* \epsilon_i - \frac{K}{n} \sum_{i \in I_\kappa} \hat{v}_{-\kappa}^* (\hat{\mu}_{-\kappa} - \mu_0)(X_i, W_i).$$

Only the first, residual-noise component is conditionally mean zero; the second has conditional mean  $-\langle \hat{v}_{-\kappa}^*, \hat{\mu}_{-\kappa} - \mu_0 \rangle_P$  and cancels the in-span plug-in error. Thus the cross-fitted correction is generally nonzero and is not “pure added noise.” For a sieve LS first stage it is first-order equivalent to the usual series linearization only under the same Gram, projection, and remainder conditions used in Theorem 5.3; exact redundancy does not follow from the training normal equations. The projected correction is especially useful when the first stage is not the LS fit on the representer’s basis — e.g. gradient boosting or random forests — which is its intended use case.

**Remark 5.5** (When the sieve cannot see the learner’s error). Assumption 5.1(e) deserves emphasis because it is the one that binds in practice. Write  $\Delta := \hat{\mu}_{-\kappa} - \mu_0$  and decompose the plug-in’s first-order error as

$$D_\mu \Theta(\mu_0)[\Delta] = \underbrace{\langle v_{K_n}^*, \Delta \rangle_P}_{\text{in-span}} + \underbrace{D_\mu \Theta(\mu_0)[(I - \Pi_{K_n})\Delta]}_{=: R_5, \text{ out-of-span}}. \quad (44)$$

The correction in (42) replaces the first term by the score  $\mathbb{E}_n[v^* \epsilon]$ ;  $R_5$  is untouched. Because  $\hat{v}_{-\kappa}^*$  and  $\hat{\mu}_{-\kappa}$  are measurable with respect to  $I_{-\kappa}$  while the oracle residual-noise score  $\mathbb{E}_n[v^* \epsilon]$  is evaluated on  $I_\kappa$ , that score is conditionally mean zero and  $\text{Cov}(\text{score}, R_5) = 0$  exactly, so

$$\text{Var}(\tilde{\Theta}) = \frac{\sigma_n^2}{n} + \text{Var}(R_5) + \text{Var}(\text{h.o.t.}) + \text{cross terms in the h.o.t.},$$

while  $\hat{\sigma}_n^2$  estimates only  $\sigma_n^2$ . This is a statement about which terms the standard error omits;

it is not a statement about the sign of the coverage error, and *v1*’s conclusion that the interval “can only under-cover, never over-cover” does not follow, for three reasons. First, the plug-in error also contains  $R_5$ ; the net change in dispersion from adding the correction is  $\sigma_n^2/n - \text{Var}(\langle v_{K_n}^*, \Delta \rangle)$ , of either sign, so the debiased estimator is not “the plug-in plus an independent perturbation” except in the degenerate case  $\Pi_{K_n}\Delta \equiv 0$ . Second, coverage depends on bias as well as dispersion, and  $\hat{\sigma}_n$  is itself a noisy band-based estimate whose own bias — upward at small  $\varepsilon_n$ , downward when the band is widened — is not sign-restricted; SE/SD ratios above one occur in several cells of Table 2. Third, the exact-orthogonality step requires the representer to be built off fold  $\kappa$ , which our current implementation does not do (Section 6.5, item (L6)).

Directly measured, the variance-addition channel is small on our design: the saved variance experiment reports  $\{\text{Var}(\text{DML}) - \text{Var}(\text{plug-in})\}/\widehat{\text{SE}}^2$  equal to  $-0.04$  (GBR),  $-0.00$  (RF) and  $+0.04$  (smooth RBF ridge) at  $n = 4000$  — not the  $\approx +1$  that the “independent added variance” reading predicts. The machine-learning undercoverage on this design is therefore predominantly residual bias, not unmodelled variance. We correct *v1*, which asserted the opposite.

What must be small in Assumption 5.1(e) is not  $\|(I - \Pi_{K_n})\Delta\|_{L^2}$  but the boundary functional evaluated at it: only the sieve’s ability to track the learner’s error on the margins matters, and global  $L^2$  richness is not the right diagnostic. This is exactly where regression trees fail. A boosted ensemble’s error is piecewise constant with jumps at adaptively chosen split hyperplanes, and boosting places its splits where the loss gradient is largest — for a CATE surface whose zero level set defines the estimand, that is near the margins themselves. So  $(I - \Pi_{K_n})\Delta$  is largest exactly on the set the derivative integrates over. Section 6 measures a global- $L^2$  proxy for this: the  $R^2$  of the boosted first-stage error on the tensor B-spline representer span is 0.21–0.34 across the four arm regressions (mean 0.26), and the projected correction removes about 31% of the plug-in’s boundary bias. Tree ensembles additionally violate the smoothness conditions in a way that is not merely quantitative:  $\nabla(\hat{\mu} - \mu_0)$  is a sum of Dirac measures on the split hyperplanes, the  $\varepsilon$ -band  $\{|\hat{\tau}| < \varepsilon\}$  is a union of whole leaf cells (of measure  $O(1)$  or exactly zero, never  $O(\varepsilon)$ ), and the estimated region is an axis-aligned staircase whose surface measure over-counts an oblique boundary by up to  $\sqrt{d}$ . The practical conclusion is stated in Section 6.4: match the representer basis to the learner’s own feature map, and prefer smooth learners.

The diagnosis is not special to two thresholds. In the single-threshold setting, the extended draft’s own Table 7 reports, for the value functional on identical samples and folds, a gradient-boosting DML estimator that is essentially unbiased ( $|\text{bias}| \leq 0.003$  at every  $n$ ) yet covers only 0.90, with SE/SD = 0.78, 0.84, 0.88 — while a smooth random-feature first



stage on the same design has  $\text{SE}/\text{SD} = 0.84, 0.94, 0.99$  and coverage rising to 0.93. The decomposition here explains that contrast, though — given the variance measurement above — the mechanism in the two-threshold design is the omitted-bias channel rather than the omitted-variance channel.

## 5.2 The doubly debiased estimator

The two devices remove different terms — the first-order projection error (DML) and the second-order diagonal (SS) — and one would like to compose them. Doing so naively does not work, and we begin by saying why, because the construction proposed in v1–v2 of this draft is invalid.

Split the sample into halves  $A, B$ ; fit generic ML first stages on each half; let  $\hat{\gamma}^A, \hat{\gamma}^B$  be the two contrast pairs (each predicted on the full covariate sample),  $\bar{\gamma}$  their average, and  $e_j := \hat{\gamma}^j - \gamma_0$ . The v1–v2 estimator added ordinary fold-specific one-step corrections to the split-sample combination:

$$\hat{\Theta}_{DD}^{v2} := \underbrace{2\Theta(\bar{\gamma}) - \frac{1}{2}(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B))}_{=: \Psi(\hat{\gamma}^A, \hat{\gamma}^B)} + \frac{1}{n} \sum_{i=1}^n \sum_{o \in \{S, Y\}} \hat{v}_{K_n, o, -\kappa(i)}^*(X_i, W_i) \hat{\epsilon}_{o, i}^{\text{of}}. \quad (45)$$

**Remark 5.6** ((45) reintroduces the diagonals it was meant to remove). *Expanding  $\Psi$  around  $\gamma_0$  and using  $\bar{e} = \frac{1}{2}(e_A + e_B)$ ,*

$$\Psi(\hat{\gamma}^A, \hat{\gamma}^B) - \Theta_0 = D\Theta[\bar{e}] + \frac{1}{2}D^2\Theta[e_A, e_B] + O(3rd), \quad (46)$$

*the own-half diagonals having cancelled — this is the whole point of SS. But the correction in (45) uses representers built at  $\hat{\gamma}^A, \hat{\gamma}^B$ , so its conditional expectation given the fits is*

$$-\frac{1}{2}D\Theta(\hat{\gamma}^A)[e_A] - \frac{1}{2}D\Theta(\hat{\gamma}^B)[e_B] = -D\Theta(\gamma_0)[\bar{e}] - \frac{1}{2}(D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]) + O(3rd).$$

*The linear terms cancel as intended, but the second line reintroduces the own-half diagonals, at  $-\frac{1}{2}$  each. The cleanest way to see it: take  $e_A = e_B = b$  deterministic. Then  $\bar{\gamma} = \hat{\gamma}^A = \hat{\gamma}^B$ ,  $\Psi = \Theta(\gamma_0 + b)$  — SS does nothing, correctly, since there is no independent replicate — and (45) has error  $\frac{1}{2}D^2\Theta[b, b] - D^2\Theta[b, b] = -\frac{1}{2}D^2\Theta[b, b]$ , which is quadratic, not third order. With independent mean-zero half-sample errors the bias is  $-\frac{1}{2}\mathbb{E}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\} = O(K_n^{1+1/d}/n)$ . Mesh-resonant designs can attain this upper rate, but cancellation may make it smaller; relative to the plug-in, the algebraic coefficient is reversed in the displayed expansion, not uniformly in every data-generating process. Theorem “DD” of v1–v2 is therefore false as*

stated. Appendix C, item 7, exhibits both facts numerically on the lens design.

The error is that (45) debiases the wrong functional.  $\Psi$  is a function of *two* arguments, and its derivative in the first is

$$D_A \Psi = D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A), \quad \text{not} \quad \frac{1}{2}D\Theta(\hat{\gamma}^A). \quad (47)$$

The two agree to first order — both are  $\approx \frac{1}{2}D\Theta(\gamma_0)$  — and differ exactly at the order that matters. Substituting (47) restores the intended cancellation: a short computation gives  $D_A \Psi[e_A] = \frac{1}{2}D\Theta_0[e_A] + \frac{1}{2}D^2\Theta[e_A, e_B] + O(3\text{rd})$  and symmetrically, so with (46)

$$\Psi - \Theta_0 - D_A \Psi[e_A] - D_B \Psi[e_B] = -\frac{1}{2}D^2\Theta[e_A, e_B] + O(3\text{rd}), \quad (48)$$

a cross-half term of the same order as SS's own leftover. It is mean zero only when its systematic bias product vanishes; see Remark 5.9.

Implementing (47) requires care:  $\bar{\gamma}$  depends on *both* halves, so no observation used to fit either is available for an independent residual-noise score. We therefore use three folds.

**Definition 5.7** (Corrected doubly debiased estimator). *Partition the sample into three folds  $I_1, I_2, I_3$  of equal size. For each cyclic assignment  $\pi$  of  $(A, B, C)$  to  $(I_1, I_2, I_3)$ : fit  $\hat{\gamma}^A$  on  $A$  and  $\hat{\gamma}^B$  on  $B$ ; form  $\bar{\gamma}$ ; build the two feasible sieve-Riesz representers  $\hat{v}^{*A}, \hat{v}^{*B}$  of the composite derivatives  $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A)$  and  $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^B)$  via (12)–(13), using only  $A \cup B$  data for the bands and the Gram matrix. In the empirical-measure case set*

$$\Psi_C(\hat{\gamma}^A, \hat{\gamma}^B) := \frac{3}{n} \sum_{i \in C} \{2m_{\bar{\gamma}}(X_i) - \frac{1}{2}m_{\hat{\gamma}^A}(X_i) - \frac{1}{2}m_{\hat{\gamma}^B}(X_i)\};$$

under known integration set  $\Psi_C := \Psi$ . Then define

$$\hat{\Theta}_{DD}^{(\pi)} := \Psi_C(\hat{\gamma}^A, \hat{\gamma}^B) + \frac{3}{n} \sum_{i \in C} \sum_{o \in \{S, Y\}} \sum_{j \in \{A, B\}} \hat{v}_o^{*j}(X_i, W_i) (O_{o,i} - \hat{\mu}_o^j(X_i, W_i)). \quad (49)$$

The corrected DD estimator is  $\hat{\Theta}_{DD} := \frac{1}{3} \sum_{\pi} \hat{\Theta}_{DD}^{(\pi)}$  over the three cyclic assignments.

Because  $\hat{v}^{*A}, \hat{v}^{*B}, \hat{\mu}^A, \hat{\mu}^B$  are all  $A \cup B$ -measurable while the score runs over  $C$ , its residual-noise component is conditionally mean zero, while the score's conditional mean is  $-\langle \hat{v}^{*A}, e_A \rangle - \langle \hat{v}^{*B}, e_B \rangle \approx -D_A \Psi[e_A] - D_B \Psi[e_B]$ , which is exactly what (48) requires.

Two definitional points that Definition 5.7 must settle, and which v3 left open. *Fold conventions for  $\Psi$* . In the empirical-measure case all three values in the composite are evaluated on the held-out fold  $C$ . Hence their residual score is independent of the fits. Averaging

the truth-level terms over the rotations gives exactly  $n^{-1} \sum_i m_{\gamma_0}(X_i)$ ; the remaining local empirical process is

$$(\mathbb{P}_C - P)\{2(m_{\bar{\gamma}} - m_{\gamma_0}) - \frac{1}{2}(m_{\hat{\gamma}^A} - m_{\gamma_0}) - \frac{1}{2}(m_{\hat{\gamma}^B} - m_{\gamma_0})\}, \quad (50)$$

which is imposed directly below. A full-sample  $\Psi$  would reuse the training observations and would not be justified by the stated cross-fitting argument. In the known-density case the issue does not arise. *Representer conditions.* Assumption 5.1(d),(d'),(e) must be read for the *composite* representer:  $\hat{v}^{*A}$  is required to satisfy them as an estimator of the Riesz representer of  $D_A\Psi = D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A)$ , not of  $\frac{1}{2}D\Theta(\gamma_0)$ . Since both terms are band constructions at  $C^1$ -close level sets, one may expect analogous bounds, but they do not follow automatically: the two estimated band vectors are dependent and are evaluated at different surfaces. They are therefore imposed directly on the *difference* rather than inherited from the ordinary representer.

Three practical points. (i) A single assignment scores on one third of the sample, so its score variance is  $3\sigma_n^2/n$ ; but  $\frac{1}{3} \sum_{\pi} \frac{3}{n} \sum_{i \in C_{\pi}} (\cdot) = \frac{1}{n} \sum_{i=1}^n (\cdot)$  over the three cyclic assignments, so the rotation recovers the full-sample score and  $\sigma_n^2/n$ , as in ordinary cross-fitting. (ii) The fits use  $n/3$  rather than  $n/2$  observations, inflating the second-order terms by a constant; this is the price of the extra fold and does not affect rates. (iii)  $\hat{v}^{*A}$  is built from a *difference* of two band vectors, at  $\bar{\gamma}$  and at  $\hat{\gamma}^A$ . Since  $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^A) \approx \frac{1}{2}D\Theta(\gamma_0)$  the difference is not a near-cancellation, so the construction is numerically benign; but the two band vectors must be computed at genuinely different level sets, which is the step an implementation is most likely to short-cut.

**Theorem 5.8** (Doubly debiased estimation, corrected). *For assignment  $\pi$ , write*

$$\hat{q}_i^{(\pi)} := \sum_{o \in \{S,Y\}} \sum_{j \in \{A,B\}} \hat{v}_o^{*j}(X_i, W_i) \hat{\epsilon}_{o,i}^j, \quad i \in C_{\pi},$$

*and, conditional on the training folds  $\mathcal{T}_{\pi} := A \cup B$ , let  $v^{*A,\pi}, v^{*B,\pi}$  be the population Riesz representer of the two actual composite derivatives and set*

$$q_i^{*,\pi} := \sum_{o \in \{S,Y\}} \{v_o^{*A,\pi} + v_o^{*B,\pi}\}(X_i, W_i) \epsilon_{o,i}, \quad \sigma_{\pi,n}^2 := \mathbb{E}[(q_i^{*,\pi})^2 \mid \mathcal{T}_{\pi}].$$

*and define the actual composite-score variance estimator*

$$\hat{\sigma}_{DD,n}^2 := \frac{1}{3} \sum_{\pi} \frac{3}{n} \sum_{i \in C_{\pi}} \{\hat{q}_i^{(\pi)}\}^2. \quad (51)$$

Suppose Assumptions 1.4–1.6, 1.9, and 5.1 hold for the ML first stages, with Assumption 5.1(f) replaced by a direct condition on the third-and-higher remainders, and in addition: (i) the level-set geometry is uniform on every segment used to expand  $\Psi, D_A\Psi, D_B\Psi$  between  $\gamma_0, \bar{\gamma}, \hat{\gamma}^A, \hat{\gamma}^B$ , w.p.  $\rightarrow 1$ ; (ii)  $\|\hat{\gamma}^j - \gamma_0\|_{C^1} = o_p(1)$  and  $\hat{\gamma}^A, \hat{\gamma}^B$  are independent; (iii) the third-order margin/corner and evaluation-point remainders in those expansions are  $o_p(\sqrt{K_n^{1/d}/n})$ ; (iv)  $D^2\Theta[e_A, e_B] = o_p(\sqrt{K_n^{1/d}/n})$ ; (v) in the empirical-measure case, uniformly over the three assignments, the displayed local empirical process (50) is  $o_p(\sqrt{K_n^{1/d}/n})$ ; (vi)

$$\max_{\pi} \|v^{*A,\pi} + v^{*B,\pi} - v_{K_n}^*\|_{sd} = o_p(\sigma_n),$$

and the ordinary-score Lyapunov condition in Assumption 1.9(c) holds. In addition, for some  $\varsigma > 0$ , the actual conditional composite scores satisfy

$$\max_{\pi} \frac{(n/3)^{-\varsigma/2} \mathbb{E}[|q_i^{*,\pi}|^{2+\varsigma} \mid \mathcal{T}_{\pi}]}{\sigma_{\pi,n}^{2+\varsigma}} \xrightarrow{p} 0;$$

and (vii)

$$\hat{\sigma}_{DD,n}^2 / \sigma_n^2 \xrightarrow{p} 1.$$

Then for  $\hat{\Theta}_{DD}$  of Definition 5.7,

$$\frac{\sqrt{n}(\hat{\Theta}_{DD} - \Theta_0)}{\hat{\sigma}_{DD,n}} \xrightarrow{d} \mathcal{N}(0, 1).$$

Condition (vii) is a genuine additional condition on the composite representers and Riesz-weighted residual errors. The ordinary two-band variance (15) may replace (51) only after ratio equivalence is proved; this theorem does not assume that unproved substitution.

The gain over Theorem 5.3 is precise but narrower than v3 claimed: the quadratic correction replaces  $D^2\Theta[\Delta, \Delta]$  — whose expectation contains the own-half stochastic diagonal, at most of order  $K_n^{1+1/d}/n$  under the general bounds and attaining that order in mesh-resonant examples — by  $D^2\Theta[e_A, e_B]$ , in which that diagonal is absent. Condition (iv) is itself a second-order condition, so it is wrong to describe the requirement as dropping “from second to third order”; what drops is the stochastic part of the second-order requirement, while the bias part survives untouched (Remark 5.9). For a learner whose error is predominantly systematic,  $D^2\Theta[e_A, e_B] \approx D^2\Theta[b, b] \approx D^2\Theta[\Delta, \Delta]$  and there is no gain at all. Note that the regular-DML rate  $\|\hat{\gamma}^j - \gamma_0\|_{\infty} = o_p(n^{-1/4})$  is not imposed: irregularity already relaxes the second-order tolerance from  $n^{-1/2}$  to  $\sqrt{K_n^{1/d}/n}$ , and the quadratic correction relaxes it again. The statement is conditional on (iii)–(vii) in the same sense as Theorem 4.6; see Appendix B,

§B.4.

**Remark 5.9** (What condition (iv) hides). *Condition (iv) is not innocuous, and it is where the limits of any split-sample device show up. Decomposing  $e_j = b_j + u_j$  into a systematic and a mean-zero part, independence of the two folds gives  $\mathbb{E} D^2\Theta[e_A, e_B] = D^2\Theta[b_A, b_B]$ : the split-sample construction removes own-half stochastic diagonals, never the bias product. For sieve LS first stages with undersmoothing this is  $O(K_n^{-(2s-1)/d}) = o(r_n^*)$  (Table 1). For a generic machine-learning first stage with a systematic bias — a tree ensemble’s CATE attenuation, for instance, measured at 13% in Section 6.4 —  $b_A \approx b_B \approx b$  is not small, and  $D^2\Theta[b, b]$  is a genuine first-order-relevant term that neither the quadratic nor the Riesz correction touches. This is the formal counterpart of the Monte Carlo finding that boosting on the rough truth is bias-limited at +0.045 regardless of what corrections are applied. No debiasing scheme in this paper repairs a biased learner; they repair the noise-induced curvature of an otherwise adequate one.*

*Proof sketch.* Combine (46), (47) and (48) (derived in Appendix B, §B.3) with the conditional-mean identity above and the conditionally mean-zero fold- $C$  residual-noise component. The leading term is  $n^{-1} \sum_i \{v_S^*(X_i, W_i)\epsilon_{S,i} + v_Y^*(X_i, W_i)\epsilon_{Y,i}\}$  after replacing each estimated composite representer first by its target and then the sum of the two targets by  $v_{K_n}^*$ . The first replacement is controlled by the composite versions of Assumption 5.1(d),(d’),(e), and the second by (vi). Its CLT is Assumption 5.1(g); the remaining terms are (iii)–(v), the out-of-span remainder  $R_5$ , and the truth-level empirical mean. The latter is  $O_p(n^{-1/2}) = o_p(\sigma_n/\sqrt{n})$  because  $K_n \rightarrow \infty$ . Condition (vii) and Slutsky’s theorem give studentization by (51).  $\square$

Theorem 5.8 answers, in the two-threshold setting, the question left open in the single-threshold draft (its Remark 7) — but the answer is more delicate than v1–v2 claimed. The cross-fitted influence-function correction and the diagonal quadratic correction are compatible only if the first is taken with respect to the *composite* functional  $\Psi$ ; correcting  $\Theta$  fold-by-fold and then adding the quadratic term does not compose, because the one-step correction is itself second-order accurate only at the point where its representer is evaluated. Two further cautions. Condition (i) *fails* for tree-ensemble first stages (Remark 4.4), so Theorem 5.8 does not cover the GBR-based DD cells of Section 6. And those cells implement (45), not (49): see Section 6.5, item (L7).

## 6 Monte Carlo Evidence Calibrated to the Graduation Experiment

### 6.1 Design

Following the calibration philosophy of [Chen and Ritzwoller \(2023\)](#) — whose generative model is itself fit to the [Banerjee et al. \(2015\)](#) “graduation” RCT in Pakistan ( $n = 854$ ; 2-year outcomes as surrogates  $S$ , 3-year consumption as  $Y$ ) — we evaluate all estimators on three data-generating processes with *known* truth:

**KRR** a kernel-ridge exact-truth oracle: per-arm conditional means of  $(S, Y)$  fit to the graduation data by kernel ridge regression are taken as the *true*  $\mu_{o,0}(\cdot, w)$ ; covariates are drawn from a Gaussian-KDE fit of the empirical covariate distribution ( $d = 2$ ), truncated to  $[-3, 3]^2$ ; residual noise is calibrated to the data with within-unit  $(S, Y)$  coupling ( $\rho_{SY} \neq 0$ ). The truth  $\theta_0 = 0.161$  and all margin/corner geometry are computable exactly. (See Section 6.5, items (L1)–(L3), for two implementation defects that displace this “exact” truth.)

**WGAN** the [Chen and Ritzwoller \(2023\)](#) three-GAN cascade  $X \rightarrow S \mid X, W \rightarrow Y \mid S, X, W$  (the **ds-wgan** design of [Athey et al., 2021](#)) trained on the same data; the fitted generators are frozen as the truth and CATEs computed by common-random-number Monte Carlo. ReLU generators make the true contrast surfaces piecewise linear — effectively  $s \approx 1$  — so this design sits *below* the smoothness threshold of every theorem in this paper and is a stress test, not a corroboration. We use two variants: the *scalar-surrogate* cascade, whose geometry has both margins binding ( $\theta_0 = 0.194$ ), and the *full-surrogate* cascade at  $d = 2$ , which is *degenerate* — conditioning  $Y$  on the full 21-dimensional surrogate vector flattens  $\tau_S$  and pushes  $\tau_Y > 0$  everywhere, so  $\theta_0 = 0$  with empty margins, violating Assumption 1.5(a) and (c). The degenerate arm is retained deliberately: it shows how the boundary-band inference behaves when the estimand sits on the boundary of the parameter space and no margin exists.

**Affine** a Gaussian–affine design with closed-form orthant truth, as a specification check of the numerical pipeline.

For each DGP we draw experiments of size  $n \in \{1000, 2000, 4000, 8000\}$  (500 replications each) and compare, at nominal 95%: (i) the sieve plug-in  $\hat{\Theta}$ ; (ii) margins-only SS  $\hat{\Theta}_{SS}^{\text{mar}}$ ; (iii) corner-aware SS  $\hat{\Theta}_{SS}$  (32); (iv) the 2-fold cross-fitted GBR plug-in; (v) (iv) + projected Riesz correction; (vi) the doubly debiased  $\hat{\Theta}_{DD}$  (49). A separate decomposition experiment

(Table 3) then varies the DML construction one axis at a time — fold count, first-stage learner (boosting, random forest, smooth random-feature ridge), and representer basis — to isolate the source of the machine-learning undercoverage. All are studentized by the two-band sieve variance (15).<sup>1</sup> Table 2 reports bias, MC standard deviation, RMSE, the mean-SE/MC-SD ratio, empirical coverage, and mean CI length; the estimated corner correction is examined separately in Section 6.2. With 500 replications the Monte Carlo standard error of a coverage figure near 0.95 is about 0.010, and near 0.90 about 0.013; differences smaller than roughly 0.03 should not be read as real.

Table 2: Calibrated Monte Carlo (500 replications per cell): bias, MC standard deviation, RMSE, mean-SE/MC-SD ratio, empirical 95% coverage, and mean CI length, by DGP, estimator, and sample size. All intervals are studentized by the two-band sieve-Riesz standard error (15) (cross-fitted residuals for the GBR family). Sieve dimension is fixed at  $K = 25$  per arm for every  $n$ .

| Estimator                                                | $n$  | Bias    | SD     | RMSE   | SE/SD | Cover | Length |
|----------------------------------------------------------|------|---------|--------|--------|-------|-------|--------|
| <i>KRR (smooth truth), <math>\theta_0 = 0.161</math></i> |      |         |        |        |       |       |        |
| Plug-in (sieve)                                          | 1000 | +0.0103 | 0.0362 | 0.0376 | 1.11  | 0.962 | 0.158  |
|                                                          | 2000 | +0.0052 | 0.0259 | 0.0264 | 1.06  | 0.958 | 0.108  |
|                                                          | 4000 | +0.0037 | 0.0185 | 0.0188 | 1.00  | 0.954 | 0.073  |
|                                                          | 8000 | +0.0031 | 0.0133 | 0.0136 | 0.94  | 0.948 | 0.049  |
| SS margins-only                                          | 1000 | -0.0058 | 0.0456 | 0.0459 | 0.88  | 0.882 | 0.158  |
|                                                          | 2000 | -0.0050 | 0.0294 | 0.0298 | 0.94  | 0.918 | 0.108  |
|                                                          | 4000 | -0.0021 | 0.0204 | 0.0205 | 0.91  | 0.932 | 0.073  |
|                                                          | 8000 | -0.0005 | 0.0135 | 0.0135 | 0.93  | 0.926 | 0.049  |
| SS corner-aware                                          | 1000 | +0.0030 | 0.0446 | 0.0447 | 0.90  | 0.894 | 0.158  |
|                                                          | 2000 | -0.0005 | 0.0293 | 0.0293 | 0.94  | 0.932 | 0.108  |
|                                                          | 4000 | +0.0004 | 0.0204 | 0.0204 | 0.91  | 0.934 | 0.073  |
|                                                          | 8000 | +0.0007 | 0.0134 | 0.0134 | 0.93  | 0.930 | 0.049  |
| Cross-fit GBR                                            | 1000 | +0.0666 | 0.0316 | 0.0737 | 1.41  | 0.680 | 0.175  |

<sup>1</sup>Implementation details: the sieve dimension is held *fixed* across  $n$  at  $5 \times 5 = 25$  tensor B-spline basis functions per arm (two knot segments per coordinate), so the design varies  $n$  at fixed  $K$  rather than along  $K_n \asymp n^{d/(2s+1)}$ ; see Section 6.5, item (L4). Bandwidths are  $\varepsilon = \delta \cdot \widehat{\text{sd}}(\hat{\tau})$  with a trimmed standard deviation and  $\delta = 0.08$  (sieve family) or 0.10 (ML family); if fewer than five evaluation points fall in a band it is widened by factors of 1.5 (at most six times) — an adaptivity that is active mainly on the degenerate arm, where the margins are empty. Ties are broken as  $\mathbb{1}\{\hat{\tau}_S \geq 0\}$  and harm as the strict  $\hat{\tau}_Y < 0$ ; the difference is measure-zero under Assumption 1.5.



| Estimator                                                       | $n$  | Bias    | SD     | RMSE   | SE/SD | Cover | Length |
|-----------------------------------------------------------------|------|---------|--------|--------|-------|-------|--------|
| + Riesz corr.                                                   | 2000 | +0.0444 | 0.0238 | 0.0504 | 1.06  | 0.578 | 0.099  |
|                                                                 | 4000 | +0.0239 | 0.0189 | 0.0304 | 0.87  | 0.666 | 0.064  |
|                                                                 | 8000 | +0.0088 | 0.0142 | 0.0167 | 0.82  | 0.820 | 0.046  |
|                                                                 | 1000 | +0.0562 | 0.0545 | 0.0783 | 0.82  | 0.704 | 0.175  |
|                                                                 | 2000 | +0.0260 | 0.0268 | 0.0373 | 0.94  | 0.784 | 0.099  |
|                                                                 | 4000 | +0.0145 | 0.0196 | 0.0244 | 0.83  | 0.808 | 0.064  |
|                                                                 | 8000 | +0.0045 | 0.0141 | 0.0148 | 0.83  | 0.882 | 0.046  |
|                                                                 | 1000 | +0.0488 | 0.0653 | 0.0815 | 0.68  | 0.706 | 0.175  |
| + quad. (DD)                                                    | 2000 | +0.0061 | 0.0361 | 0.0366 | 0.70  | 0.830 | 0.099  |
|                                                                 | 4000 | -0.0046 | 0.0273 | 0.0277 | 0.60  | 0.752 | 0.064  |
|                                                                 | 8000 | -0.0109 | 0.0195 | 0.0223 | 0.60  | 0.690 | 0.046  |
|                                                                 |      |         |        |        |       |       |        |
| <i>WGAN scalar (rough truth), <math>\theta_0 = 0.194</math></i> |      |         |        |        |       |       |        |
| Plug-in (sieve)                                                 | 1000 | +0.0645 | 0.1008 | 0.1195 | 1.34  | 0.956 | 0.527  |
|                                                                 | 2000 | +0.0592 | 0.0884 | 0.1063 | 1.15  | 0.928 | 0.398  |
|                                                                 | 4000 | +0.0470 | 0.0618 | 0.0776 | 1.23  | 0.936 | 0.298  |
|                                                                 | 8000 | +0.0317 | 0.0453 | 0.0552 | 1.13  | 0.952 | 0.200  |
| SS margins-only                                                 | 1000 | +0.0435 | 0.1396 | 0.1461 | 0.96  | 0.886 | 0.527  |
|                                                                 | 2000 | +0.0373 | 0.1236 | 0.1290 | 0.82  | 0.866 | 0.398  |
|                                                                 | 4000 | +0.0275 | 0.0782 | 0.0828 | 0.97  | 0.918 | 0.298  |
|                                                                 | 8000 | +0.0152 | 0.0537 | 0.0557 | 0.95  | 0.940 | 0.200  |
| SS corner-aware                                                 | 1000 | +0.0655 | 0.1370 | 0.1518 | 0.98  | 0.876 | 0.527  |
|                                                                 | 2000 | +0.0519 | 0.1213 | 0.1318 | 0.84  | 0.868 | 0.398  |
|                                                                 | 4000 | +0.0363 | 0.0780 | 0.0859 | 0.98  | 0.910 | 0.298  |
|                                                                 | 8000 | +0.0206 | 0.0535 | 0.0573 | 0.96  | 0.930 | 0.200  |
| Cross-fit GBR                                                   | 1000 | +0.0452 | 0.0434 | 0.0626 | 2.17  | 0.960 | 0.369  |
|                                                                 | 2000 | +0.0463 | 0.0359 | 0.0585 | 1.18  | 0.840 | 0.166  |
|                                                                 | 4000 | +0.0469 | 0.0318 | 0.0567 | 1.03  | 0.726 | 0.128  |
|                                                                 | 8000 | +0.0415 | 0.0276 | 0.0499 | 0.97  | 0.656 | 0.105  |
| + Riesz corr.                                                   | 1000 | +0.0492 | 0.1242 | 0.1335 | 0.76  | 0.910 | 0.369  |
|                                                                 | 2000 | +0.0411 | 0.0417 | 0.0585 | 1.02  | 0.840 | 0.166  |
|                                                                 | 4000 | +0.0456 | 0.0335 | 0.0566 | 0.98  | 0.718 | 0.128  |
|                                                                 | 8000 | +0.0416 | 0.0280 | 0.0501 | 0.95  | 0.632 | 0.105  |
| + quad. (DD)                                                    | 1000 | +0.0609 | 0.1413 | 0.1537 | 0.67  | 0.772 | 0.369  |
|                                                                 | 2000 | +0.0446 | 0.0703 | 0.0832 | 0.60  | 0.656 | 0.166  |
|                                                                 | 4000 | +0.0336 | 0.0570 | 0.0661 | 0.57  | 0.660 | 0.128  |
|                                                                 | 8000 | +0.0209 | 0.0446 | 0.0492 | 0.60  | 0.702 | 0.105  |

| Estimator                                                              | $n$  | Bias    | SD     | RMSE   | SE/SD | Cover | Length |
|------------------------------------------------------------------------|------|---------|--------|--------|-------|-------|--------|
| <i>WGAN full-surrogate (degenerate), <math>\theta_0 = 0.000</math></i> |      |         |        |        |       |       |        |
| Plug-in (sieve)                                                        | 1000 | +0.0604 | 0.0432 | 0.0742 | 1.60  | 0.922 | 0.271  |
|                                                                        | 2000 | +0.0329 | 0.0243 | 0.0409 | 1.52  | 0.940 | 0.144  |
|                                                                        | 4000 | +0.0173 | 0.0113 | 0.0207 | 1.73  | 0.936 | 0.076  |
|                                                                        | 8000 | +0.0085 | 0.0061 | 0.0105 | 1.73  | 0.930 | 0.042  |
| SS margins-only                                                        | 1000 | +0.0140 | 0.0669 | 0.0683 | 1.03  | 0.866 | 0.271  |
|                                                                        | 2000 | +0.0019 | 0.0362 | 0.0362 | 1.02  | 0.894 | 0.144  |
|                                                                        | 4000 | +0.0011 | 0.0188 | 0.0188 | 1.04  | 0.886 | 0.076  |
|                                                                        | 8000 | -0.0002 | 0.0092 | 0.0092 | 1.15  | 0.874 | 0.042  |
| SS corner-aware                                                        | 1000 | +0.0314 | 0.0643 | 0.0715 | 1.07  | 0.888 | 0.271  |
|                                                                        | 2000 | +0.0101 | 0.0347 | 0.0361 | 1.06  | 0.906 | 0.144  |
|                                                                        | 4000 | +0.0050 | 0.0183 | 0.0190 | 1.06  | 0.898 | 0.076  |
|                                                                        | 8000 | +0.0018 | 0.0091 | 0.0093 | 1.16  | 0.898 | 0.042  |
| Cross-fit GBR                                                          | 1000 | +0.1742 | 0.0357 | 0.1778 | 1.94  | 0.126 | 0.271  |
|                                                                        | 2000 | +0.1608 | 0.0300 | 0.1636 | 1.17  | 0.008 | 0.138  |
|                                                                        | 4000 | +0.1341 | 0.0231 | 0.1360 | 1.02  | 0.000 | 0.092  |
|                                                                        | 8000 | +0.0962 | 0.0175 | 0.0977 | 0.92  | 0.000 | 0.063  |
| + Riesz corr.                                                          | 1000 | +0.1732 | 0.1099 | 0.2051 | 0.63  | 0.146 | 0.271  |
|                                                                        | 2000 | +0.1598 | 0.0361 | 0.1638 | 0.97  | 0.026 | 0.138  |
|                                                                        | 4000 | +0.1342 | 0.0241 | 0.1363 | 0.98  | 0.000 | 0.092  |
|                                                                        | 8000 | +0.0966 | 0.0176 | 0.0982 | 0.91  | 0.000 | 0.063  |
| + quad. (DD)                                                           | 1000 | +0.1184 | 0.1203 | 0.1687 | 0.57  | 0.422 | 0.271  |
|                                                                        | 2000 | +0.0858 | 0.0523 | 0.1005 | 0.67  | 0.376 | 0.138  |
|                                                                        | 4000 | +0.0461 | 0.0353 | 0.0580 | 0.67  | 0.508 | 0.092  |
|                                                                        | 8000 | +0.0071 | 0.0200 | 0.0212 | 0.80  | 0.866 | 0.063  |
| <i>Affine (exact truth), <math>\theta_0 = 0.209</math></i>             |      |         |        |        |       |       |        |
| Plug-in (sieve)                                                        | 1000 | +0.0042 | 0.0546 | 0.0547 | 0.97  | 0.916 | 0.208  |
|                                                                        | 2000 | +0.0017 | 0.0339 | 0.0339 | 1.07  | 0.958 | 0.142  |
|                                                                        | 4000 | +0.0012 | 0.0249 | 0.0249 | 1.00  | 0.940 | 0.098  |
|                                                                        | 8000 | -0.0013 | 0.0171 | 0.0172 | 1.00  | 0.954 | 0.067  |
| SS margins-only                                                        | 1000 | +0.0023 | 0.0680 | 0.0680 | 0.78  | 0.858 | 0.208  |
|                                                                        | 2000 | +0.0007 | 0.0392 | 0.0391 | 0.93  | 0.934 | 0.142  |
|                                                                        | 4000 | +0.0003 | 0.0267 | 0.0267 | 0.93  | 0.934 | 0.098  |
|                                                                        | 8000 | -0.0022 | 0.0178 | 0.0179 | 0.96  | 0.930 | 0.067  |
| SS corner-aware                                                        | 1000 | +0.0025 | 0.0681 | 0.0680 | 0.78  | 0.860 | 0.208  |
|                                                                        | 2000 | +0.0005 | 0.0393 | 0.0393 | 0.92  | 0.928 | 0.142  |

| Estimator     | $n$  | Bias    | SD     | RMSE   | SE/SD | Cover | Length |
|---------------|------|---------|--------|--------|-------|-------|--------|
| Cross-fit GBR | 4000 | +0.0003 | 0.0268 | 0.0268 | 0.93  | 0.932 | 0.098  |
|               | 8000 | -0.0022 | 0.0178 | 0.0179 | 0.96  | 0.932 | 0.067  |
|               | 1000 | +0.0277 | 0.0378 | 0.0468 | 1.00  | 0.862 | 0.148  |
|               | 2000 | +0.0161 | 0.0284 | 0.0326 | 0.94  | 0.878 | 0.105  |
|               | 4000 | +0.0097 | 0.0238 | 0.0257 | 0.85  | 0.886 | 0.079  |
|               | 8000 | +0.0036 | 0.0174 | 0.0178 | 0.86  | 0.900 | 0.059  |
| + Riesz corr. | 1000 | +0.0255 | 0.0416 | 0.0487 | 0.91  | 0.870 | 0.148  |
|               | 2000 | +0.0140 | 0.0291 | 0.0322 | 0.92  | 0.896 | 0.105  |
|               | 4000 | +0.0077 | 0.0241 | 0.0253 | 0.84  | 0.876 | 0.079  |
|               | 8000 | +0.0022 | 0.0182 | 0.0184 | 0.83  | 0.886 | 0.059  |
|               | 1000 | +0.0032 | 0.0604 | 0.0604 | 0.63  | 0.764 | 0.148  |
|               | 2000 | -0.0024 | 0.0441 | 0.0441 | 0.60  | 0.778 | 0.105  |
| + quad. (DD)  | 4000 | -0.0014 | 0.0325 | 0.0325 | 0.62  | 0.780 | 0.079  |
|               | 8000 | -0.0036 | 0.0241 | 0.0243 | 0.63  | 0.778 | 0.059  |



Figure 1: Empirical 95% coverage by sample size, DGP (columns), and first-stage family (rows). Dashed line: nominal 0.95.

## 6.2 Corner anatomy

Because the KRR oracle’s contrast surfaces are known in closed form, the corner set and its geometry are computable: the zero curves of  $\tau_S$  and  $\tau_Y$  cross at ten points, with gradient norms  $\|\nabla\tau_S\| \in [10.3, 135.8]$ ,  $\|\nabla\tau_Y\| \in [13.9, 56.4]$ ,  $|\cos\omega| \leq 0.90$  (transversality holds:

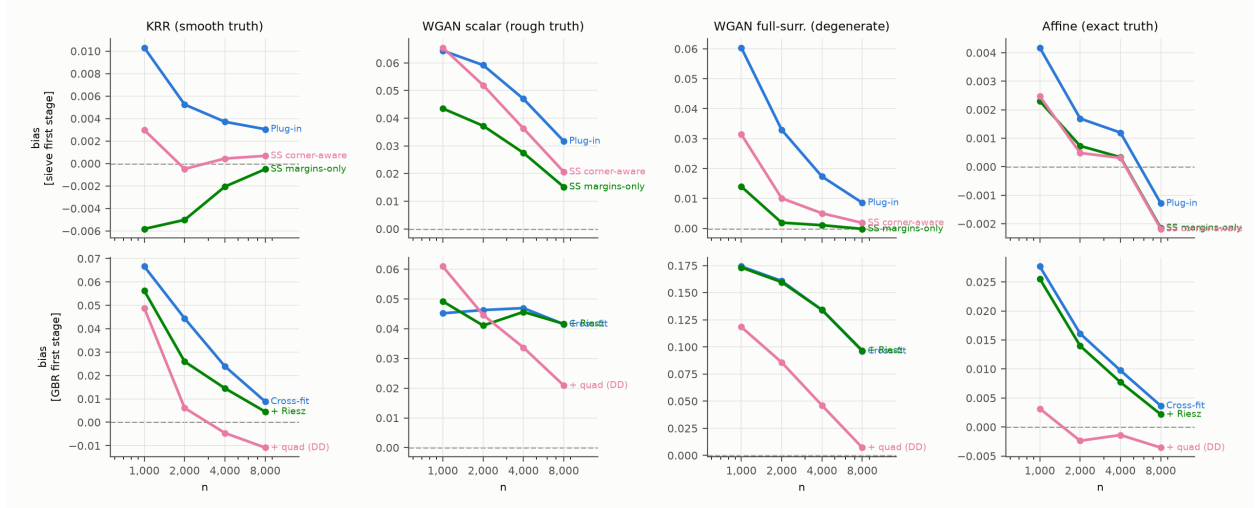


Figure 2: Monte Carlo bias by sample size, DGP, and estimator. Dashed line: zero.

Table 3: Decomposition of the machine-learning undercoverage on the KRR design ( $n = 4000$ , 300 replications). “Plug-in bias” is the uncorrected cross-fit bias; “bias”, “SD”, “SE/SD”, and “coverage” refer to the Riesz-corrected estimator, studentized by (15). The tensor-sieve plug-in is the reference. Varying folds, learner, and representer basis one at a time isolates each mechanism of Remark 5.5. MC standard error of a coverage figure is about 0.013 at 300 replications.

| Configuration                 | Bias    | Plug-in bias | SD     | SE/SD | Coverage |
|-------------------------------|---------|--------------|--------|-------|----------|
| tensor-sieve plug-in          | +0.0034 | +0.0034      | 0.0178 | 1.03  | 0.943    |
| gbr K=2 riesz=sieve           | +0.0216 | +0.0317      | 0.0186 | 0.96  | 0.760    |
| gbr K=5 riesz=sieve           | +0.0106 | +0.0160      | 0.0191 | 0.94  | 0.903    |
| gbr K=10 riesz=sieve          | +0.0079 | +0.0127      | 0.0196 | 0.92  | 0.913    |
| rf K=5 riesz=sieve            | +0.0119 | +0.0453      | 0.0195 | 1.01  | 0.907    |
| krr K=5 riesz=sieve           | +0.0038 | +0.0039      | 0.0184 | 0.90  | 0.917    |
| gbr K=5 riesz=rf              | +0.0078 | +0.0160      | 0.0203 | 0.96  | 0.937    |
| krr K=5 riesz=rf              | +0.0047 | +0.0039      | 0.0181 | 0.98  | 0.937    |
| gbr(high-cap) K=5 riesz=sieve | +0.0439 | +0.0461      | 0.0131 | 1.00  | 0.077    |
| gbr K=5 no correction         | +0.0160 | +0.0160      | 0.0189 | 0.94  | 0.857    |

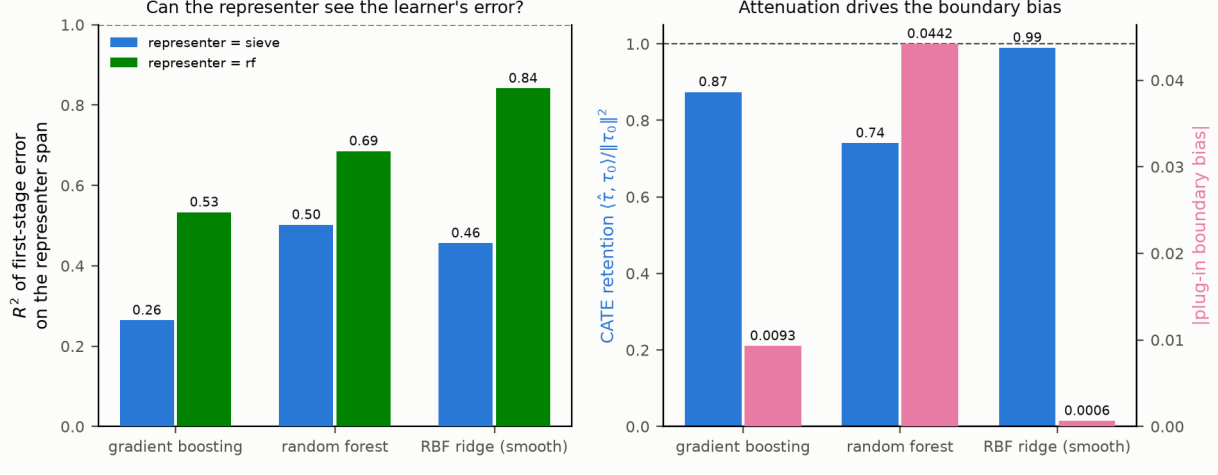


Figure 3: Why boosting-based DML undercovers. *Left*: the fraction ( $R^2$ ) of the first-stage error that the sieve-Riesz representer can span in global  $L^2$ , measured against the oracle arm means; a boosted-tree error is largely invisible to a smooth representer, a random-feature ridge’s error is not. *Right*: tree learners attenuate the CATE (no-intercept projection coefficient of fitted on true,  $\hat{\tau}'\tau/\tau'\tau < 1$ ), which displaces the decision boundary and biases the plug-in; the smooth learner barely attenuates and is nearly unbiased. Both panels average five simulated datasets and are indicative rather than precise.

Table 4: Corner anatomy on the KRR-calibrated oracle (300 reps, fixed sieve dimension). “Total” is the implied plug-in corner bias  $\sum_{x^*} \mathbb{E}[\Delta_g \Delta_h] w / (\|\nabla g_0\| \|\nabla h_0\| |\sin \omega|)$  in  $\theta$  units; “SS estimate” is the mean corner cross correction  $\widehat{\text{corner}}$  of (35), which by split-sample independence targets the *stochastic* ( $K/n$ ) part of the diagonal; “remainder” is their difference, the bias-product ( $K^{-2s/d}$ ) part. Sign convention: the anatomy script measures  $\mathbb{E}[\Delta \hat{\tau}_S \Delta \hat{\tau}_Y]$  (positive here, as  $\rho_{SY} > 0$ ); the table reports the corner term in  $\theta$  units, where the harm-share convention  $h_0 = -\tau_Y$  flips its sign.  $\overline{\text{corr}}$  averages  $\text{Corr}(\Delta_g(x^*), \Delta_h(x^*))$  over the ten corners. The column implements the corner *cross* term  $\mathcal{B}_C^\times$  of Proposition 4.10 — the first term of the bracket in (27) — which by (37) is exactly the quantity that margins-only debiasing leaves behind and that  $\widehat{\text{corner}}$  targets, so the comparison across columns is like-for-like. It is *not* the corner’s full contribution to the plug-in bias, which also contains the two  $\cos \omega$ -weighted diagonals of (27) (non-negligible here, since  $|\cos \omega|$  reaches 0.90) but which margins-only debiasing already removes. The column also uses an unnormalized covariate density (Section 6.5, item (L2)).

| $n$  | Total corner bias | SS estimate | Remainder | $\overline{\text{corr}}(\Delta_g, \Delta_h)$ | $\rho_{SY}$ |
|------|-------------------|-------------|-----------|----------------------------------------------|-------------|
| 2000 | −0.0080           | −0.0044     | −0.0036   | 0.50                                         | 0.524       |
| 4000 | −0.0048           | −0.0022     | −0.0025   | 0.46                                         | 0.524       |
| 8000 | −0.0038           | −0.0012     | −0.0026   | 0.45                                         | 0.524       |

$|\sin \omega| \geq 0.45$ ), and within-unit residual correlation  $\rho_{SY} = 0.524$ . Table 4 traces the corner mechanism of Section 3.2 across 300 replications per sample size (sieve first stage, fixed  $K$ ).

Three predictions of Section 3.2 are consistent with the table. (i) The pointwise correlation of the two surface errors at the corner tracks the within-unit residual correlation (0.45–0.50 against  $\rho_{SY} = 0.524$ ), which is the mechanism behind the  $V_{gh}$  term in (27). (ii) The stochastic part of the corner diagonal halves as  $n$  doubles at fixed  $K$  ( $-0.0044 \rightarrow -0.0022 \rightarrow -0.0012$ ) — the  $K/n$  scaling — and is tracked by the SS corner correction, which by construction cancels bias parts across the two independent halves. (iii) The remainder is roughly constant in  $n$  at fixed  $K$  ( $-0.0036$  at  $n = 2000$ , then  $\approx -0.0025$ ), as befits the  $K^{-2s/d}$  bias product, which is controlled by undersmoothing  $K$  rather than by debiasing — and which, per Proposition 4.10, the SS statistic is not supposed to capture. The measured corner bias (0.4–0.8 percentage points of a  $\theta_0 = 0.161$  target) is material relative to the plug-in standard error, and its sign matches the Monte Carlo finding that the corner-aware and margins-only SS estimates differ by  $+0.004$  to  $+0.005$  at  $n = 2000$ . This validates the cross-corner component of (27) — the component that Proposition 4.10 says is at stake in the margins-only comparison. The two  $\cos \omega$  components of (27) are not tested here, because both debiasing variants remove them; testing them would require comparing the *plug-in* against a corner-corrected plug-in.

### 6.3 A fixed- $K$ diagnostic for the cross-half off-diagonal

Table 2 contains, without any further computation, a diagnostic for the  $n$ -dependence of the off-diagonal bound derived in Appendix B, §B.2. Because the design holds  $K$  fixed, it cannot test the exponent on  $K$  that enters Theorem 4.6(ii)’s window (39).

The SS estimator’s error differs from the plug-in’s by removing the stochastic diagonal and adding the mean-zero cross-half term  $\frac{1}{2}D^2\Theta[u^A, u^B]$ . By §B.2 that term has variance  $\asymp n^{-2}K_n^{(d+3)/d}$ , against a score variance  $\sigma_n^2/n \asymp K_n^{1/d}/n$ . Since the two estimators use the same observations and the same standard error, this predicts

$$\frac{\text{SD}(\hat{\Theta}_{SS})}{\text{SD}(\hat{\Theta})} = \sqrt{1 + c \frac{K_n^{(d+2)/d}}{n}} (1 + o(1)), \quad c \text{ constant in } n, \quad (52)$$

which at the design’s fixed  $K = 25$ ,  $d = 2$  is  $\sqrt{1 + 625c/n}$ . Writing  $R_n$  for the observed ratio  $\text{SD}_{SS}/\text{SD}_{\text{plug-in}}$ , the implied  $c = n(R_n^2 - 1)/K^2$  should be flat in  $n$ :

| Design                                    | $n = 1000$ | 2000 | 4000  | 8000  | spread |
|-------------------------------------------|------------|------|-------|-------|--------|
| Affine (exact truth; assumptions hold)    | 0.89       | 1.10 | 1.01  | 1.07  | 1.2×   |
| KRR (smooth)                              | 0.83       | 0.90 | 1.38  | 0.19  | 7.2×   |
| WGAN scalar ( $s \approx 1$ ; violates)   | 1.36       | 2.83 | 3.80  | 5.05  | 3.7×   |
| WGAN degenerate (empty margins; violates) | 1.94       | 3.33 | 10.39 | 15.69 | 8.1×   |

On the affine design — the only one with an exactly closed-form truth and all maintained assumptions satisfied —  $c$  is flat within a factor 1.2 and centered on 1, across a factor of eight in  $n$ , with no fitted parameter beyond the  $\asymp$  constant itself. On KRR the first three cells agree; the  $n = 8000$  cell is uninformative, since there  $(\text{SD}_{SS}/\text{SD}_{\text{plug-in}})^2 - 1 = 0.016$  is of the order of the Monte Carlo error of a ratio of two standard deviations estimated from 500 replications (we have not computed that error, which is smaller than an independent-samples calculation would give, since the two use *common* replications). The two designs that violate the maintained assumptions show  $c$  increasing monotonically by factors of 3.7 and 8.1, flatly inconsistent with (52). The test therefore discriminates across these designs rather than merely fitting a constant, but it remains a fixed- $K$  check of the  $1/n$  component only.

Two corollaries. First, this numerically accounts for the SS standard error’s calibration. Because the implementation computes one standard error from the full-sample fit and applies it to both estimators, SE/SD for SS must equal (SE/SD for the plug-in) divided by the SD ratio, as an identity; and indeed the reported values agree to within 0.003–0.008 on all four designs and all four sample sizes. The SS SE/SD deficit of 0.78–0.96 is thus consistent with the variance of the quadratic correction, which the first-order standard error omits by construction — not a failure of the two-band variance formula, and not the corner.

Second, the finite-sample formula mirrors a condition in Theorem 4.6(ii). The inflation factor  $K_n^{(d+2)/d}/n$  vanishes precisely when  $a < d/(d+2)$ , which is the *upper edge of the window* (39). This algebra explains why that edge appears, but the fixed- $K$  experiment does not test it. At the design’s fixed  $K = 25$  the factor is 0.63, 0.31, 0.16, 0.08 — large at  $n = 1000$ , still 8% at  $n = 8000$  — and a growing- $K_n$  implementation inside (39) is exactly what would drive it away. A standard error for  $\hat{\Theta}_{SS}$  that included the correction’s own variance would fix the calibration at fixed  $K$ ; we do not currently have one.

## 6.4 Results

Table 2 and Figures 1–2 report all cells; Section 6.5 states which conclusions they can and cannot support.

(1) *Smooth truth (KRR): the plug-in is already good; corner-aware SS removes what bias*



there is. The plug-in covers at 0.948–0.962 at all  $n$  with a small decaying bias ( $+0.010 \rightarrow +0.003$ ) and a well-calibrated standard error (SE/SD 0.94–1.11, tightening toward one with  $n$ ). SS debiasing removes the bias essentially completely ( $+0.003 \rightarrow +0.0007$ ; a 3.5-fold bias reduction at  $n = 1000$  and a tenfold reduction at  $n = 2000$ ), at a variance cost that is 23% of the standard deviation at  $n = 1000$  and falls to 10% at  $n = 4000$  and about 1% at  $n = 8000$  — Section 6.3 shows this is the cross-half term  $\frac{1}{2}D^2\Theta[u^A, u^B]$ , decaying as  $1/n$  predicts at the fixed  $K$  used here, and that it accounts for the whole of the SS coverage shortfall. The *margins-only* correction overshoots to negative bias ( $-0.006$  to  $-0.0005$ ) at every  $n$ : it removes the (positive) margin diagonals — and, per (37), the two  $\cos\omega$  corner diagonals as well — but not the (negative) corner *cross* term. The gap between the corner-aware and margins-only estimates equals the estimated corner correction ( $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$ ) and matches the anatomy of Table 4 to the fourth decimal ( $+0.0045$  at  $n = 2000$  against minus the anatomy’s SS corner estimate of  $-0.0044$ ). The raw corner cross form  $q_{\text{full}} - q_g - q_h$  (eight times the corner correction) halves as  $n$  doubles ( $-0.070 \rightarrow -0.036 \rightarrow -0.020 \rightarrow -0.0095$ ) at fixed  $K$ , consistent with the  $1/n$  component of (27); it does not test the exponent on  $K$ . Coverage of the SS intervals is 0.894–0.934 on this design: an improvement in centering bought at a 2–5 point coverage cost relative to the plug-in, which we do not describe as honest coverage.

(2) *Rough truth (scalar WGAN,  $s \approx 1$ ): debiasing accelerates the bias decay.* This design violates the smoothness hypothesis of every theorem above ( $s \approx 1 < (d+1)/2 = 1.5$ ), so it is a robustness probe. The plug-in bias is large and decays at only  $\approx n^{-0.34}$  ( $+0.064 \rightarrow +0.032$ ), and its near-nominal coverage rests on a conservative SE (SE/SD 1.1–1.3). SS debiasing accelerates the empirical decay to  $\approx n^{-0.55}$  ( $+0.066 \rightarrow +0.021$ ) with SE/SD 0.83–0.98 and coverage  $0.87 \rightarrow 0.93$ . Because  $K$  is fixed across  $n$  here, these exponents describe the finite-sample behavior of the estimators at  $K = 25$ ; they are not estimates of the asymptotic rate, which would require  $K_n$  to grow (Section 6.5, item (L4)). On this rough truth the margins-only variant is slightly better centered than the corner-aware one — with  $s \approx 1$  the corner correction is itself noisily estimated — a useful caution: corner-awareness is a *bias-decomposition* necessity, but its sampled version adds noise when the surfaces are rough.

(3) *Machine-learning first stages: the projected Riesz correction works where the learner is adequate; the quadratic correction is not a free lunch.* On the smooth truth the GBR bias falls monotonically along the ladder cross-fit  $\rightarrow +\text{Riesz} \rightarrow +\text{quadratic}$  at every  $n$  (at  $n = 4000$ :  $+0.024 \rightarrow +0.015 \rightarrow -0.005$ ), consistent with the two corrections targeting distinct error terms. But the composition has two finite-sample costs: the SS-on-GBR quadratic correction inflates the dispersion (by a factor of two at  $n = 1000$ , 1.4 thereafter), and the first-order SE does not account for it (SE/SD  $\approx 0.6$ ), so DD intervals undercover (0.69–0.83); and

at  $n = 8000$  on KRR the quadratic correction overshoots ( $-0.011$ ). *This ladder must now be read with item (L7) of Section 6.5 in hand*: the DD column implements the superseded construction (45), whose second-order bias  $-\frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$  predicts exactly this over-correction and this dispersion inflation, on smooth and rough designs alike. v2 attributed the overshoot to Remark 4.4 — (32) being a fourth-order finite difference of a step function for tree first stages — which is a real effect but cannot explain the same pattern on the affine design. On the rough truth GBR is bias-limited ( $+0.045$  nearly flat in  $n$ ; coverage falls to 0.64): no first-order correction can rescue a nuisance that misses the kinks. The degenerate arm is the extreme case: GBR manufactures a spurious harm share of  $+0.17$  (coverage 0), and there the quadratic correction *does* rescue it ( $+0.007$ , coverage 0.87 at  $n = 8000$ ).

(4) *The two-band SE is well calibrated for the sieve family.* The variance estimator (15) includes the empirical-measure term  $\hat{\zeta}^2/n = \hat{\theta}(1 - \hat{\theta})/n$  — the sampling error of the average given the surfaces, the analogue of the  $\text{Var}([h_0]_+)$  term the single-threshold unknown-density welfare theorem adds — which is 6–12% of the SE at these sample sizes and, if omitted, produces a spurious SE/SD of 0.87–0.95. With it included, SE/SD is 0.90–1.11 for plug-in and SS on the KRR and affine designs, and the studentized intervals reach 0.93–0.95 by  $n = 8000$ ; on the degenerate arm ( $\theta_0 = 0$ , empty margins) the sieve family degrades gracefully.

(5) *Machine-learning first stages: why the sieve-DML interval undercovers, and when it does not.* The cross-fitted GBR estimators cover only 0.58–0.88 on the smooth design, in sharp contrast to the sieve plug-in on the *same samples* — so the shortfall is a property of the first stage, not the DGP or the variance formula. The diagnosis is Remark 5.5. Measured against the oracle arm means, the  $R^2$  of the boosted first-stage error on the tensor B-spline representer span is 0.21–0.34 across the four arm regressions (mean 0.26); the projected correction removes 31% of the plug-in’s boundary bias. The direct variance test reported in Remark 5.5 shows that the remaining 69% is *bias*, not an independent added variance: the leading explanation is that the correction cannot see the part of the learner’s error that lives on the margins, so it leaves the boundary displaced. Three levers move coverage: (i) *folds* —  $K = 2 \rightarrow 5 \rightarrow 10$  lifts GBR coverage  $0.76 \rightarrow 0.90 \rightarrow 0.91$  by shrinking the own-observation bias trained on  $(K - 1)/K$  of the sample; (ii) *representer basis* — matching a 400-feature random-feature representer to the boosted learner raises the projected  $R^2$  from 0.26 to 0.53 and coverage from 0.90 to 0.94; (iii) *learner smoothness* — a random-feature (RBF) ridge, whose error the representer *can* span ( $R^2 = 0.84$ ), attenuates the CATE by only about 1% (against boosting’s 13%) and has a plug-in bias of  $+0.0006$  (against  $+0.016$ ), so there is almost nothing left to correct and, with a matched representer, it too reaches 0.94. Higher-capacity

boosting is counterproductive — it sharpens the axis-aligned staircase, driving coverage toward zero. The pattern reproduces the parent draft’s own single-threshold Table 7. The residual 0.01–0.02 shortfall of even the best machine-learning cell below the sieve plug-in’s coverage is within two Monte Carlo standard errors and should not be over-interpreted.

## 6.5 What the current Monte Carlo does and does not establish

The tables above are reproduced from v1 without change. Auditing the code against the theory turned up seven implementation gaps. None of them affects Theorem 3.1 or the order accounting of Section 3.2, which are analytic; all of them bear on how much weight the simulation evidence can carry, and all are scheduled for repair before the next revision. We list them explicitly rather than leave them implicit.

### (L1) The KRR “oracle truth” is displaced by uncentered residual resampling.

The sampler draws outcome noise by resampling the fitted per-arm KRR residual pools, which are not centered (ridge shrinkage leaves arm-specific residual means). The generated conditional means therefore differ from the nominal KRR surfaces by a constant per arm, shifting the generated CATEs by  $\Delta\tau_S = +0.126$  and  $\Delta\tau_Y = +0.182$  (the latter includes the short→long coupling). The reported truth  $\theta_0$  moves from 0.16115 to 0.15998 — immaterial for the bias columns — but the ten corner locations, gradients, angles, and first-stage errors used in Table 4 are all computed against the *wrong* surfaces. Fix: center each arm’s residual pool before resampling, or define the oracle truth to include the shifts.

### (L2) The corner density is unnormalized.

Covariates are drawn from the KDE conditional on  $X \in [-3, 3]^2$  by rejection sampling, but the density accessor returns the untruncated KDE. The KDE mass of the box is 0.807, so the corner integrals in Table 4 should be scaled by  $1/0.807 \approx 1.24$  before any comparison with (27). Together with the omission of the  $\cos\omega$  terms noted in the table caption, the “Total” column is an incomplete implementation of the theory it is meant to check.

### (L3) The KRR design violates Assumption 1.5(d).

The KDE density is positive at the truncation boundary, and a direct scan of the four box edges finds twelve sign crossings of the two contrast surfaces ( $\tau_S$ : three on  $x_1 = -3$ , two on  $x_1 = +3$ , three on  $x_2 = -3$ ;  $\tau_Y$ : two on  $x_1 = -3$ , two on  $x_2 = +3$ ). The margins therefore terminate at the support edge, and by Remark 1.12 the correct

second derivative for this DGP carries additional support-edge  $\mathcal{H}^{d-2}$  terms that (18) omits. Fix: taper  $w$  near  $\partial\mathcal{X}$ , or shrink the box so that all margins close inside it.

**(L4) The main Monte Carlo holds  $K$  fixed.**

Every sample size uses two knot segments per coordinate, i.e.  $K = 25$  basis functions per arm at every  $n$ . The design therefore does not implement  $K_n \asymp n^{d/(2s+1)}$ , and the estimated log-rate slopes reported in Section 6.4(2) are not estimates of the asymptotic rate. At fixed  $K$  the design can reveal the predicted  $1/n$  decline of the corner diagonal, but it cannot test the exponent on  $K$  in the  $K/n$  bound (Table 4).

**(L5) The variance bandwidth does not satisfy Proposition 2.4.**

The implementation sets  $\varepsilon = \delta \cdot \widehat{\text{sd}}(\hat{\tau})$  with  $\delta$  fixed and then widens  $\varepsilon$  by factors of 1.5 until at least five observations enter the band. Since  $\widehat{\text{sd}}(\hat{\tau})$  converges to a positive constant,  $\varepsilon_n \not\rightarrow 0$ , contradicting Proposition 2.4 and Assumption 5.1(b). On the degenerate arm the widening manufactures a nonempty derivative band where the population margin is empty. Fix:  $\varepsilon_n = \delta_n \cdot \widehat{\text{sd}}(\hat{\tau})$  with  $\delta_n \rightarrow 0$  at a rate satisfying  $n\varepsilon_n/K_n \rightarrow \infty$ .

**(L6) The coded Riesz correction is not fold-specific.**

Both the cross-fitted DML estimator and the DD estimator build a single full-sample Gram matrix and band representer from pooled out-of-fold predictions, whereas (42) and Theorem 5.3 require an off-fold representer for each fold. In the implementation, observation  $i$ 's outcome can influence the global band through the model trained on  $i$ 's own fold, which breaks the conditional mean-zero step used in Remark 5.5. Fix: build  $\hat{v}_{-\kappa}^*$  inside the fold loop.

**(L7) The DD rows implement the superseded construction (45).**

Every “+ quad. (DD)” cell of Table 2 evaluates  $\hat{\Theta}_{DD}^{\text{v2}}$ , which by Remark 5.6 carries a second-order bias  $-\frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$  that the corrected estimator of Definition 5.7 does not. These cells therefore say nothing about Theorem 5.8. They are, however, *consistent* with the diagnosis: since the uncorrected plug-in bias is positive on every design, the corresponding  $D^2\Theta[e, e]$  diagonal has the same sign, and the extra minus coefficient reverses it, predicting an over-correction into negative territory, which is what the smooth designs show ( $-0.0046$  and  $-0.0109$  at  $n = 4000, 8000$  on KRR;  $-0.0024$  to  $-0.0036$  on Affine) together with the dispersion inflation ( $\text{SE}/\text{SD} \approx 0.6$  throughout). v2 attributed this over-correction to the non-smoothness of tree first stages (Remark 4.4); the structural defect is the better explanation, because it is present for the smooth (KRR, affine) designs as well, where the tree explanation does not apply. Fix: implement Definition 5.7 with three folds.

Taken together: Theorem 3.1 is derived analytically and checked numerically (Appendix C); the  $1/n$  component and the sign and magnitude of the corner correction are consistent with Tables 4 and 2 at  $d = 2$  and fixed  $K$ , but the  $K$  exponent is not tested. These readings are subject to (L1)–(L3). The cross-half off-diagonal order  $n^{-1}K_n^{(d+3)/(2d)}$  of Appendix B, §B.2 — which sets the upper edge of the SS window (39) — has its fixed- $K$   $1/n$  implication supported by Section 6.3 on the designs that satisfy the assumptions; its  $K_n$  exponent remains untested. The asymptotic rate claims of Theorems 2.2, 4.6, 5.3 and 5.8 are *not* tested by the present design, because  $K$  does not grow with  $n$ ; the DD cells test a construction we have since shown to be invalid, (L7); and the mechanism behind the machine-learning undercoverage is identified as residual boundary bias rather than unmodelled variance, by the direct variance test in Remark 5.5.

*Practical guidance.* We state it narrowly, because Section 6.5 limits what the experiments support and Section 5 limits what the theory establishes.

For sieve first stages, the corner-aware SS estimator (32) with the two-band SE is the recommendation: it is tuning-free and removes the second-order stochastic diagonal including the corner. In our designs its centering is clearly better than the plug-in’s and its coverage is 2–5 points worse (0.894–0.934 on KRR), so it buys bias reduction at a coverage cost; we would not call its intervals honest. Undersmoothing relative to  $K_n \asymp n^{d/(2s+1)}$  is required for the interval to be centered at all — window (39) — and the fixed- $K$  Monte Carlo does not exercise this.

For a machine-learning first stage — needed when  $d$  is too large for a tensor sieve — prefer a *smooth* learner and *match the representer basis to the learner’s own feature map*, so the projected correction is near-exact; avoid tree ensembles, whose piecewise-constant error is largest precisely on the margins the representer must span; and use at least five folds for the nuisance cross-fit. (This concerns the number of *nuisance* folds and does not conflict with the two-half construction of  $\hat{\Theta}_{SS}$ : there, the two halves furnish the independent replicates that the polarization identity (31) requires, and  $K = 2$  is part of the definition of the quadratic correction, not a cross-fitting choice. The corrected DD estimator of Definition 5.7 needs a third fold on top of that.) When a tree first stage is unavoidable, pair it with a full-refit bootstrap rather than the analytic SE.

We do *not* currently recommend the doubly debiased estimator in applied work. The construction evaluated in Table 2 is invalid (Remark 5.6); the corrected construction of Definition 5.7 has not been implemented or simulated; and even for it, Theorem 5.8 is conditional on remainder bounds we have not established and excludes tree first stages outright.

The comparison of the corner-aware and margins-only corrections is a useful specification

diagnostic in its own right: their gap estimates the corner cross term, whose magnitude flags whether the two-threshold structure is quantitatively binding.

## A Derivation of the Second Derivative (Theorem 3.1)

Write  $T_g(t) := \int_{\{g_t=0\} \cap \{h_t \geq 0\}} \varpi_t d\mathcal{H}^{d-1}$  with weight  $\varpi_t := u w / \|\nabla g_t\|$  (we reserve  $\omega(x)$  for the transversality angle throughout), the first term of (10) along  $\gamma_t$ ; the second term is symmetric. Let  $\mathbf{X}(\cdot, t)$  be the normal diffeomorphism carrying  $\mathcal{M} := \{g_0 = 0\}$  to  $\{g_t = 0\}$  with velocity  $\dot{\mathbf{X}} = -(u / \|\nabla g_0\|) n_g$ , and pull the integral back to  $\mathcal{M}$ :

$$T_g(t) = \int_{\mathcal{M}} \mathbb{1}\{h_t(\mathbf{X}(x, t)) \geq 0\} \varpi_t(\mathbf{X}(x, t)) J_{\mathbf{X}}(x, t) d\mathcal{H}^{d-1}(x).$$

Differentiating at  $t = 0$  yields four contributions. (1) *Integrand*:  $\partial_t \varpi_t = -u w (n_g \nabla u) / \|\nabla g_0\|^2$ . (2) *Transport of  $\varpi$  and area*: by the surface-transport formula (Delfour and Zolésio, 2001, Ch. 9),  $\partial_t \{\varpi(\mathbf{X}) J_{\mathbf{X}}\} = -[(u / \|\nabla g_0\|) n_g \cdot \nabla \varpi + \varpi (u / \|\nabla g_0\|) H_g]$ , the moving-level-set terms of Chen and Gao (2026, Lem. PathD–Level) with weight  $\varpi$ . (3) *Direct motion of the cut*: the indicator  $\mathbb{1}\{h_t \geq 0\}$  evaluated on the (fixed,  $t = 0$ ) manifold defines the region  $\{h_0 + tv \geq 0\} \cap \mathcal{M}$ ; by the within-manifold Leibniz rule its derivative is an integral over the relative boundary  $\mathcal{C}$  with normal velocity  $v / \|\nabla_{\mathcal{M}} h_0\|$ , where  $\|\nabla_{\mathcal{M}} h_0\| = \|P_T \nabla h_0\| = \|\nabla h_0\| |\sin \omega|$ , giving  $+\int_{\mathcal{C}} \varpi_0 v / (\|\nabla h_0\| |\sin \omega|) d\mathcal{H}^{d-2}$  — the cross term. (4) *Induced motion of the cut*: the composition  $h_0(\mathbf{X}(x, t)) = h_0(x) - t(u / \|\nabla g_0\|) n_g \cdot \nabla h_0 + o(t) = h_0(x) - t u \|\nabla h_0\| \cos \omega / \|\nabla g_0\| + o(t)$  perturbs the cut function by an effective  $v_{\text{eff}} = -u \|\nabla h_0\| \cos \omega / \|\nabla g_0\|$ ; the same within-manifold Leibniz rule then contributes  $-\int_{\mathcal{C}} \varpi_0 u \cos \omega / (\|\nabla g_0\| |\sin \omega|) d\mathcal{H}^{d-2}$  — the diagonal corner term of (19). Collecting (1)–(4), and adding the symmetric derivative of the  $\mathcal{M}_h$ -term (which contributes the mirror-image margin form, the second copy of the cross term, and the  $v^2$  diagonal corner term), gives (18)–(19). Transversality bounds every corner integrand; twice-differentiability of  $g_0, h_0$  and Assumption 1.4(d) bound the curvature and gradient terms.  $\square$

## B Orders of the Remainder Terms

The margin components of all remainder bounds are adaptations of the single-threshold arguments — Chen and Gao (2026, Thms. 5, 8 and the SS/LOO constructions) for the quadratic terms, and the extended single-threshold draft (sieve-DML theorem and its Assumption 5) for the cross-fitted first-order correction — applied margin by margin with the truncation

indicators carried as bounded weights. This appendix records the two order calculations that are specific to the two-threshold geometry and that v1 got wrong.

## B.1 Why the margin diagonal is $K_n^{1+1/d}/n$

Substituting the stochastic part (21) into  $Q_g$  and writing  $\zeta_{g,i}(x) := \sum_w q_w s_{w,i}(x) \epsilon_{S,i}$ , then separating  $i = j$  from  $i \neq j$ ,

$$Q_g[u_g, u_g] = \underbrace{\frac{1}{n^2} \sum_i Q_g[\zeta_{g,i}, \zeta_{g,i}]}_{T_{\text{diag}}} + \underbrace{\frac{1}{n^2} \sum_{i \neq j} Q_g[\zeta_{g,i}, \zeta_{g,j}]}_{T_{\text{off}}}.$$

The leading integrand of  $Q_g$  carries one gradient, so  $\mathbb{E} Q_g[\zeta_{g,i}, \zeta_{g,i}]$  involves an arm-wise sum of terms

$$\psi(x)' R_w^{-1} \Omega_{SS}^w R_w^{-1} \nabla \psi(x),$$

whose modulus is  $\lesssim K_n^{1+1/d}$  by (22), whence  $\mathbb{E} T_{\text{diag}} = O(n^{-2} \cdot n \cdot K_n^{1+1/d}) = O(K_n^{1+1/d}/n)$ . This is the *bound* stated by Chen and Gao (2026) for the single-threshold value functional, and it is the bound that Table 1 records. The one-sided form is deliberate: the quantity is signed, and Remark 3.4 shows that its surface integral can be an order smaller. The curvature part of  $Q_g$ , which carries no gradient, contributes  $O(K_n/n)$ .

The corner forms in (18)–(19) carry *no* gradient of  $u$  or  $v$ . Their diagonals therefore involve  $\mathcal{V}_{gg,n}(x)$ ,  $\mathcal{V}_{hh,n}(x) \asymp K_n$  and the signed  $\mathcal{V}_{gh,n}(x) = O(K_n)$  from (23), and are  $O(K_n/n)$ : bounded by  $K_n^{-1/d}$  times the margin bound. v1’s assertion that the corner diagonal is “of the same order  $K_n/n$  as the margin diagonals” conflated a bound with an exact order.

Two asymmetries between the two statements should be recorded. (i) The two point-wise corner variance building blocks are  $\asymp K_n/n$ , while the cross-covariance block is only  $O(K_n/n)$  and can vanish. Every *integrated signed* corner coefficient remains only an upper bound unless Assumption 1.7 is imposed. (ii) The margin order is a bound only. In the homoskedastic, constant-propensity special case,  $\mathbb{E}[\zeta_{g,i} \partial_n \zeta_{g,i}]$  is proportional to  $\frac{1}{2} \partial_n \mathcal{V}$  for the corresponding effective leverage function, and  $\mathcal{V}$  oscillates at the mesh scale with amplitude  $\asymp K_n$ , so while  $\|\nabla \mathcal{V}\|_\infty \asymp K_n^{1+1/d}$  pointwise, the surface integral in (29) can cancel. Expanding  $\mathcal{V} = \bar{\mathcal{V}} + \tilde{\mathcal{V}}$  with  $\tilde{\mathcal{V}}$  mesh-periodic and mean zero, and writing  $\tilde{\mathcal{V}}(x) = \sum_{j \neq 0} c_j e^{2\pi i j \kappa^{-1} x}$ , the integral of  $\partial_n \mathcal{V}$  along a margin of slope  $\alpha$  (in  $d = 2$ ) picks up a non-decaying contribution only from pairs  $(j, l)$  with  $j + l\alpha = 0$  exactly, or from near-resonant pairs whose small denominator offsets Fourier decay. Every rational  $\alpha = p/q$  has exact pairs  $(j, l) = (-pr, qr)$ ; a larger  $q$  suppresses their coefficients but does not change that fact. Irrational slopes require a separate Diophantine bound, and phase or endpoint cancellation can remove individual



modes. Appendix C, item 5, is therefore only finite-mesh evidence: it shows much larger normalized integrals for  $\alpha \in \{0, 1\}$  than for the other tested slopes, but it does not establish two-sided asymptotic orders for any row. The upper margin bound can be attained by suitably phased mesh-resonant margins — in particular for  $d = 1$ , where  $\mathcal{M}_g$  is a point and no surface averaging occurs — but need not be attained for a fixed higher-dimensional geometry.

## B.2 Off-diagonal (cross-half) orders

$T_{\text{off}}$  is a degenerate second-order  $U$ -statistic. Stack the two arm bases, their signs, and the outcome covariance matrices into a block-diagonal design with Gram  $G_{\text{arm}} := \text{diag}(R_0, R_1)$ . Every quadratic form in (18) then evaluates to  $a'_i M a_j$  for an arm-stacked coefficient  $a_i$  and a symmetric matrix  $M$  built from the relevant boundary integral. Uniform overlap and bounded conditional covariance give the upper bound

$$\text{Var}(T_{\text{off}}) \lesssim n^{-2} \text{tr}\{(M G_{\text{arm}}^{-1})^2\}.$$

There are only two arms, so stacking changes constants but not exponents. For tensor B-splines with mesh  $\varkappa = K_n^{-1/d}$ , a normalized basis function has height  $\asymp \varkappa^{-d/2}$  on a cell of volume  $\varkappa^d$ , so for a  $p$ -dimensional integration set  $\mathcal{S}$  and  $\ell$  gradient factors,

$$\#\{k : \text{supp}(\psi_k) \cap \mathcal{S} \neq \emptyset\} \asymp \varkappa^{-p}, \quad M_{kk} \asymp \varkappa^{-d} \varkappa^p \varkappa^{-\ell} = \varkappa^{p-d-\ell},$$

and therefore  $\text{tr}(M^2) \lesssim \varkappa^{-p} \varkappa^{2(p-d-\ell)} = \varkappa^{p-2d-2\ell} = K_n^{(2d+2\ell-p)/d}$ . These are upper bounds; matching lower bounds require noncancellation. Substituting:

| Term                   | $(p, \ell)$ | $\text{tr}(M^2)$ | $T_{\text{off}}$          |
|------------------------|-------------|------------------|---------------------------|
| margin, gradient part  | $(d-1, 1)$  | $K_n^{(d+3)/d}$  | $n^{-1} K_n^{(d+3)/(2d)}$ |
| corner                 | $(d-2, 0)$  | $K_n^{(d+2)/d}$  | $n^{-1} K_n^{(d+2)/(2d)}$ |
| margin, curvature part | $(d-1, 0)$  | $K_n^{(d+1)/d}$  | $n^{-1} K_n^{(d+1)/(2d)}$ |

The first line reproduces the  $n^{-1} \sqrt{K_n^{(d+3)/d}}$  bound of [Chen and Gao \(2026\)](#), which is a useful check on the calculus. Two consequences. First, all three are  $O(r_n^*)$  at  $K_n \asymp n^{d/(2s+1)}$  under  $s \geq (d+1)/2$  (the corner needs only  $s \geq d/2$ ), so the off-diagonals never bind before the terms in Table 1's third block do. Second, and correcting v1: it is *not* true in general that a lower-dimensional integration set gives a smaller cross-half term. Reducing  $p$  by one *raises*  $\text{tr}(M^2)$  by  $K_n^{1/d}$ , because there is less averaging, not more. Comparing like with like, the

codimension-two corner has a *larger* generic off-diagonal than a hypersurface with the same integrand. The corner off-diagonal ends up smaller than the leading margin off-diagonal only because the margin form carries a gradient factor ( $\ell = 1$ ) that costs  $K_n^{2/d}$  in  $\text{tr}(M^2)$ , which more than offsets the corner's  $K_n^{1/d}$  dimension gain; it remains *larger* than the margin's curvature part.

### B.3 The composite split-sample score

Write  $\Psi(\gamma^A, \gamma^B) := 2\Theta(\frac{\gamma^A + \gamma^B}{2}) - \frac{1}{2}\Theta(\gamma^A) - \frac{1}{2}\Theta(\gamma^B)$ . Differentiating in the first argument,  $D_A \Psi = 2 \cdot \frac{1}{2}D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\gamma^A) = D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\gamma^A)$ , which is (47). Evaluating along  $e_A$  and expanding both derivatives at  $\gamma_0$ ,

$$\begin{aligned} D_A \Psi[e_A] &= D\Theta_0[e_A] + D^2\Theta[\bar{e}, e_A] - \frac{1}{2}D\Theta_0[e_A] - \frac{1}{2}D^2\Theta[e_A, e_A] + O(3\text{rd}) \\ &= \frac{1}{2}D\Theta_0[e_A] + \frac{1}{2}D^2\Theta[e_A, e_B] + O(3\text{rd}), \end{aligned}$$

using  $D^2\Theta[\bar{e}, e_A] = \frac{1}{2}D^2\Theta[e_A, e_A] + \frac{1}{2}D^2\Theta[e_A, e_B]$ . Adding the symmetric expression and subtracting from (46) gives (48). By contrast, the v1–v2 correction subtracts  $\frac{1}{2}D\Theta(\gamma^A)[e_A] + \frac{1}{2}D\Theta(\gamma^B)[e_B] = \frac{1}{2}D\Theta_0[e_A + e_B] + \frac{1}{2}\{D^2\Theta[e_A, e_A] + D^2\Theta[e_B, e_B]\}$ , whose second bracket has no counterpart in (46) and therefore survives with a minus sign. The difference between the two scores is  $D\Theta(\bar{\gamma}) - D\Theta(\gamma^A) = -\frac{1}{2}D^2\Theta[\Delta, \cdot] + O(\|\Delta\|^2)$ , i.e. exactly of the order the quadratic correction operates at — which is why the two look interchangeable at first order and are not.

### B.4 What is not proved

The joint CLT for the two-band linear term follows from the Cramér–Wold device and the Lindeberg–Lyapunov condition of Assumption 5.1(g) applied to the summed per-unit influence  $v_S^*(X_i, W_i)\epsilon_{S,i} + v_Y^*(X_i, W_i)\epsilon_{Y,i}$ ; the within-unit correlation  $\rho_{SY}$  is absorbed into  $\sigma_n^2$  by construction (15). Beyond the order calculations of B.1–B.3, complete proofs of Theorems 4.6, 5.3 and 5.8 are not written. Seven items are not merely mechanical, and each is stated as an assumption above rather than derived:

- (1) **The Taylor and evaluation-point remainders.** Assumption 1.9(a), the remainder half of Assumption 5.1(f), and condition (iii) of Theorem 5.8 are imposed directly. Version 4's proposed  $C^1$  modulus is false, as shown by Remark 1.10. The optional  $C^2$  modulus (7) is a conservative sufficient route, not a proved bound over the low-smoothness parameter space. *This is the single most important open item in the paper.*

- (2) **Series linearization.** Equation (21) separates the population-Gram influence term from  $(r_g, r_h)$ ; it is not an exact empirical-LS identity. Assumption 1.9(d) imposes the required functional and second-moment bounds for full- and split-sample fits. A primitive empirical-Gram argument establishing those bounds uniformly is not included.
- (3) **Uniform geometry and fourth-order path control.** Assumption 4.1(c)–(e) is high-level.  $C^1$  consistency can stabilize regularity and transversality under compact separation, but does not preserve a no-support-edge condition without additional separation or tapering. Moreover,  $C^4$  sieve membership guarantees existence of the fourth path derivative, not the  $K_n$ -uniform  $C^1$  bound in (d). Both issues fail outright for non-smooth learners (Remark 4.4).
- (4) **Band, Gram, and weighted-residual control.** Proposition 2.4(b)–(c'') and Assumption 5.1(b),(d') require an empirical-process argument for a  $K_n$ -dimensional band average and for  $\hat{v}^*$  in the standard-deviation norm. The Riesz-weighted residual condition cannot be replaced by global unweighted  $L^2$  consistency. These are nontrivial parts of a sieve variance proof; we assume them.
- (5) **Attainment of the margin bound.** Remark 3.4 exhibits mesh-resonant examples in which the margin diagonal bound  $K_n^{1+1/d}/n$  can be attained, and finite-mesh probes in which the integral is much smaller. Those probes do not establish a generic lower order. The exact order for fixed curved margins, general  $d$ , and heteroskedastic errors is open; settling it could sharpen (or weaken) the  $s \geq d$  of Theorem 2.2(i). Until then  $K_n^{1+1/d}/n$  is used only as an upper bound.
- (6) **Stochastic equicontinuity.** Assumption 1.9(b) is supplied by the global VC bound (8), which yields the window edge  $a < d/(3d - 2)$ . A local-complexity argument — only the  $O(\varkappa^{-(d-1)})$  basis functions meeting the margins are active — would plausibly improve it and relax that edge for  $d \geq 3$ . We have not attempted it.
- (7) **Composite-score conditions for Theorem 5.8.** The representer conditions are stated for the differences  $D\Theta(\bar{\gamma}) - \frac{1}{2}D\Theta(\hat{\gamma}^j)$  rather than inherited from the ordinary case. Target-score equivalence and a conditional Lyapunov condition are imposed in (vi), and ratio consistency of the actual composite-score variance (51) is imposed in (vii). No equivalence with (15) is asserted.

The corner parts of the argument, by contrast, require only the transversality bound, the dimension count  $\dim \mathcal{C} = d - 2$ , and Assumption 1.7 for the lower-bound direction.

## C Numerical Verification of Theorem 3.1

Theorem 3.1 was checked against exact area computations in  $d = 2$  on four designs, chosen so that between them every term of (18)–(19) is activated in isolation. In each case  $\Theta(t) = \int \mathbb{1}\{g_0 + tu \geq 0\} \mathbb{1}\{h_0 + tv \geq 0\} w$  has a closed form;  $\Theta''(0)$  is obtained by a Richardson-extrapolated central second difference and compared with the right-hand side of (18).

1. *Radially weighted disk.*  $g_0 = 1 - \|x\|^2$ , no second threshold,  $w = \rho(\|x\|)$ ,  $u \equiv a$ . Only the transport and curvature terms of  $Q_g$  are active, and (19) reduces to  $\pi a^2 \rho'(1)/2$ . Relative error  $6 \times 10^{-10}$ .
2. *Disk with a non-constant direction.*  $g_0 = 1 - \|x\|^2$ ,  $w \equiv 1$ ,  $u(x) = \|x\|^2$ , so  $\{g_t \geq 0\}$  is the disk of radius  $(1 - t)^{-1/2}$  and  $\Theta(t) = \pi/(1 - t)$ . This activates the  $u(n_g \cdot \nabla u)$  term. Formula and truth both give  $D\Theta = \pi$ ,  $D^2\Theta = 2\pi$  to ten digits.
3. *Disk cut by an oblique chord.*  $g_0 = 1 - \|x\|^2$ ,  $h_0 = x_1 - \kappa$ ,  $w \equiv 1$ ,  $u \equiv a$ ,  $v \equiv c$ . Both margin forms vanish, isolating the two corner diagonals and the corner cross term at  $\cos \omega = -\kappa$ . Checked at  $\kappa \in \{0.6, 0.2, -0.5\}$  (both signs of  $\cos \omega$ ) and  $(a, c) \in \{(1, 1), (1.7, -0.9), (0.4, 2.1)\}$ ; maximum relative error  $2 \times 10^{-10}$ .
4. *Lens of two disks.*  $g_0 = 1 - \|x - p\|^2$ ,  $h_0 = 1 - \|x - q\|^2$  with  $\|p - q\| = D$ ,  $w \equiv 1$ ,  $u \equiv a$ ,  $v \equiv c$ ; both surfaces curved and obliquely intersecting, with  $\cos \omega = 1 - D^2/2$ . The prediction  $\{ac - \frac{1}{2}(a^2 + c^2) \cos \omega\} / |\sin \omega|$  matches the exact lens area's second derivative at  $D \in \{0.8, 1.2, 1.7\}$  and the same three  $(a, c)$  pairs; maximum relative error  $2 \times 10^{-9}$ .

Designs 3 and 4 are the substantive ones. Varying  $(a, c)$  over three non-proportional pairs identifies the quadratic form in  $(a, c)$  separately, so they confirm all three coefficients

$$\underbrace{\frac{2}{\|\nabla g_0\| \|\nabla h_0\| |\sin \omega|}}_{\text{cross}}, \quad \underbrace{\frac{-\cos \omega}{\|\nabla g_0\|^2 |\sin \omega|}}_{Q_g \text{ diag.}}, \quad \underbrace{\frac{-\cos \omega}{\|\nabla h_0\|^2 |\sin \omega|}}_{Q_h \text{ diag.}},$$

and their relative signs, at both signs of  $\cos \omega$ .

The sharpest single check is design 4 with a *one-sided* perturbation,  $v \equiv 0$ : there the cross term is identically zero, so anything nonzero must come from the  $Q_g$  corner diagonal. At  $D = 0.8, 1.2, 1.7$  ( $\cos \omega = +0.680, +0.280, -0.445$ ) and  $a = 1$  the exact lens curvatures are  $-0.463713017, -0.145833333, +0.248456065$ , against formula values  $-0.463713017, -0.145833333, +0.248456064$  — including the sign reversal as  $\cos \omega$  crosses zero. This is the term that v1's order discussion omitted and that (27) restores: the corner contributes to the

plug-in bias even when the two contrast errors are uncorrelated, provided the margins do not meet orthogonally.

**5. Polarization and mesh resonance.** Two further checks support statements made in the body.

(a) *The polarization identity* (37). On all thirteen ( $D$  or  $\kappa$ ,  $a$ ,  $c$ ) configurations of designs 3 and 4, the numerically differenced  $q_{\text{full}} - q_g - q_h$  agrees with  $2C[a, c]$  to full double precision, while the sum of the two  $\cos \omega$  diagonals — which is nonzero and often larger in magnitude — is absorbed entirely into  $q_g$  and  $q_h$ . For instance at  $D = 1.2$ ,  $a = c = 1$ :  $q_{\text{full}} = 0.750000$ ,  $q_g = q_h = -0.145833$ , so  $q_{\text{full}} - q_g - q_h = 1.041667 = ac/|\sin \omega| = 2C$ , against a corner-diagonal sum of  $-0.291667$ . This is what licenses Proposition 4.10: margins-only debiasing removes the  $\cos \omega$  diagonals and leaves the cross term.

(b) *Mesh resonance in* (29). With periodic uniform cubic tensor B-splines on the torus in  $d = 2$  ( $K_n = m^2$ , mesh  $\varkappa = 1/m$ ), we computed  $\mathcal{V} = \mathcal{V}_1 \otimes \mathcal{V}_1$  in closed form — the diagnostic  $\int_0^1 \mathcal{V}_1 = m$  holds exactly and  $\|\mathcal{V}_1\|_\infty = 1.226 m^2$  to four digits for  $m = 16, 32, 64$ , confirming the pointwise scalings (22) — and evaluated  $I_m(\alpha) := \int_{\mathcal{M}_\alpha} \partial_n \mathcal{V} \, d\mathcal{H}^1$  along fixed straight margins of slope  $\alpha$  for  $m = 16, 32, 64, 128$ . The table reports  $|I_m(\alpha)|/K_n$ ; constancy or linear growth in  $m$  would be *diagnostic* of the two candidate scalings, but four mesh sizes cannot establish either asymptotic order:

| $\alpha$                                     | $m = 16$    | 32          | 64          | 128         |
|----------------------------------------------|-------------|-------------|-------------|-------------|
| 0 (mesh-aligned)                             | 16.1        | 20.7        | 66.6        | 27.4        |
| 1 (resonant)                                 | 3.6         | 5.2         | 15.4        | 6.9         |
| 1/2, 8/5 (rational, $q \geq 2$ )             | $\leq 0.12$ | $\leq 0.20$ | $\leq 0.09$ | $\leq 0.59$ |
| $1/\pi, \sqrt{2}, \sqrt{5} - 1$ (irrational) | $\leq 0.43$ | $\leq 0.65$ | $\leq 0.40$ | $\leq 0.48$ |

The first two rows are much larger and strongly phase-sensitive; the last two remain below 0.65 over the tested grid. The resulting finite-mesh spread is evidence that cancellation can be substantial, not a two-sided rate result. In particular, the rational slopes 1/2 and 8/5 do possess exact resonant Fourier pairs; their small values here can reflect small coefficients or phase cancellation and must not be read as  $I_m = O(K_n)$  asymptotically.

**6. The support-edge term of Remark 1.12.** Take the unit disk  $g_0 = 1 - \|x\|^2$  with the *fixed* support  $\mathcal{X} = \{x_1 \geq \kappa\}$ ,  $w \equiv 1$ ,  $u \equiv a$ , and no second threshold. The exact curvature of the circular-segment area matches  $-2a^2 \cos \omega_\partial / (4|\sin \omega_\partial|)$  with  $\cos \omega_\partial = -\kappa$  at  $\kappa \in \{0.6, 0.2, 0, -0.5\}$  and  $a \in \{1, 1.7\}$  to ten digits, including the exact vanishing at  $\kappa = 0$  (orthogonal meeting). This confirms both the form of the edge term and that it carries no cross component.

**7. The doubly debiased score (Remark 5.6).** On the two-disk lens with  $D = 1.2$  and direction  $\eta = (a, c) = (1, 0.8)$ , write  $\phi(t) := \Theta(\gamma_0 + t\eta)$ , so  $\phi''(0) = D^2\Theta[\eta, \eta] = +0.5942$ , and set  $e_j = t_j\eta$ . Three estimators are compared:  $\Psi$  (pure SS), the v1–v2 estimator (45), and the composite-score estimator (49), all evaluated exactly.

(a) *The counterexample*  $e_A = e_B = b$ .  $\Psi$  reduces to  $\Theta(\gamma_0 + b)$  and (45) has error  $-2.679 \times 10^{-3}, -7.044 \times 10^{-4}, -1.808 \times 10^{-4}, -4.580 \times 10^{-5}$  at  $t = 0.1, 0.05, 0.025, 0.0125$  — quartering as  $t$  halves, i.e.  $O(t^2)$ , and matching  $-t^2\phi''(0)/2$  ( $-2.971 \times 10^{-3}, \dots, -4.642 \times 10^{-5}$ ) to the accuracy of the third-order term. The error is quadratic, not cubic.

(b) *Independent mean-zero errors.* With  $t_A, t_B$  i.i.d.  $\mathcal{N}(0, \varsigma^2)$  and  $4 \times 10^5$  draws:

| $\varsigma$ | $\mathbb{E}[\Psi - \Theta_0]$ | $\mathbb{E}[(45) - \Theta_0]$ | $\mathbb{E}[(49) - \Theta_0]$ | $-\varsigma^2\phi''(0)$ |
|-------------|-------------------------------|-------------------------------|-------------------------------|-------------------------|
| 0.08        | $3.4 \times 10^{-5}$          | $-3.86 \times 10^{-3}$        | $1.7 \times 10^{-5}$          | $-3.80 \times 10^{-3}$  |
| 0.04        | $-1.0 \times 10^{-4}$         | $-9.56 \times 10^{-4}$        | $2.1 \times 10^{-6}$          | $-9.51 \times 10^{-4}$  |
| 0.02        | $3.0 \times 10^{-5}$          | $-2.38 \times 10^{-4}$        | $1.7 \times 10^{-7}$          | $-2.38 \times 10^{-4}$  |
| 0.01        | $-3.9 \times 10^{-6}$         | $-5.96 \times 10^{-5}$        | $1.2 \times 10^{-9}$          | $-5.94 \times 10^{-5}$  |

$\Psi$  is unbiased at this order (its leftover  $\frac{1}{2}D^2\Theta[e_A, e_B]$  is mean zero); (45) has bias  $-\varsigma^2\phi''(0)$  to three significant digits, confirming Remark 5.6; and (49) is unbiased with a residual decaying like  $\varsigma^4$ .

**8. The two polarization families (Remark 4.5).** On the lens design with  $D = 1.2$ ,  $\eta = (a, c) = (1, 0.8)$  and half-sample errors  $e_j = t_j\eta$ , all four objects of (34) and (35) are computable in closed form. The two *within-family* identities  $\hat{\Theta}_{SS}^{\text{mar}} - \text{corner} - \hat{\Theta}_{SS}$  and  $\hat{\Theta}_{SS}^{\text{mar, jack}} - \widehat{\text{corner}} - \hat{\Theta}_{SS}^{\text{jack}}$  evaluate to *exactly* 0 (machine zero) at every  $(t_A, t_B)$  tried. The *cross-family* gap  $\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS}$ , which v3 asserted was zero, is not:

| $t_A$ | $t_B$ | $\hat{\Theta}_{SS}^{\text{jack}} - \hat{\Theta}_{SS}$ | $-\phi^{(4)}/384$       | ratio |
|-------|-------|-------------------------------------------------------|-------------------------|-------|
| 0.40  | 0.16  | $-8.943 \times 10^{-6}$                               | $-8.896 \times 10^{-6}$ | 1.01  |
| 0.20  | 0.08  | $-8.674 \times 10^{-7}$                               | $-8.646 \times 10^{-7}$ | 1.00  |

confirming Lemma 4.2’s constant  $1/384$  to one percent. (Smaller  $t_j$  are not reported: the five-point stencil for  $\phi^{(4)}$  amplifies rounding by  $h^{-4}$  and the comparison becomes noise-limited below  $\sim 10^{-7}$ .) This is the numerical content of Remark 4.5: each family is internally exact, and only the cross-family statement carries the fourth-order error.

The scripts for items 1–8 sit alongside this document:

`verify_D2_theorem.py`, `verify_polarization.py`, `verify_mesh_resonance.py`,  
`verify_dd_score.py`, `verify_offdiag_order.py`, `verify_two_families.py`.

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